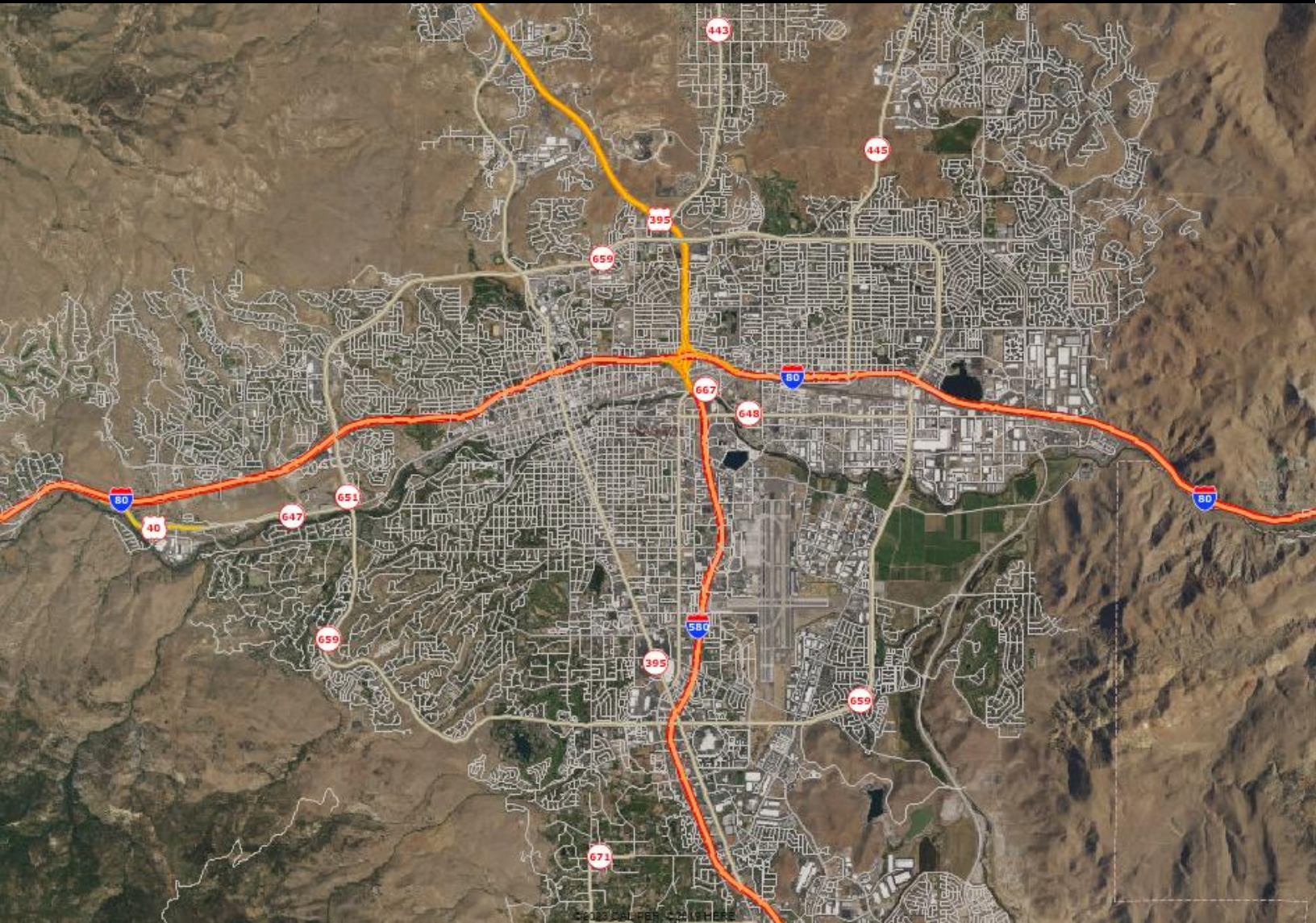


Technical Documentation for RTC Travel Model Development

Submitted by
Caliper Corporation



rtcwashoe.com



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Introduction

The Regional Transportation Commission (RTC) undertook a major update to their existing travel demand model (RTCWCM) beginning in September 2023. Caliper Corporation, with support from Wood Rodgers, were selected to develop a travel demand model that would run within the TransCAD Version 10 environment with a base year of 2022/2023. Like the previous version of the model, the model has a hybrid trip-based/tour-based architecture. This latest version of the model includes a number of advances over RTC's existing hybrid model. Machine-learning was used for trip production models as and nested destination choice models replaced the traditional destination choice models. The new design includes full spatial and modal linkage between home-based and non-home-based trips and consistency with tours and trip-chaining as well as robust treatment of special travel markets including non-motorized, transit, visitor, freight, university, and airport travel.

Data

The update of the model to the new, post-pandemic base year was supported by and made use of a number of surveys undertaken by RTC to study travel behavior and patterns in the Reno region as well as big data from connected vehicles supplied by Caliper together with traffic counts for the region. The surveys were all collected by ETC Institute on behalf of RTC and included:

- 2023 Regional Transportation Commission of Washoe County On-board Transit Survey
- 2024 Washe County Household Travel Survey
- 2024 Regional Transportation Commission of Washoe County Visitor Travel Survey

The household survey is supported the development of most of the demand components of the regional model, while the transit survey and visitor survey contributed to those specific components of the model. The 2024 Household Travel Survey collected information for a random sample of 1,998 households across the region shown below in Figures 1 and 2.

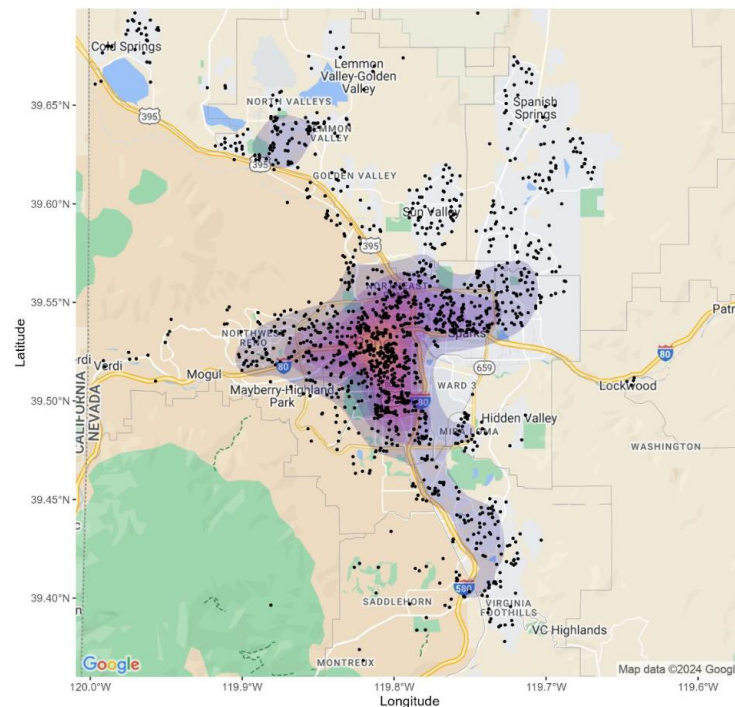


Figure 1. Location of surveyed households (source: 2024 Washoe County Household Travel Survey Final Report)

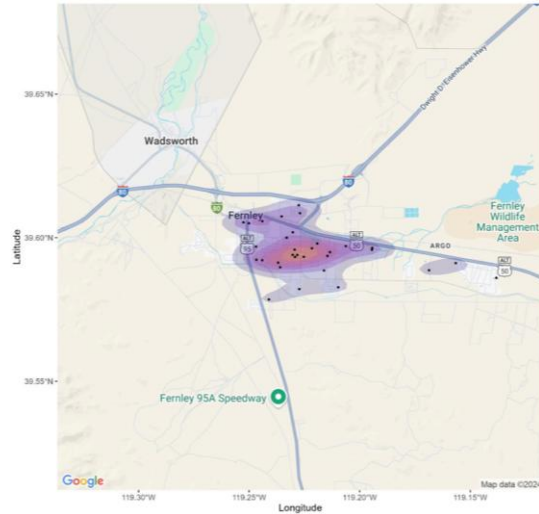


Figure 2. Location of surveyed households, Fernley detail (source: 2024 Washoe County Household Travel Survey Final Report)

The survey was collected through a variety of mediums including smartphone app (13.4% of respondents), phone (44.1% of respondents), and website (42.5% of respondents) and collected information on 20,492 trips. For more information refer to the [2024 Washoe County Household Travel Survey Final Report](#).

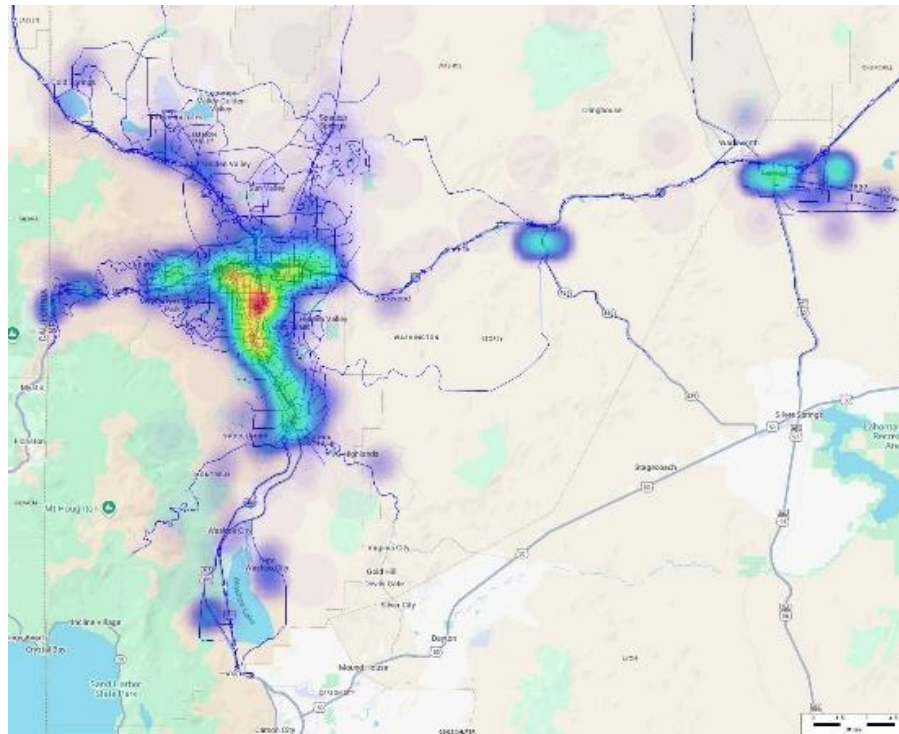
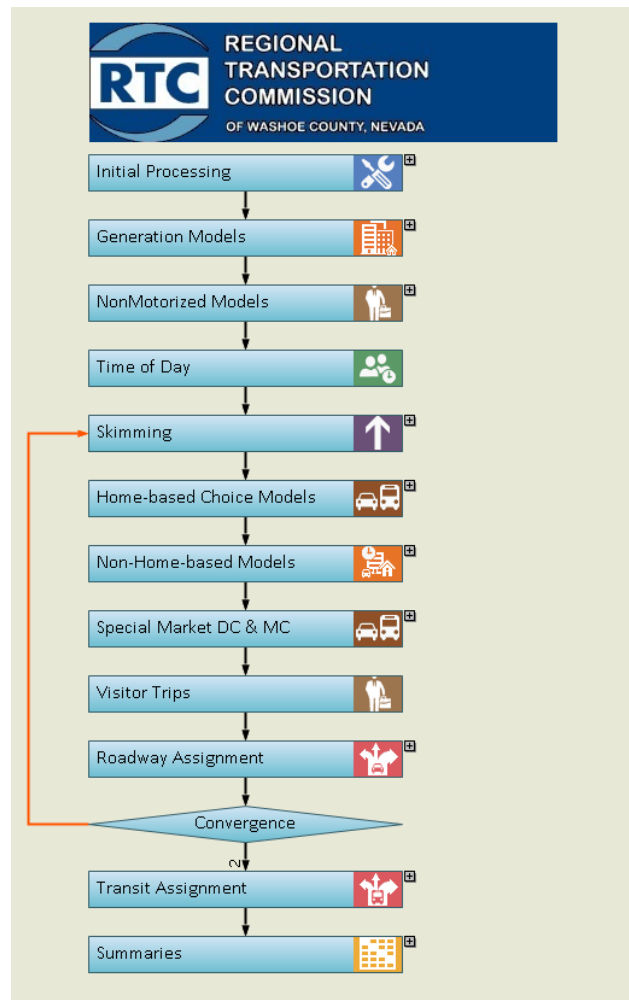


Figure 3. Heat Map of Internal Attractions of External Trips from Transography

Caliper's Transography data was also used to supplement the survey data, particularly for external trips not covered by the surveys and to support destination choice model development. The Transography data included observation of 1,354,706 connected vehicle trips made over the course of twenty weekdays in April 2022.

Using the Model

The new model also greatly improves on the user experience, making use of the flowchart interface available in more recent versions of TransCAD. Model installation is greatly simplified; after the model is unzipped, the user only needs to drag and drop “rtwc.model” into TransCAD. The model’s flowchart will open and all the relevant code is compiled. The flowchart clearly illustrates the modeling process, and each step can be expanded to see more detail. The model is designed to be intuitive to use without heavy reading of this documentation.



Improved user experience also extends to model maintenance and regional collaboration. The model is housed in a GitHub repository. This allows RTC and its partners (NDOT, TMRPA, Fernley, etc.) to coordinate and contribute to the model in an organized manner. It also makes it easy for model users and regional stakeholders to obtain the model, and they can also provide feedback using GitHub issues.

TransCAD and Computer Information

The new Reno model (RTCWCM v1.2) uses TransCAD 10 build 40715. Later versions may require different builds. See the release notes for which build is required. If your version of TC is too old, a warning message will appear when opening the flow chart.

Caliper achieved reported processing times using the Dell machines with 12-th gen Intel CPUs.

Dell XPS Desktop 8950

12th Gen Intel(R) Core(TM) i9-12900K processor (16-Core, 30MB Cache, 3.2GHz to 5.2GHz)

Performance CPU liquid cooling (optional)

64GB DDR5, 2x 32GB, at 4400MHz

2TB M.2 PCIe NVMe solid state drive

Windows 11 Pro (this is key for optimum performance)

Downloading the Model

The model can be downloaded in two ways:

- Access the latest release zip file here:
https://github.com/Caliper-Corporation/Reno_TDM/releases
- Clone the open source model repository from here:
https://github.com/Caliper-Corporation/Reno_TDM.git

The model downloaded from the repository will be the latest official version of the model. The model can be cloned or unzipped into any directory you choose.

Opening the Model

Find the 'rtwc.model' file, drag and drop it into TransCAD (or from TC, use File -> Open).

All model scripts are compiled automatically when the flowchart is opened. This simplifies model installation and also ensures that the latest scripts are used.

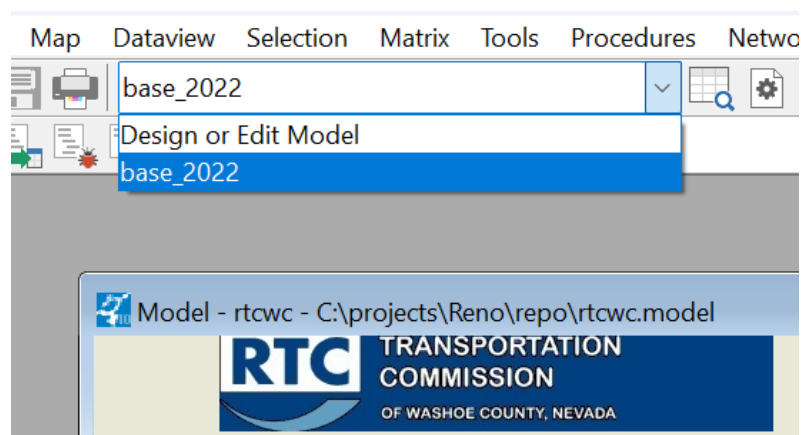
Creating and Managing Scenarios

Models are used to predict travel behavior given various scenarios of land use and transport capacity. Base validation year scenarios are used to verify that model predictions are reasonable while future year scenarios are used to make planning decisions. The model package includes validation year, the MTP year, plus interim year scenarios.

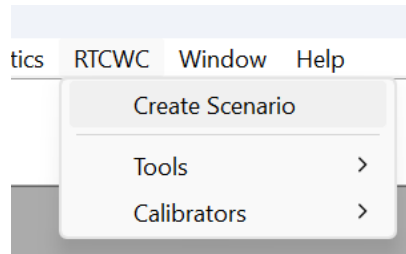
Base Year Scenario

The model comes with the validation year scenario folder included. Find it under the "scenarios" directory. Note that it includes both the highway and transit project lists.

To select it in the flowchart, use the scenario dropdown menu as shown below.



Once the scenario is selected in the drop-down menu, check the input parameters as noted in section 6 below. Once confirmed, choose "Create Scenario."



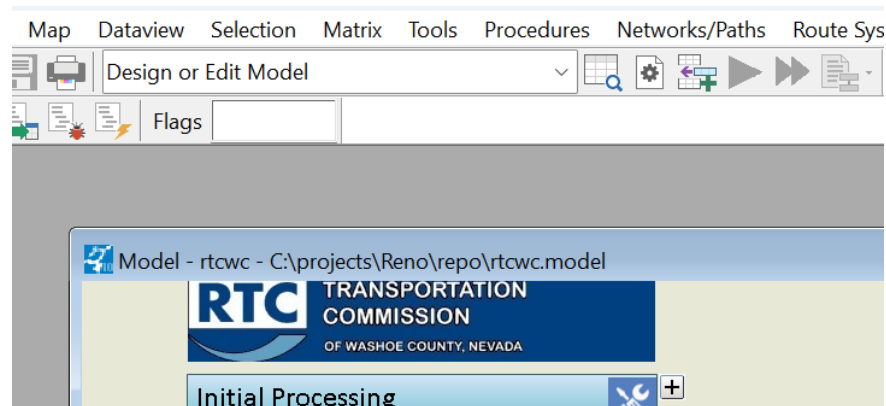
This will build the scenario from the master files and include the appropriate transit routes and roadway projects. Note that this step is REQUIRED prior to running the model and must be run without error. If an error message is returned, after fixing the issue, rerun the "Create Scenario" step.

Creating New Scenarios

Custom scenarios can be created using the following steps.

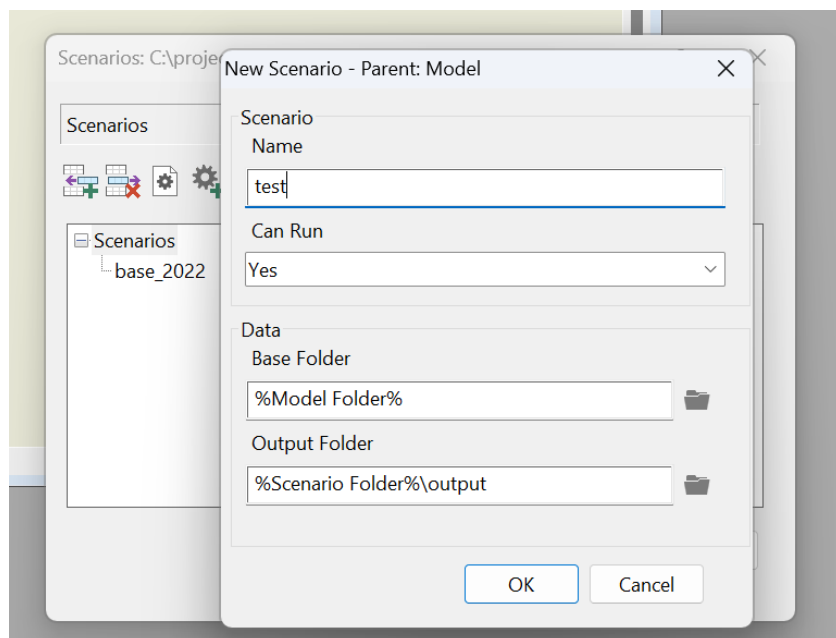
1. Create a folder in the "scenarios" directory and include the roadway and transit project lists with the appropriate project IDs.
2. New scenarios can inherit parameters from other scenarios becoming child scenarios or be created as a new scenario. In the "Scenario Selection" dropdown, Select "Design or Edit Model" to create a new scenario that uses the master files or the scenario to inherit from.

For an example, we set up a new scenario without inheriting anything by highlighting "Design or Edit Model" first.



This will have the same parameters as the base year scenario, but importantly, any changes made to the base scenario's parameters will not affect the test scenario. For child scenarios, changes made to parent scenarios are inherited from the child scenario.

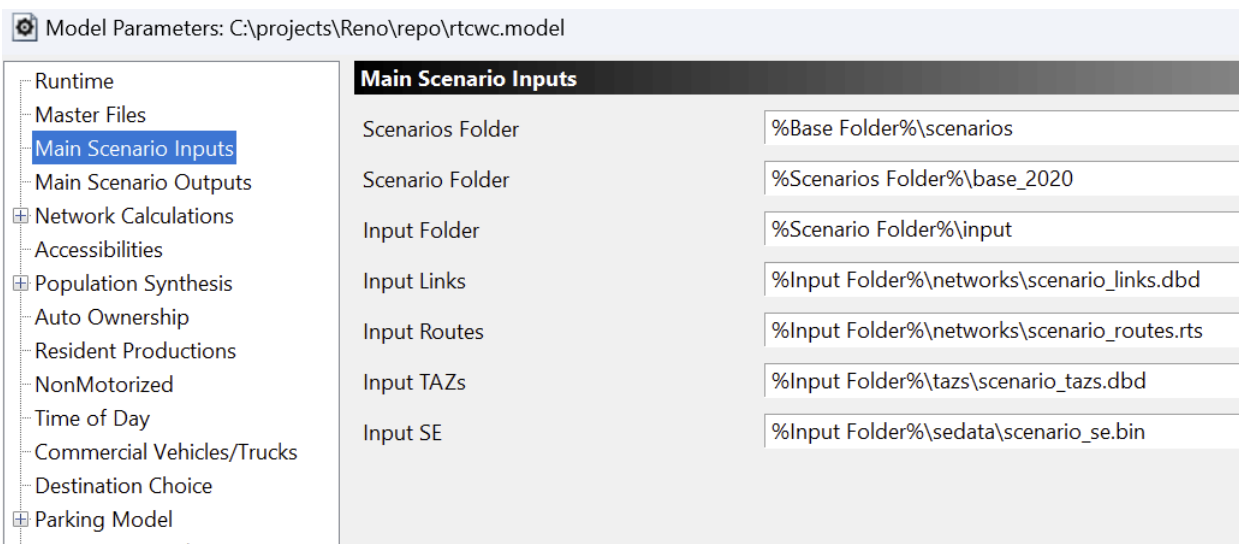
3. From the "Add/Remove Scenario" menu item dropdown, choose the "Add a New Scenario" button.



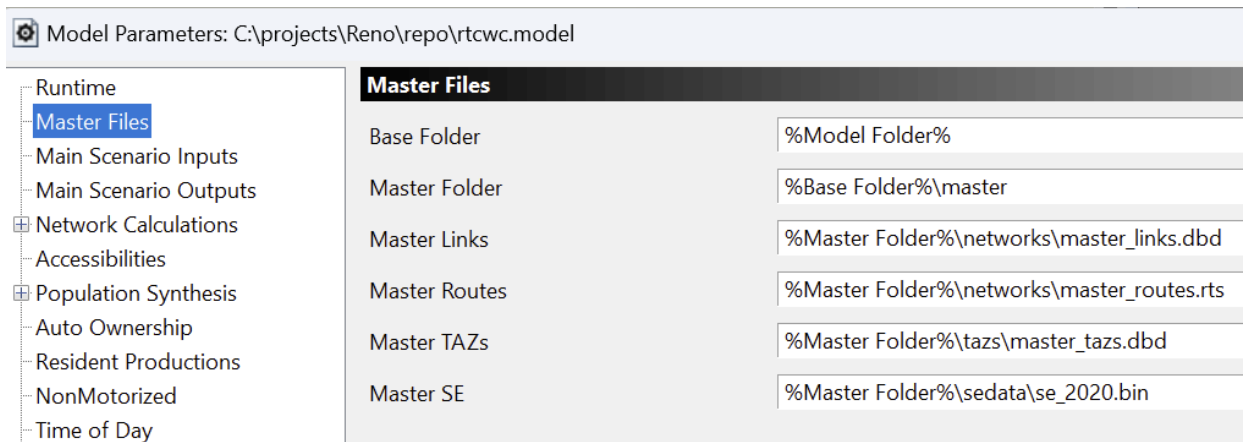
4. Name the new scenario, accepting all other defaults, and click "OK".
5. Click "OK" again to close the scenario dialog box.
6. Check/modify common scenario input parameters:

Select the created scenario (e.g. test) from the scenario dropdown box and double-click empty space in the flowchart to open the parameter editor, or select the "Modify Parameters" menu option.

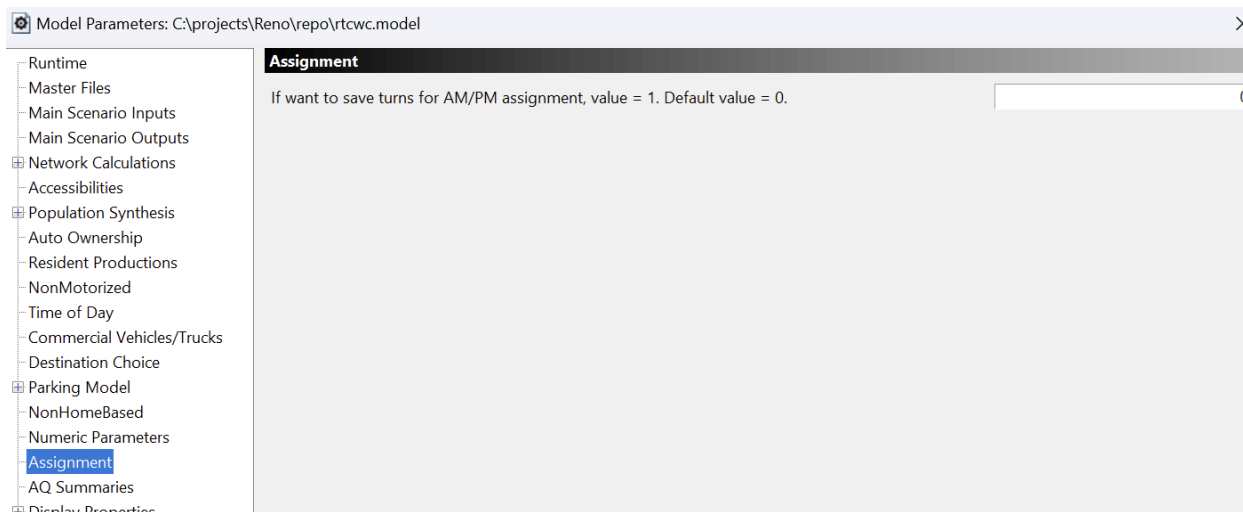
Make sure that the Scenario Folder is pointing to the correct location. To change it, click on the folder icon on the right. In the pop-up, select the folder location in the "Value" section and browse to the correct folder location in the dropdown.



If se data alterations are involved in this scenario (e.g. you're running the 2050 scenario), right click in the model panel beside the flow chart, select "master files" and then "Master SE". Click the icon to the right of the file name to select the edited se_xxxx.bin input file. Note that this is the se_xxxx.bin file that will be copied to the scenario folder (_name) as "scenario_se.bin". The SE data file is located here:



Other parameters are unlikely to be changed for most model applications. However, if you'd like to save turns for the AM/PM assignment, you can change the save turn flag to 1. Find this under Modify Parameters / Assignment:



Many of the model's parameters are stored in CSV files to make viewing them in GitHub easier. The flowchart can be used to browse to those files or open them in TransCAD. Currently, CSV files cannot be edited in TransCAD, but this is under active development.

7. Select the new scenario from the drop-down, and select "Create Scenario" from RTCWC menu. Note that this step is **REQUIRED** and must be run without error. If an error message is returned, after fixing the issue, rerun the "Create Scenario" step.

Running Scenarios

The scenario can be run using the green play arrow in the toolbar. This is the most common method. Alternatively, individual model steps can be run by right-clicking them in the flowchart and choosing the relevant option under the "Run" context menu. Note that "Run Model to Here" will run the model to this step but will not run this step.

For more information on running the model and managing networks to develop scenarios, refer to the model's latest user's guide, available on the model's GitHub site at https://github.com/Caliper-Corporation/Reno_TDM/wiki.

Model Structure

As mentioned in the introduction, the RTCWM is an advanced hybrid model. Hybrid travel models are the most recent family of model designs to evolve and generally developed as an attempt to compromise between practical and theoretical concerns or trip-based and activity-based designs. More recently, some hybrid models have also been developed as an incremental step on the way to the development of an activity-based model. (Vyas, 2017) The hybrid designation is a somewhat loose one in that it describes a wide range of model designs that offer some blend of trip-based and activity-based features. They can be distinguished from activity-based models in that they have at least some aggregate components and generally (but not always) do not employ Monte Carlo simulation. They can be distinguished from trip-based models in that they make some connections between trips, offer improved consistency with tours, and generally have at least some disaggregate components.

Most but not all hybrid models have at least some disaggregate component models at the front end of the model stream which use individual travelers or more commonly households as agents in choice models in a database or table-based framework similar to activity-based models. These component models may in some cases be mathematically identical to corresponding component models in activity-based models. In many, but not all, hybrid models, they are implemented differently, however, and applied to estimate expected values rather than realizations of probabilities using Monte Carlo simulation. However, this approach cannot be extended beyond the spatial or destination choices because the size of the data would explode due to the need to keep track of the probabilities for every candidate destination (and mode, etc.). Therefore, all hybrid models ultimately use aggregate trip matrices like traditional trip-based models at the end of their model stream.

Although it can be hard to generalize about hybrid models to some extent due to their diversity, they all, to varying degrees do offer some compromise between traditional trip-based and full activity-based models. All offer some improved consistency with tours and consistency between travelers' choices and most offer at least some reduced aggregation bias, but none as much as fully disaggregate models. Because, like activity-based models more of their components tend to be discrete choice (logit) models, most offer at least some improved sensitivity to additional factors such as urban form, walkability, etc., and some hybrid models may be just as sensitive as activity-based models to these factors. Because of the lack of fully disaggregate data, none of the hybrids can conduct the type of equity analyses that full activity-based models make possible, and most hybrid models treat time more simplistically than activity-based models, although in some cases still more robustly than traditional trip-based models.

The RTCWM is a hybrid model in four regards:

- **Population Synthesis & Disaggregate Components** – it begins by synthesizing a disaggregate population for the region and the trip generation, non-motorized mode choice, and time-of-day choices are disaggregate, person-level models.
- **Trips Segmented by Tour Type** – although the model generates trips rather than day patterns or tours, it segments trips based on whether they are on a work or non-work tours so that trips on different tour types reflect the distinct patterns and characteristics of those tours.
- **Advanced, Behavioral Component Models** – like an activity-based model, many of the Reno model's components are logit discrete choice models which are grounded in the behavioral theory of random utility maximization and capture the sensitivity of traveler's choices to more factors than traditional trip-based models such as accessibilities and the mixture of land uses. The Reno model also includes decision tree models, but specifically ones that have been audited for behavioral realism.
- **Linkage of Home-based and Non-home-based Trips / Consistency with Tours** – unlike traditional four-step models which can imply physically impossible travel patterns with people

making non-home-based trips between zones they never travel to on a home-based trip, hybrid models like Reno's enforce the conservation of travelers by connecting non-home-based trips and home-based trips.

The overall structure of the resident passenger components of the model (which account for the vast majority of travel in the region and comprise the majority of the model itself) are displayed in Figure 4 below.

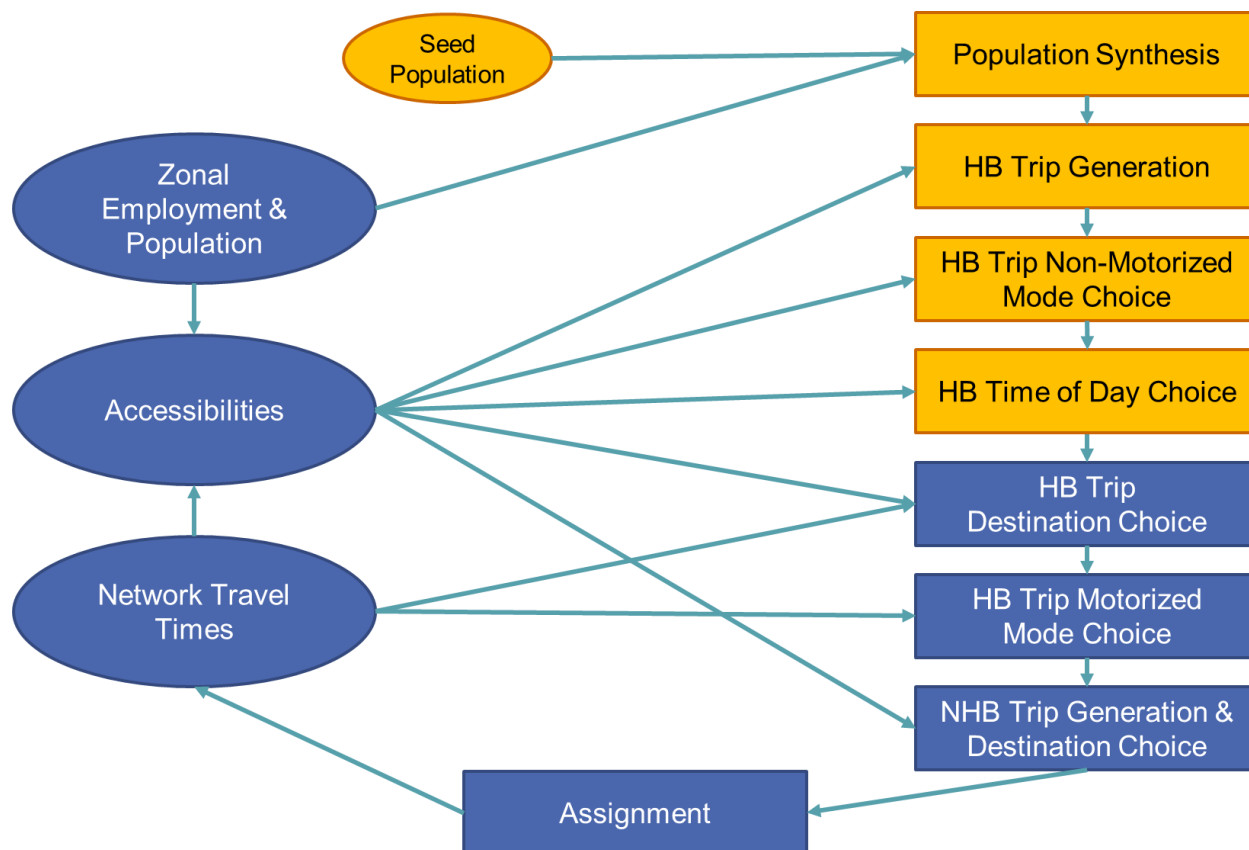


Figure 4. The Reno Hybrid Model Structure (disaggregate components shown in gold; aggregate components shown in blue)

Data sets are shown to the left in ovals, while model components are shown in boxes. Disaggregate components are shown in gold and aggregate components are shown in blue. The remainder of this report steps through and documents the various model components shown above, along with the auxiliary demand models for external trips, trucks, and visitors.

Model Networks

Introduction

The highway and transit networks are critical inputs for any travel model. They provide travel time, accessibility, and other impedance information. This documentation uses the term “network” specifically to refer to the machine-readable files (.net and .tnw) TransCAD uses for shortest pathing and assignment. The GIS files that planners and modelers review and edit are referred to as (GIS) layers. This language is something Caliper is trying to promote to clarify which data component is being referenced.

Roadway Layer

Like the prior version of the model, the new Reno regional model uses a master layer designed to contain all projects stakeholders might consider during transportation plan development (including competing ideas). This allows each scenario layer to be exported from the master with little or no manual editing. Information on using the project management tool is covered in the user guide. The Caliper team also updated the master layer to reflect a 2022/2023 base year, the target year for final model delivery. In other words, if you export a scenario layer without any projects, it will reflect the 2022/2023 road system. This was done by permanently modifying the base attributes of the master layer with any projects scheduled to be complete by the base year date.

Update to 2022/2023 Conditions

The most significant change to the roadway network is the addition of the Fernley area to the east of the prior model boundary. Caliper integrated the added network for the area already created for a special scenario into the master network.

Caliper then reviewed the network and generally found the model’s roadway network coverage to be reasonable and sufficient. However, a small number of additional roadways were identified for inclusion in the new base network:

- Crystal Canyon Blvd.
- Magenta Dr. / Lightning Dr.
- Tracy Rd.

Some alignments were also updated or corrected:

- Copper Canyon Pkwy.
- Lear Blvd.
- Estates Dr.
- Kiley Pkwy.
- Meadowood Mall Way
- Ridgeview Dr.
- SB 580 Ramps at Meadowood
- Windmill Farms Blvd.
- TRI Connector

Greiger Grade was also extended to the south.

Together with the redesign of the model zones, the centroid connectors were also updated for the network. As the zones increased, the number of centroid connectors did as well, from 1,858 to 2,029. However, they did not increase at the same rate. The increased number of zones allowed for slightly smaller zones on average, and these smaller zones were often more focused on a single loading point. The number of connectors per zone therefore decreased from 2.1 in the old model to 1.7 in the new model. Generally, a lower ratio indicates a better model and a better correspondence between the zones and the network.

Traffic counts were updated to reflect the new 2022/2023 base year. RTC staff provided traffic counts for 2022 and 2023. Daily traffic counts were incorporated into the network from 808 stations. Truck counts were also incorporated but were only available for 73 locations.

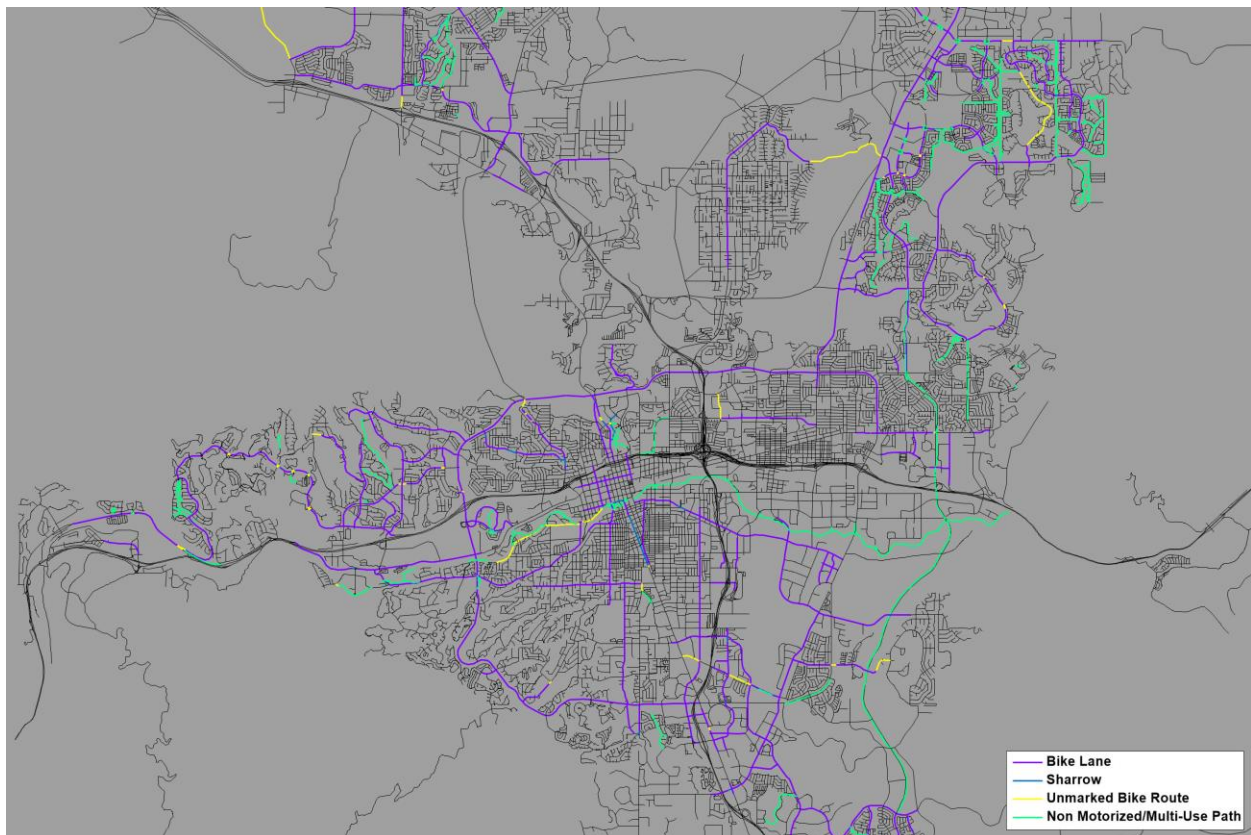
The network attributes were also changed to use a more recent master network schema that allows the user to specify additional changes as part of projects, such as changes in median type. For details on the network attributes and use of the master network, see the user's guide.

Count Data

RTC provided traffic counts for 2023 for model validation. After filtering to identify count stations on the assigned roadway network, a total of 855 counts were able to be attached to the roadway network. Of these, 110 included truck counts.

Incorporation of All-Streets for Walk and Bike Modeling

As part of the update to v1.2, Caliper upgraded the road layer to include all local streets and non-motorized, multi-use paths. These additional streets provide a much better representation of the walk, bike, and transit paths travelers actually take. Attributes related to biking were also incorporated including presence of bike lanes and sharrows.



While these added streets improve walk and bike network accuracy, they are not used by the model in assigning motorized vehicles to the network. The DTWB field clarifies which basic modes can be used on any link:

- D: Drive
- T: Transit
- W: Walk
- B: Bike

A freeway link would have a value of “DT”, which means only drive and transit are allowed. Most links have a value of “DTWB” meaning all modes are allowed.

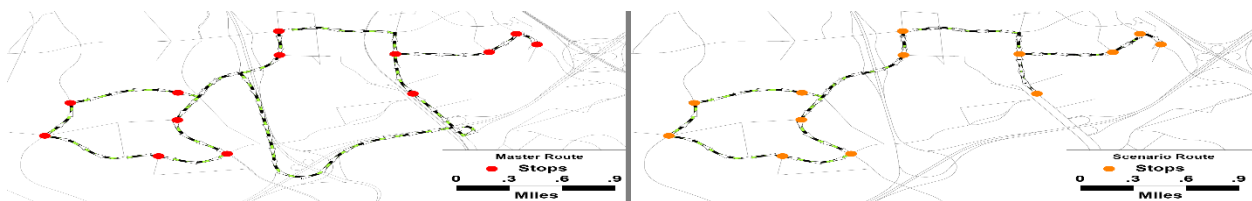
The minor local streets not used for the vehicular assignment are coded “TWB” and do not necessarily have accurate attribute information (such as posted speeds) and for that reason should not be included for use by vehicle travel. They will only be used by non-motorized and transit travelers. This means the added streets do not burden model users with additional attribute maintenance.

Transit Layer

The new Reno regional model adds a master layer for transit routes along with a project manager specially designed to transfer routes accurately between line layers. This enhancement greatly reduces the burden of coding and managing projects between various base and future year scenarios.

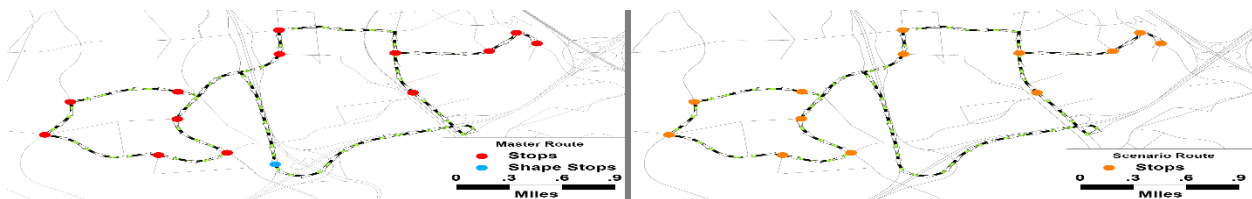
More detail on how to use the new tools are included in the user’s guide, but information on the creation of the master route system requires a basic understanding of the tool. In short, the transit project manager finds the shortest path between route stops to reconstruct the route on a new layer. This approach struggles when route stops are not frequent enough to get an accurate path, but the Reno regional model solves this problem using “shape stops”. These are extra stops along the route that are used to improve alignment, but are then removed.

The maps below give a good example of a potential problem when transferring routes between layers using only stop locations only. The shortest path between stops is not always the route a bus takes.



Transferring without shape points

The maps below show how a single shape stop solves the alignment problem while maintaining the correct number of stops.



Transferring with shape points

Each time a scenario is created, the transit project manager writes out a file next to the route system detailing how accurate the transfer went. This is achieved by comparing lengths before and after as well as checking for any missed stops. Shape stops can be added until the match is exact, but minor differences in route lengths do not have a measurable impact on model results. The table below provides a sample of the comparison table.

projid	master_route_id	scenario_route_id	master_length	scenario_length	diff	pct_diff	missing_stops
1	250	99	9.2	9.0	-0.1	-1.2	0
1	278	106	9.7	9.6	0.0	-0.2	0
2	328	132	14.4	14.3	-0.1	-0.8	0
2	329	133	14.3	14.2	-0.1	-0.7	0
31	48	10	5.5	5.5	0.0	0.0	0
31	49	11	4.2	4.2	0.0	0.0	0

Transit Network Checking

Transit networks are the files TransCAD uses for shortest path, assignment, and other procedures (extension: “tnw”). During the model run, the route system file is converted into many different networks. In the Reno regional model, a separate transit network file is created for each combination of

- Time of Day
 - AM
 - MD
 - PM
 - NT
- Transit Mode
 - Local Bus
 - Express Bus
 - Bus Rapid Transit (if present)
 - Commuter Rail (if present)
 - Light Rail (if present)
- Access Mode
 - Walk
 - Kiss-and-Ride (aka drop off)
 - Park-and-Ride

Creating separate files facilitates the critical step of auditing the networks. The model translates the route, stop, link, and node layers into a transit network, and it is important to make sure the networks are behaving appropriately. In the Reno regional model, for example, the AM and PM PNR networks behave differently. In the AM, where trips are primarily from home to destinations like work, drive access works in the PA direction:

1. Drive to a PNR lot
2. Walk to the bus access stop
3. Ride the bus to the egress stop
4. Walk to the final destination

In the PM network, trips are primarily back to home. For this reason, the drive access flow is reversed.

1. Walk to the access stop
2. Ride the bus to the egress stop
3. Walk to the PNR lot
4. Drive home

TransCAD can handle this behavior, but it is always important to check that the model code is setup correctly. Separate network files allowed the two maps below to be created quickly to confirm proper behavior.

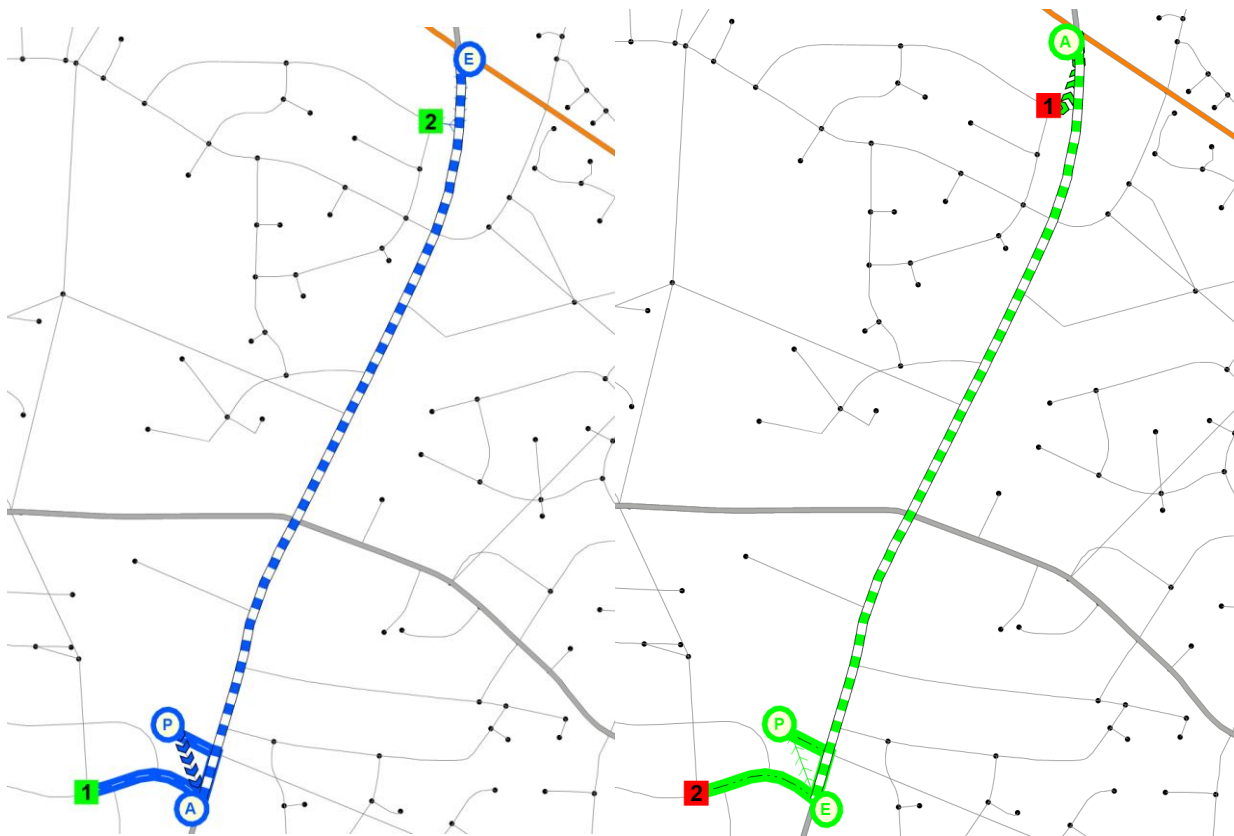


Figure 5. AM vs PM transit skims

TAZ and SE Data

Introduction

In the Reno regional model, as in most travel models, the geographic area being analyzed is split into many smaller spatial units or zones. These *TAZs* (which stands for travel, transportation, or traffic analysis zones) have several uses, including: storing information about the people and places in each zone, serving as origins and destinations of trips, and calculating travel times between (and within) zones.

TAZ Boundary Updates

As with the network, the first major modification of the zone system was the addition of zones to the east to cover the Fernley area. Beyond this, Caliper undertook an extensive review of the Reno regional TAZ system. Factors considered in the review included:

- Avoiding concavities in the model area boundary
 - The overall model study area boundary now is largely convex. Small concavities remain, but no large ones remain which would likely cause travel originating from and destined to model zones to travel outside the study area before returning to it.
- Minimizing zones with overly large population (>2,000) or employment (>2,500)
 - It is desirable to minimize the number of these zones as they are apt to cause overloading of the network.
 - Zones with population over 2,000 decreased from 33 in the old model to 16 in the new model.
 - Zones with population employment over actually increased from 16 to 19, but this was due to the growth of several large employers such as hospitals and casinos which should not be split.
- Separating homogenous residential areas from homogenous commercial areas
 - It is desirable to make residential neighborhoods distinct origins/destinations from commercial developments particularly if they access the network at different locations.
 - Comparing the employment of the zone to the sum of employment and households, the percentage of zones which were homogenous (<10% or >90% employment) increased from 42% to 58%.
- Maximizing consistency with census block and block group boundaries
 - It is advantageous in the estimation of socio-economic attributes for zones if the zones conform to census boundaries. Conformity with property boundaries did not allow perfect conformity with census boundaries as well, but the number of zones violating census boundaries was reduced.
- Elimination of unnecessary small zones
 - Additional unneeded zones simply add to the model's runtime (which is a function of the number of zones squared). Therefore, small zones with little population or employment (or prospect thereof) were combined to created larger zones.

The prior model included 896 zones. Eighteen zones were added for Fernley in the special scenario version of the old model bringing the total to 914 zones. As a result of the review, there was a net increase of 250 zones to bring the total in the new model to 1,064.

Socioeconomic Data Update

The socioeconomic data for all model scenario years was stored as attributes on the TAZ layer in the previous version of the model. In the new model, the TAZ layer contains only basic, scenario-independent attributes while scenario-specific socioeconomic data is contained in scenario-specific .bin tables. The socioeconomic variables used in the new model are also different than those in the prior model. Both models use the number of households, the population in households, and the number of households by household size. The new model uses the median household income rather than a breakout of the number of low-, medium-, and high-income households. The new model also uses the percent of workers (versus people) in households in each zone; the prior model did not take the number of workers as a direct input. The new model also uses the percent children and the percent seniors in place of the population by age category though these are very similar and related variables. On the economic side, the new model takes the number of employees by the 25 two-digit NAICS categories in place of the six custom industry categories in the old model. The actual socioeconomic data was prepared by Truckee Meadows Regional Planning Agency. Please contact Damien Kerwin or Jeremy Smith for more information on the preparation of the socioeconomic data.

Accessibility

Network travel times and zonal socioeconomic variables can be combined to calculate accessibility variables. Accessibility is a measure of a traveler's ability to reach locations where they can engage in necessary and desired activities.

Accessibility can be measured in various ways and different measures are useful for different purposes. It is not uncommon for planners to use accessibility measures such as the number of jobs accessible within a 30-minute commute because measures such as this are readily understandable. However, metrics like this are not as good for analysis and modeling because they can exhibit significant discontinuities or "cliff-effects" where, for example, a large employer is 31 versus 29 minutes away. For this reason, in travel models, more continuous but sometimes less intuitive accessibility measures are used.

Another way accessibility can be measured, more similarly to the way it is commonly measured in models is the average cost (usually in terms of minutes) of a trip from a given location, perhaps for a particular purpose, by a particular mode, and/or at a particular time of day.

Accessibilities, thought of as the average or expected cost of a trip, affect many aspects of travel behavior and are therefore included in many components of behaviorally realistic advanced travel models. The average cost of trips (in miles) from a residence location clearly affects auto ownership because, for instance if most activity destinations can be reached within a mile, a car is not necessary. Residents of dense, highly-accessible areas also can afford to make more trips, because their trips are shorter on average, and so they can accomplish more trips in the same travel time budget. Thus, downtown dwellers make more, shorter trips than the regional average and rural residents make fewer, but longer trips. Although some older trip-based models partially reflected this through the stratification of trip generation rates by area type, the use of continuous accessibility variables in generation models is clearly preferable both in terms of statistical support and model response properties. To fully and accurately reflect this fundamental difference between travel patterns by residents of different parts of the region, accessibility must also be incorporated in destination choice. The accessibility of the home location scales residents' willingness to travel (sensitivity to travel time) because when a person chooses their residence location, they are also choosing how far they are generally willing to travel. The accessibility of destinations is also a factor in destination choice because it measures the cost of an additional (non-home-based) trip from that location.

A number of accessibility measures are therefore used in the Reno regional model to measure these various effects of accessibility on travel behavior and patterns.

Logsum Accessibilities

Logsums can be considered as a weighted average accessibility. While logsums are most often discussed in relation to mode choice and destination choice models, the logsums used for these accessibility measures are based on a simpler gravity formulation as a proxy and to avoid endogeneity problems from simultaneously estimating destination choice models including logsums of those same and other destination choice models.

$$A_i = \ln\left(\sum_j S_j t_{ijm}^\beta e^{\gamma t_{ijm}}\right)$$

Where:

- A_i : Accessibility of zone i
- t_{ijm} : Travel time between zone i and j by mode m
- S_j : Attractions in zone j (size term)

Accessibility logsums are useful for many steps of the model. One example is the non-home-based production models. A non-home-based walk trip is more likely to occur in areas with high walk accessibility, and they are less likely in area with poor access.

The Reno regional model makes use of some generic functions for calculating multiple accessibility variables, based loosely on NCHRP 365, etc. The table below presents three of the most widely used as examples. The first four rows detail which skims, cores, and decay parameters to use. The remaining rows show the attraction rates for the size variable calculation.

Param	access_general_sov	access_nearby_sov	access_transit
skim	SOV	SOV	transit
core	FFTime	FFTime	Total Time
b	-0.93	-2.05	-2.05
c	-0.09	-0.06	-0.06
HH	1.9	0.5	1.9
K12	1.5	0	1.5
College	5.7	0	5.7
StudGQ_UNR	5.7	0	5.7
Retail	18.7	4.1	18.7
Service_RateLow	5.6	1.2	5.6
Service_RateHigh	5.6	1.2	5.6
Office	3	0.5	3
Industry	3	0.5	3

Population Synthesis

Background and Overview

Creating a synthetic population, or in other words, enumerating a complete set of households and persons (with key attributes) for a modeling region is a necessary first step for the application of advanced travel models such as hybrid trip-based models and Activity Based Models (ABM).

The synthesized household and person characteristics reflect the distribution of key variables in the study region and match observed marginal distributions at an aggregate level.

Having access to detailed person and household characteristics in the model opens the door for model estimation to include a range of variables that define trip and travel making behavior. Even for traditional trip-based models, development of a synthetic population and household database can help generate various cross-tabulations at desired geographic levels that may not be available otherwise.

For the Reno regional model, a population synthesis forms the core basis for applying a state-of-the-art hybrid travel demand model. The aim of the synthesis is focused on matching both household and person marginals at the Traffic Analysis Zone (TAZ) level. Another crucial requirement is a solution that is practical to apply and runs in minutes without sacrificing accuracy of the results. To that end, the updated population synthesis procedure in TransCAD 10.0 is employed.

This documentation briefly summarizes the basic idea of population synthesis, some of the advancements made in recent years and points the reader to published literature for additional details. It also details the creation of disaggregate curves, which convert aggregate zonal data into targets for the synthesis process. The technical approach for population synthesis in the Reno regional model is briefly illustrated, following details regarding the application. Finally, the results of the application are demonstrated.

Basic Idea (Simple Iterative Proportional Fitting / IPF)

The basic idea of population synthesis is best illustrated by Figure 1. In simple forms of population synthesis, there are two inputs: disaggregate household (HH) and person seed tables and HH marginals summarizing the distribution of key characteristics of HH at some aggregate level of geography.

The seed data tables consist of a sample of household (HH) and person records, typically obtained from a survey of the study region, and tagged to a high-level geography such as Public Use Microdata Areas (PUMAs). Public Use Microdata Sample (PUMS) data from the American Community Survey (ACS) typically forms this seed, although this sample can be enhanced by incorporating data from a travel survey and reweighing the combined records.

Various aggregate marginals, typically ACS marginals at the census block or block group level in the base year or sometimes at higher levels of geography for future years provide control totals that the synthesized databases must match. It is also a well-accepted practice to fit curves using the census marginal distributions, which can then be applied to generate marginals at a TAZ level for example. These TAZs are usually smaller than census block groups but larger than census blocks.

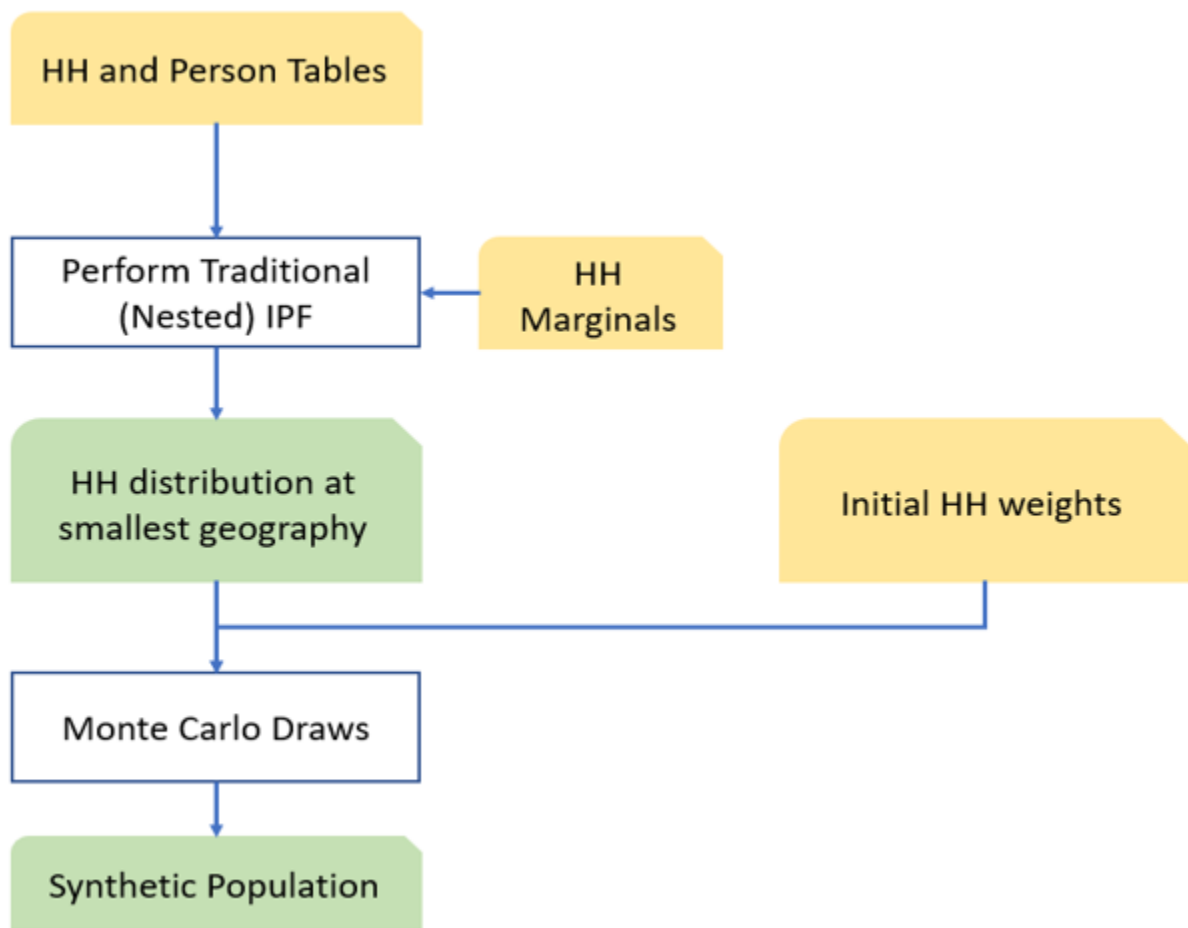


Figure 1 Simple population synthesis process flow diagram

An Iterative Proportional Fitting (IPF) step then estimates the number of each type of household (identified in the seed) to be generated for each of the smallest geographic entities. This is achieved by identifying the number of households of each type that will add up to various household marginals. This is best illustrated for a single block group using the schematic in Figure 2. In this example, the two classifications of interest are HH by Size and HH by Number of Vehicles. The row and column marginals highlighted in yellow are the input control totals and the goal of the process is to fill in the cells of the matrix (i.e. the values TS1_V0 etc.) such that the row and column totals are respected. The value TS1_V0 represents the number of single person, zero car HHs that need to be synthesized for this block group.

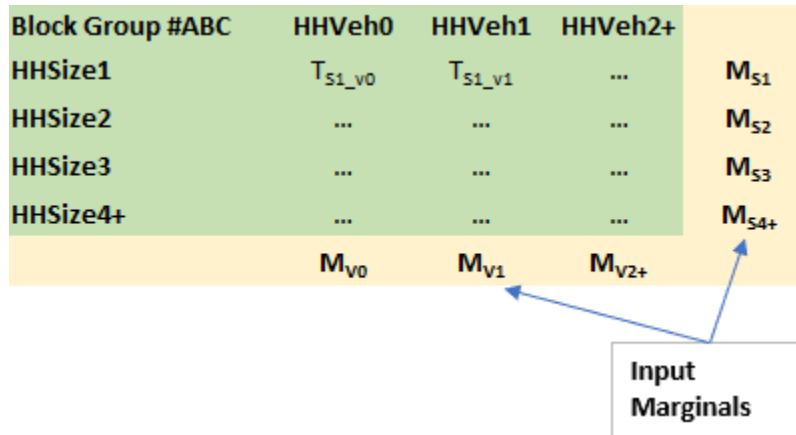


Figure 2 IPF

The term “Nested” in Figure 1 indicates that the household marginals may themselves be specified at different (but nested) geographies: for example, the vehicle ownership marginals may be at the traffic analysis zone (TAZ) level while the income marginals may be at the level of Census blocks or block groups. Nested synthesis will drill all the marginals down to a common geography (usually the smallest/finest among all input geographies).

Finally, for each smallest geographic entity, the required number of households of each type determined by IPF are sampled or drawn from the surveyed (seed) households using externally supplied (initial) weights (generally based on the survey sampling). The selected households are accumulated in a HH output file, and all persons living within each sampled household are automatically copied into a person output file. Together, the HH and person files comprise the synthetic population once the sampling is complete.

The most important drawback of this process is the lack of control on person marginals. Since the algorithm only controls for household variables, the obtained distribution of person attributes in any given zone is largely a lottery. There is no guarantee that the chosen households will yield, for instance, the correct distribution of age, gender, etc., in the synthetic population. This can have adverse effects from a modeling application perspective, for example if the distributions of working-age persons, students or seniors are incorrect.

Person Marginals and Iterative Proportional Updating (IPU)

The Iterative Proportional Updating (IPU) process (Xe et al, 2009) has been proposed and applied (most notably in PopGen) as a way of matching person marginals while essentially retaining the simplicity of the IPF framework. In this approach, before sampling households for a particular geographic entity (zone) using the initial weights in the seed, these sampling weights are first adjusted (or raked) to explicitly match both household and person marginals for the zone. In this process, note that the same seed household(s) may be weighted differently for different zones, which adds precision compared to the use of *a priori* weights supplied externally. Further, this household re-weighting may be applied at a different (usually more aggregate) geography than that used for simulating the population.

Figure 3 shows the general schematic of the IPU enhancement over IPF. Note the additional inserted steps which creates the loop. As mentioned earlier, the HH seed weights are adjusted for each zone to match HH and person marginals for that zone.

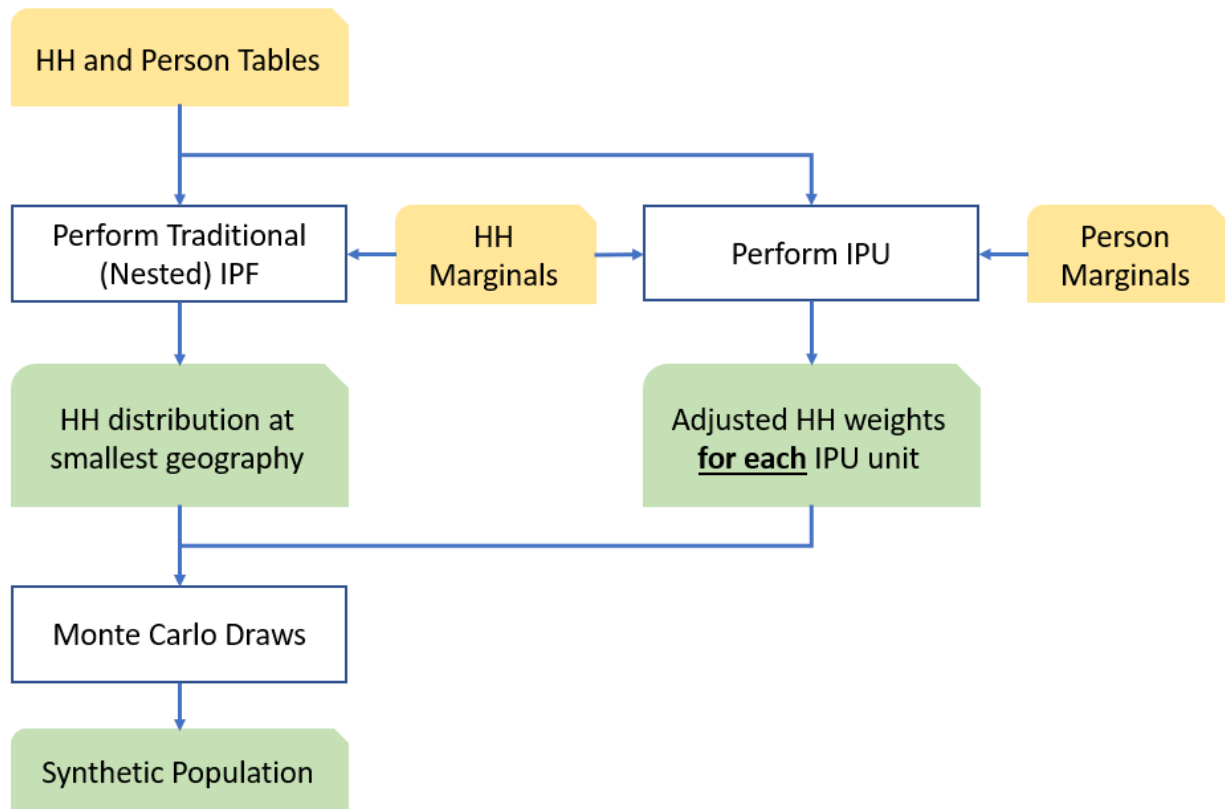


Figure 3: Population Synthesis with IPU

The originally published IPU method in the literature was implemented in the PopGen software. In this original IPU algorithm, two separate IPFs are run, one on the HH marginals and one on the person marginals. The results of the second IPF on the person marginals are used to populate detailed person-level columns in the incidence table that specify the full, joint distribution of person attributes. So, if there are multiple marginals to match such as HHs by Size (1,2,3,4+), number of vehicles (0,1,2,3+) and number of workers (1,2,3,4+) and Persons by Age (assume 5 categories) and Gender, then the incidence table has 64 HH columns, one for each combination of size, vehicles and workers and 10 person columns, one for each age category and gender combination. Since only one-dimensional marginals are typically known, the marginal values in the incidence tables come from the IPF run and the second IPF must be performed on the person characteristics to populate the person columns.

This version of IPU has several drawbacks. One, the process is slow, especially if there are a lot of marginals and the process can take hours. Second, the solution introduces a couple of other issues that must be dealt with and these issues have garnered significant attention in literature. One of these issues is a zero-cell problem, where the seed data is stretched thin and certain combinations may not be present in the seed. The second issue is a zero-marginal problem, but this can be solved by replacing the zero in the marginal row with epsilon values. There are several techniques to address these issues, which have been implemented in PopGen, but addressing these issues adds further runtime.

Therefore, a different variant of IPU was developed for and implemented in TransCAD. For want of minimizing these issues and at the behest of maintaining fast run times, only the one-dimensional marginals are used to form the incidence table. Thus, in the hypothetical example, the incidence table has 12 HH columns (as opposed to 64) and 7 person columns (as opposed to 10). The enhanced TransCAD IPU method has been applied successfully for the Las Vegas RTC model, the California Central Coast ABM model, as well as the Triangle Regional Model in North Carolina. In all cases, not only was the match to the one-dimensional marginals (especially the persons) very close but the joint distribution observed from the synthesis was also very reasonable. These applications take a fraction of time compared to that reported on similar sized problems with the original IPU method or optimization methods. For more details on TransCAD's IPU algorithm, see Balakrishna, R., S. Sundaram and J. Lam. ["An Enhanced and Efficient Population Synthesis Approach to Support Advanced Travel Demand Models."](#) Presented at the 99th TRB Annual Meeting.

Other Techniques

Several optimization methods have also been published in the literature. For a general literature review, see Ramadan and Sisipiku, 2019. While these methods might tackle certain issues of the IPU approach, these methods tend to be customized. Owing to the nonlinear and non-convex nature of the resulting formulation, the performance can be problem dependent and can result in long run times, local optima, and sensitivity to the scale of the objective function.

Univariate Marginal Distributions

The first step in synthesizing the population of residents in the Reno regional model is to develop univariate or one-dimensional marginal distributions describing the population in each zone based on the limited demographic variables forecast for each zone, typically averages. The distribution of households by household size is provided by the user as an input to the Reno regional model, but distributions for households by income group and number of workers must be developed by the model based on the simple zonal average (income and worker) data provided. In other words, the model starts by determining the number of single worker, two worker, three worker, etc., households in each zone based on the average workers per household in the zone.

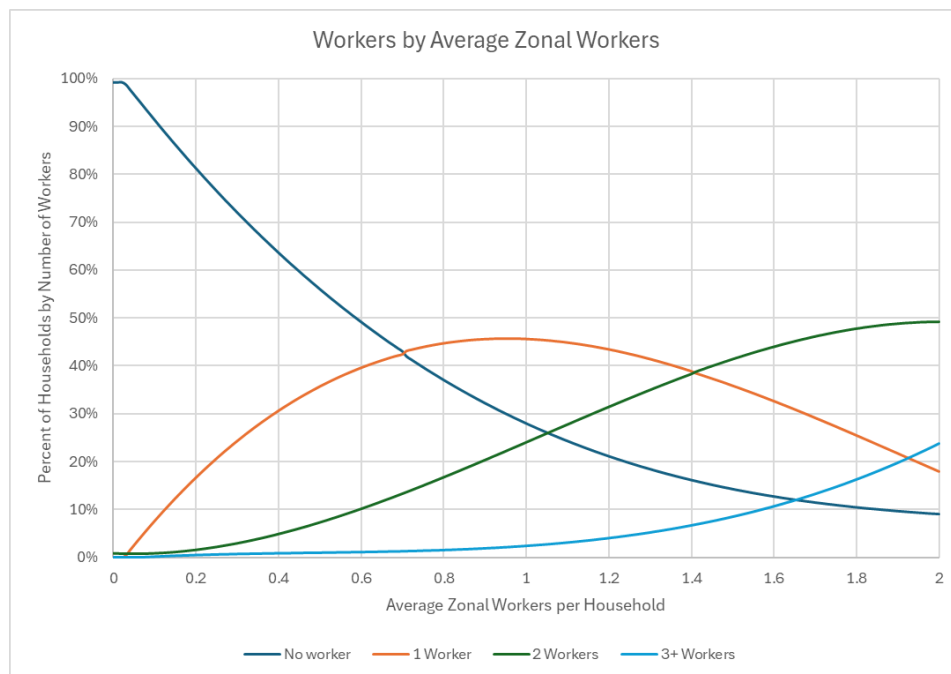
Stratification curves simply encode information on the extent to which, for example, a zone with 2.5 workers-per-household on average will have a high percentage of 2- and 3+-worker households and a zone with an average of 1.5 persons-per- household will have more 1- and 2-worker households.

In the Reno region, the marginal distribution stratification curves used are:

- Income
 - Low: \$0 - \$24,999
 - Medium-Low: \$25,000 - \$74,999
 - Medium-High: \$75,000 - \$149,999
 - High: \$150,000 and above
- Workers (0-3+)

Workers

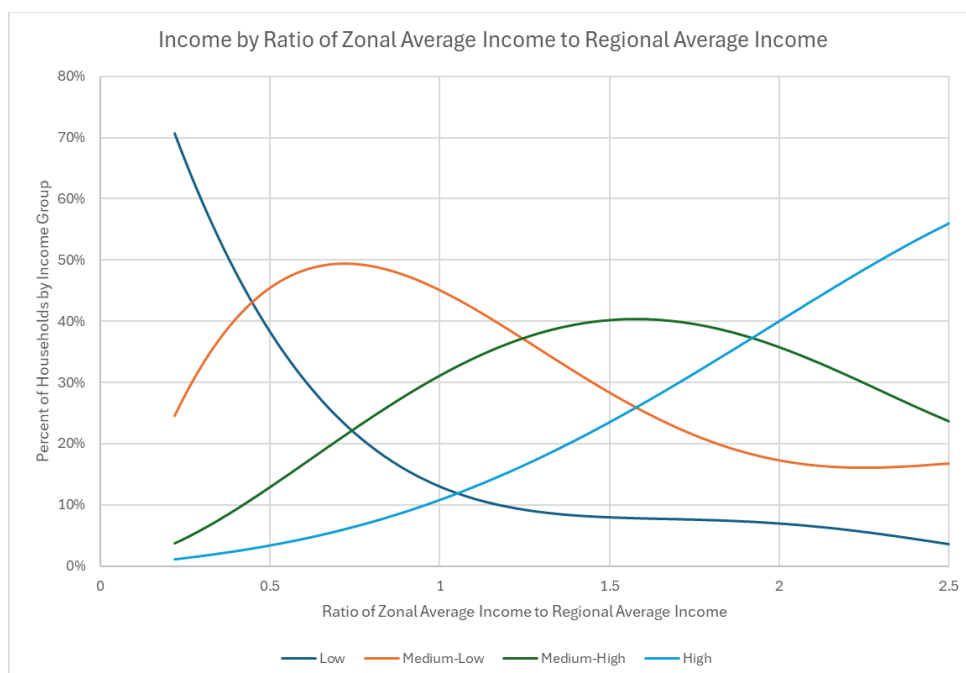
The chart below shows the stratification curves used in the Reno regional model to break out households by number of workers based on the average number of workers per household in the zone.



Income

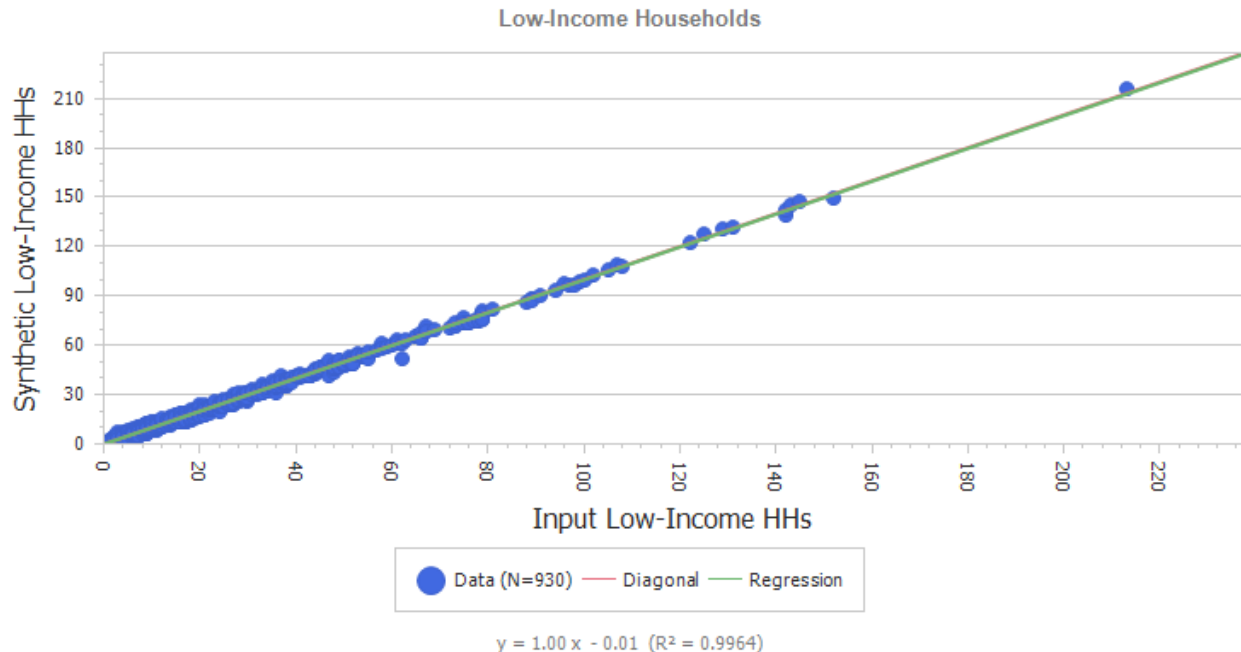
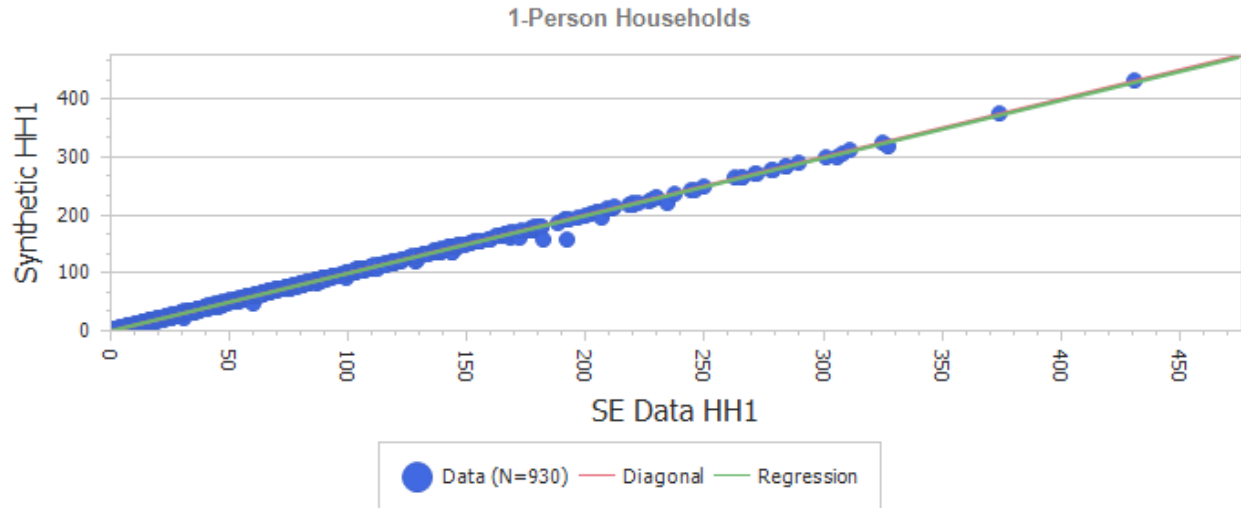
The chart below shows the stratification curves used in the Reno regional model to break out households by income group based on the ratio of the average zonal income per household in the zone to the average income per household in the Reno region. The model uses four income groups defined as follows:

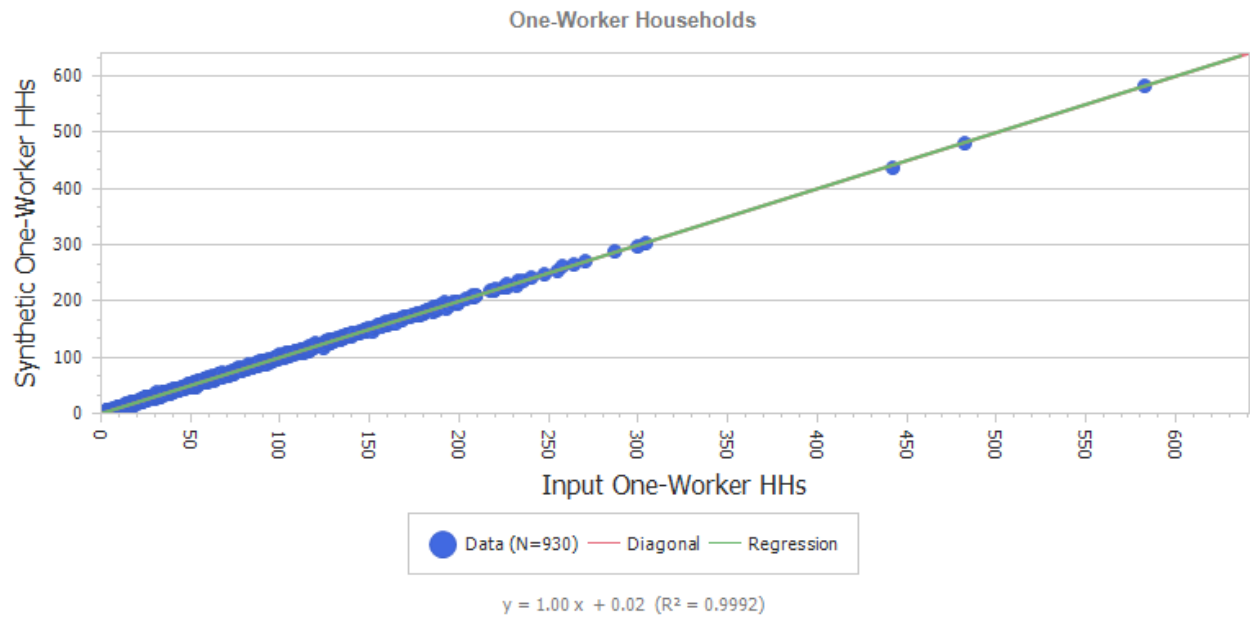
- Low Income: Less than \$25,000 per household per year
- Medium-Low Income: \$25,000 – 75,000 per household per year
- Medium-High Income: \$75,000 – 150,000 per household per year
- High Income: Greater than \$150,000 per household per year



Checking Synthesis Results

The results of the population synthesis are compared back to the input marginals to ensure a close match. In the example charts below, each dot represents a TAZ. The x-axis is the input marginal (e.g. number of one-person households) and the y-axis is the corresponding metric from the synthetic population. Note that households by size are given in the input SE Data. Other input marginals come from the disaggregate curves. The synthetic population closely matches the input marginal distributions.





Auto Ownership

Introduction

Auto ownership is a long-term decision that directly impacts daily mode choice. The choice is influenced by household factors like income and number of workers, but also by where people choose to live. The auto ownership model allows the Reno regional model to be sensitive to these factors and respond to changes in the future.

Auto ownership is predicted using a discrete choice model. For more details on general model form, click [here](#).

Model Specification

The auto ownership model in the Reno regional model makes use of variables from the synthetic population and zonal accessibility to make predictions. The model structure is a simple multinomial logistic (MNL) regression model with five alternatives:

- 0 Vehicles
- 1 Vehicle
- 2 Vehicles
- 3 Vehicles
- 4+ Vehicles

The table below shows the utility terms and model coefficients.

	No Vehicles	1 Vehicle	2 Vehicles	3 Vehicles	4+ Vehicles
Constant	0	4.79	2.49	1.80	0.74
Workers		0.72	3.33	4.34	5.34
Non-working Adults		0.00	2.08	0.00	0.00
Seniors		0.08	0.08	2.83	3.65
Walk Accessibility		-0.37	-0.62	-0.62	-0.72
Transit Accessibility		-0.09	-0.15	-0.20	-0.20
Nearby Accessibility		-0.38	-0.53	-0.83	-1.14
Walkability		-0.53	-0.53	-0.53	-0.53
Low Income HHs		-0.56	-1.80	-1.84	-3.20
Med-Low Inc HHs		1.60	1.60	1.60	1.60
Med-Hi Income HHs		1.82	2.61	2.75	2.75
High Income HHs		2.23	3.70	4.15	4.51

The coefficients all have the right sign and the relative sizes are intuitive. For example, higher-income households are more likely to own more cars. One particularly encouraging result of this model is that households with strong walk and transit accessibility are less likely to own a vehicle and even less likely to own multiple vehicles. This adds another dimension of model sensitivity to transit investments. New transit routes will affect long-term household decisions about auto ownership, which further influence their daily transportation decisions.

Looking in more detail, the trend for coefficients across alternatives is also intuitive. Large numbers of workers in a household have a small positive impact on the utility of owning 1 auto, but a large impact on owning 2 or more. High income households are more likely to own more vehicles.

Resident Productions

Productions are predicted at the person level using decision trees for the machine learning stack. Compared to traditional cross-classification matrices, they use the survey samples more efficiently and can consider more variables. While state of the art decision trees methods including bagging and boosting achieve slightly higher prediction accuracy, they are opaque. The models below are simple trees that are easy to understand and perform nearly as well.

Trip purposes in the model are first stratified by the tour type they occur on (work vs non-work), and are laid out below:

- Trips on work tours
 - Home-based work (W_HBW)
 - Home-based other (W_HBO)
- Trips on non-work tours
 - Home-based social/recreational (N_HBSR)
 - Home-based other (N_HBO)
 - Home-based school (N_HBSCH)
 - Home-based shop (N_HBSHP)

For most purposes, separate decision trees were estimated for each market segment. The market segments are:

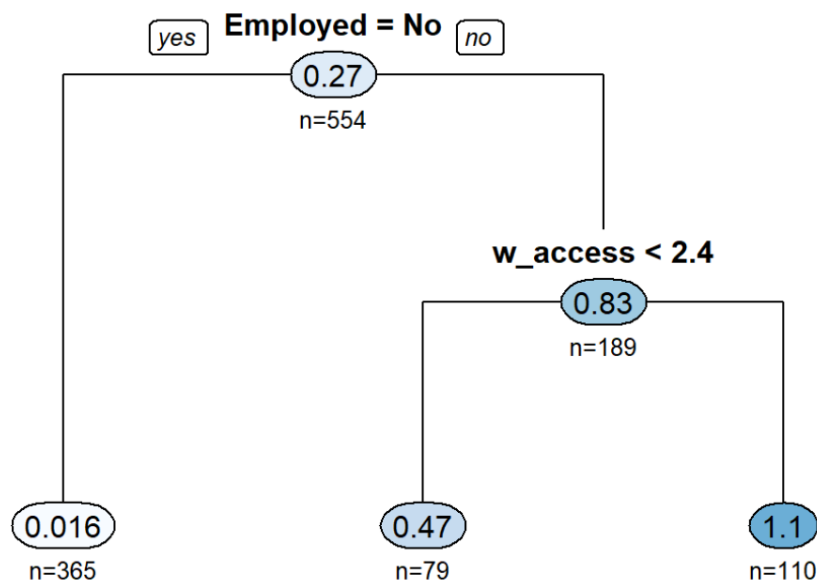
- Zero vehicle (v0)
- Vehicle insufficient (vi)
- Vehicle sufficient (vs)

W_HBW

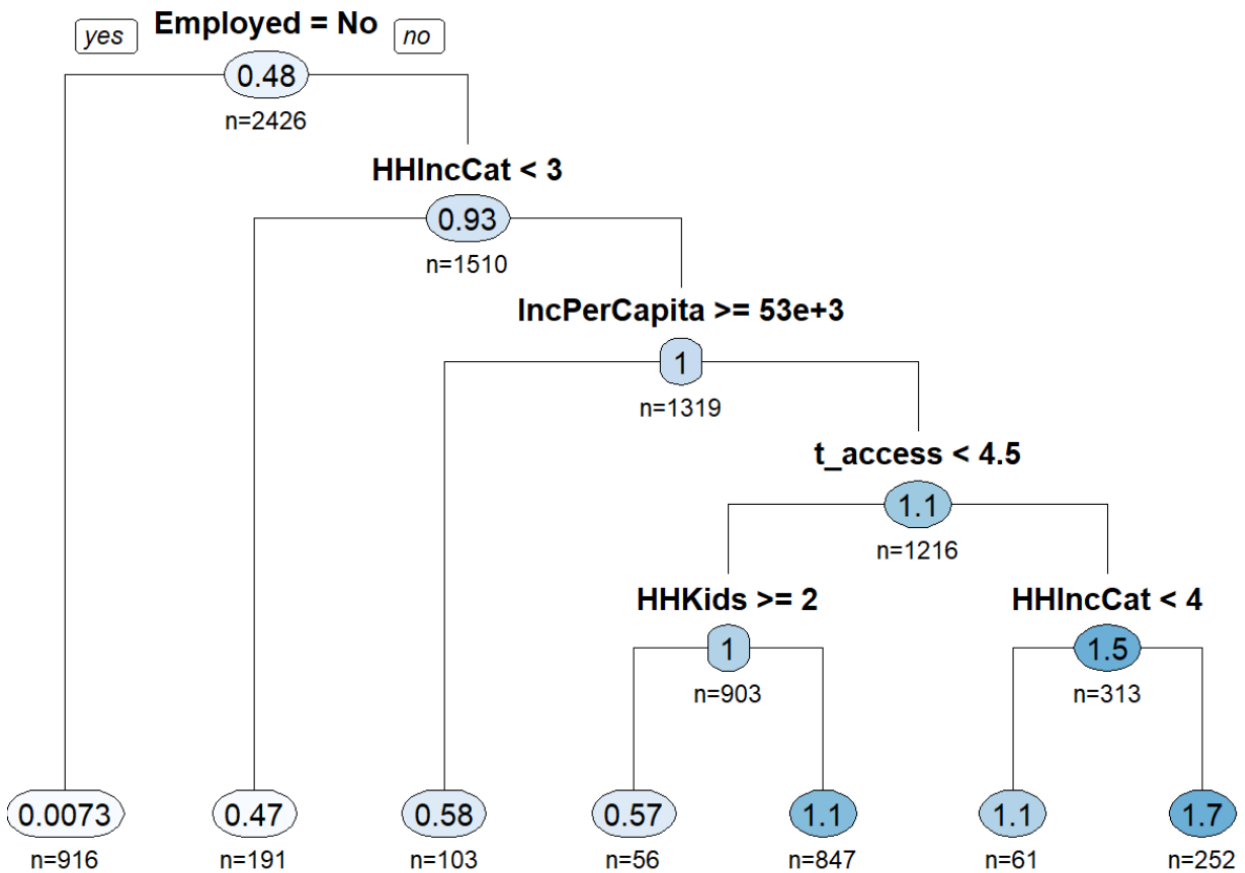
The home-based work purpose is sensitive to the usual variables like employment status of the person, their age, and their households income per capita (income divided by size). The models are also sensitive to various measure of accessibility:

- Transit accessibility (t_access)
- General/auto accessibility (g_access)
- Walk accessibility (w_access)

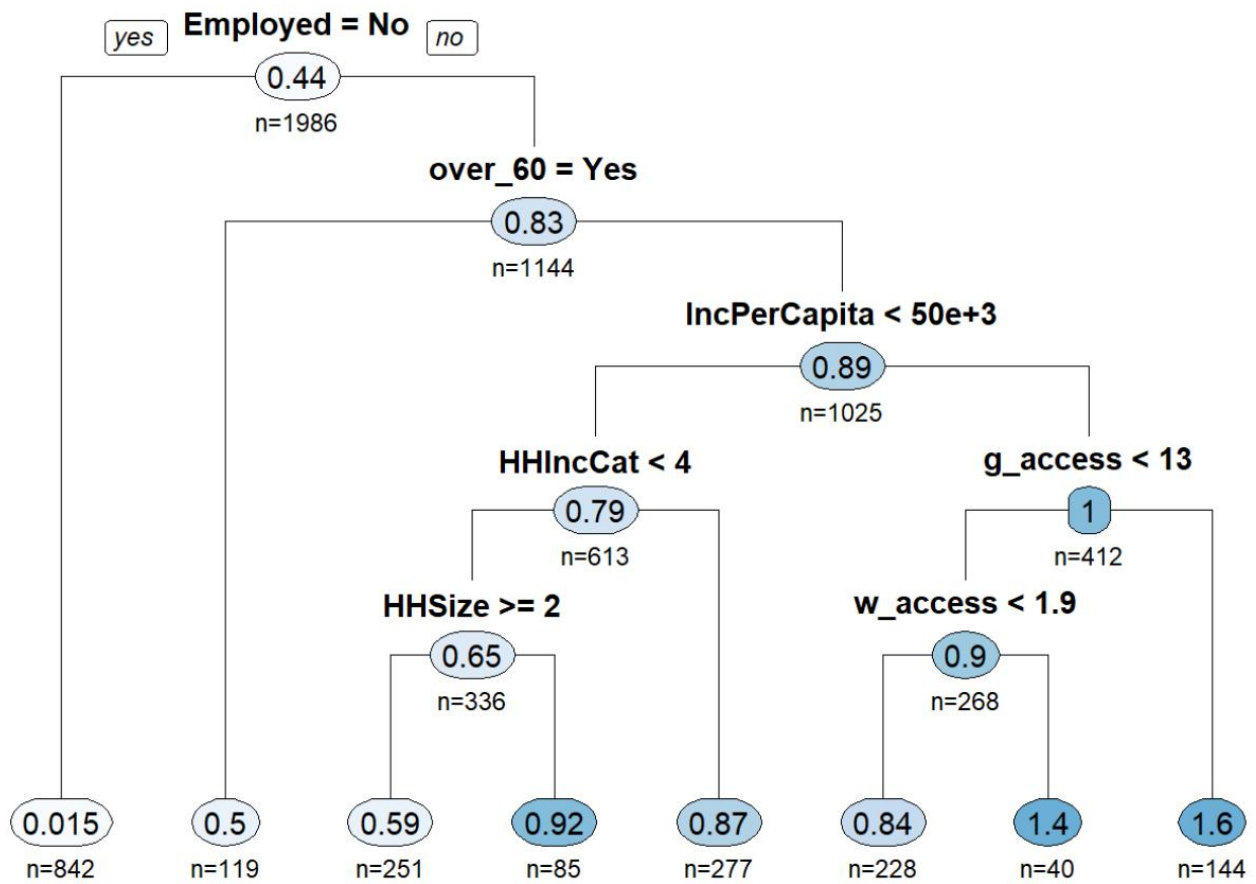
v0



vi



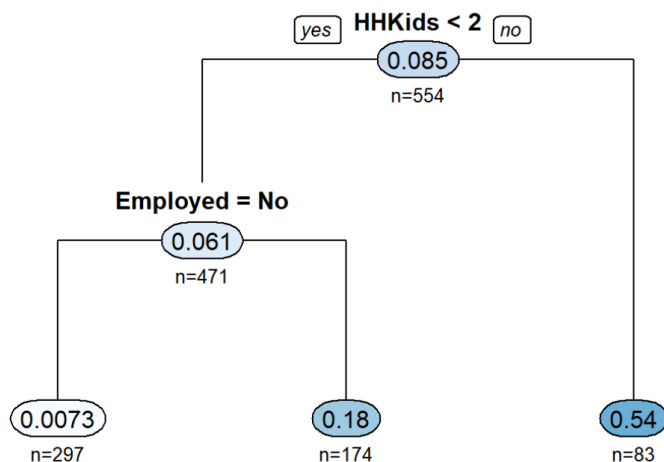
VS



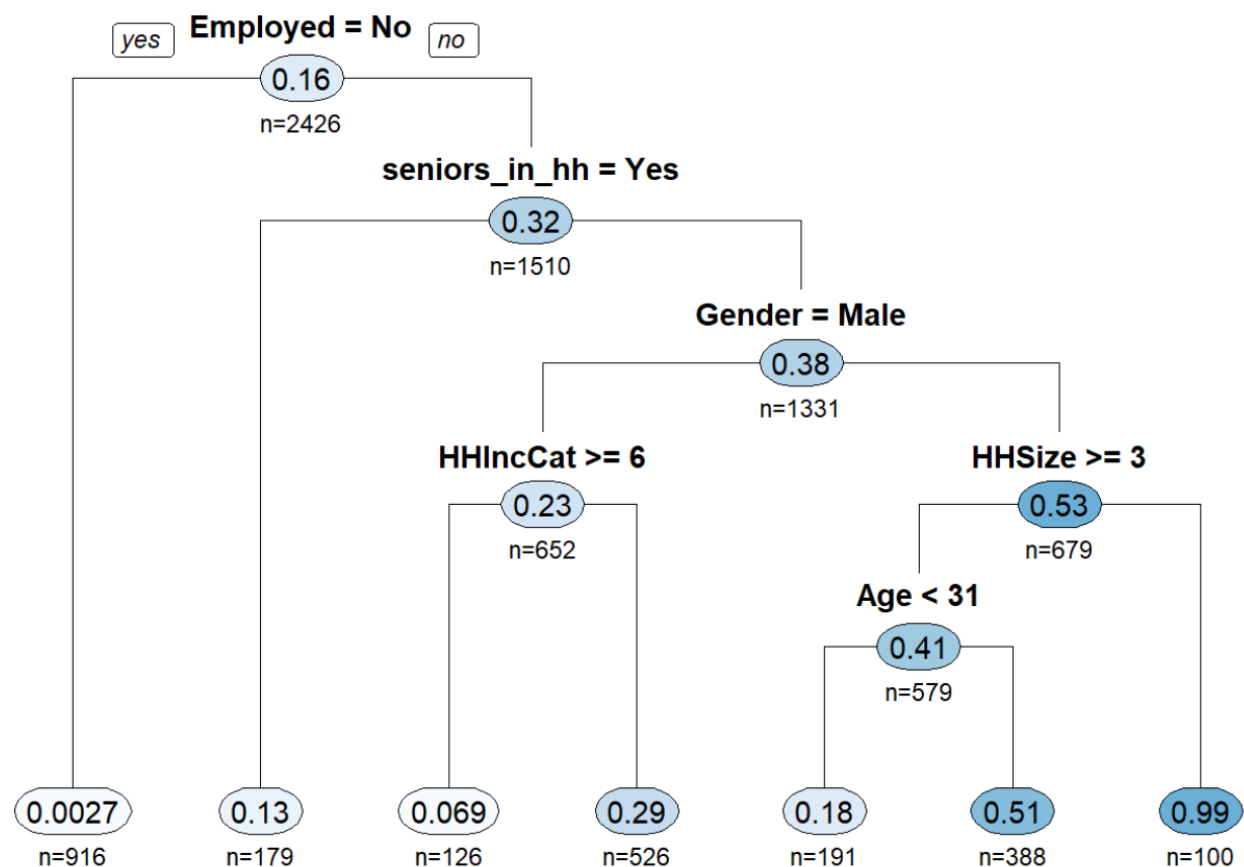
W_HBO

This purpose is sensitive to variables like the presence of seniors, gender, size, income, and age.

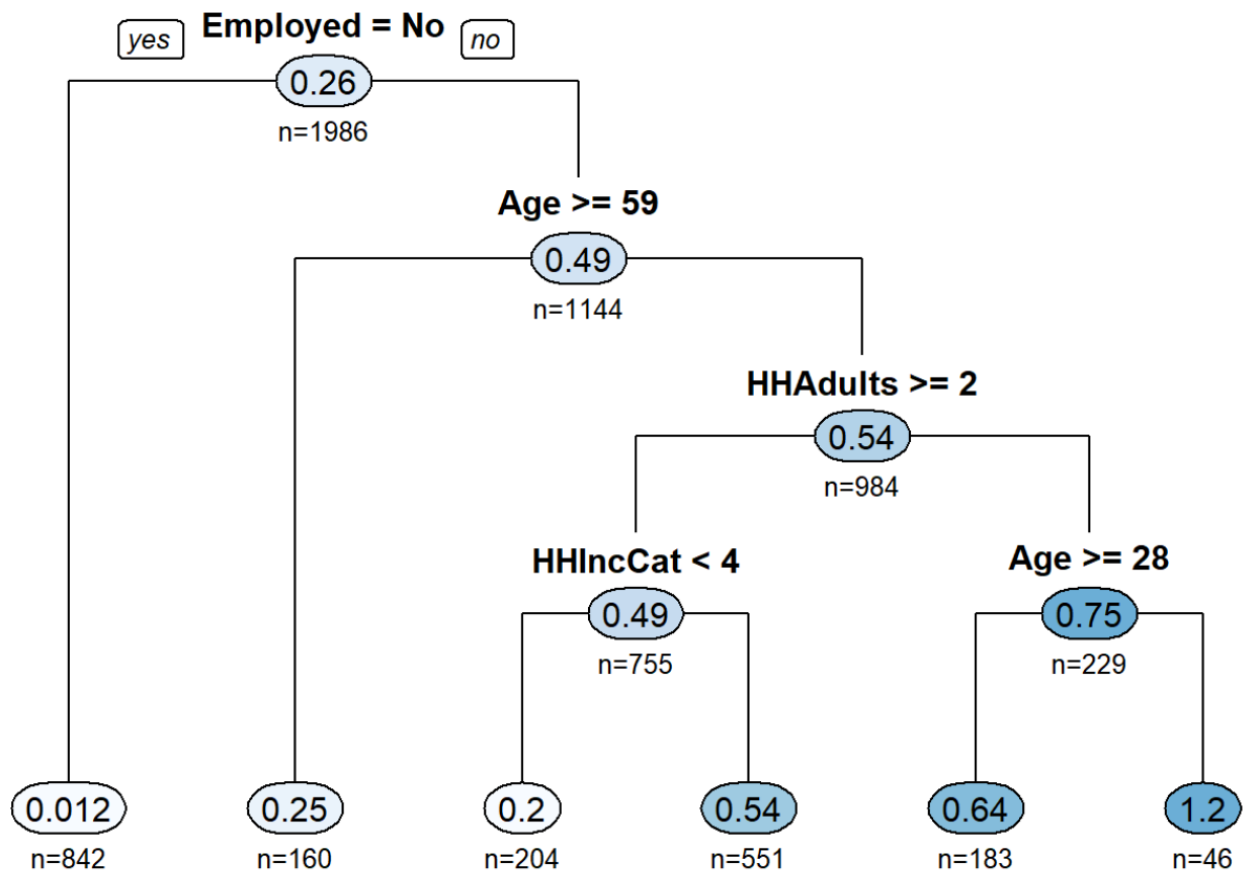
v0



vi



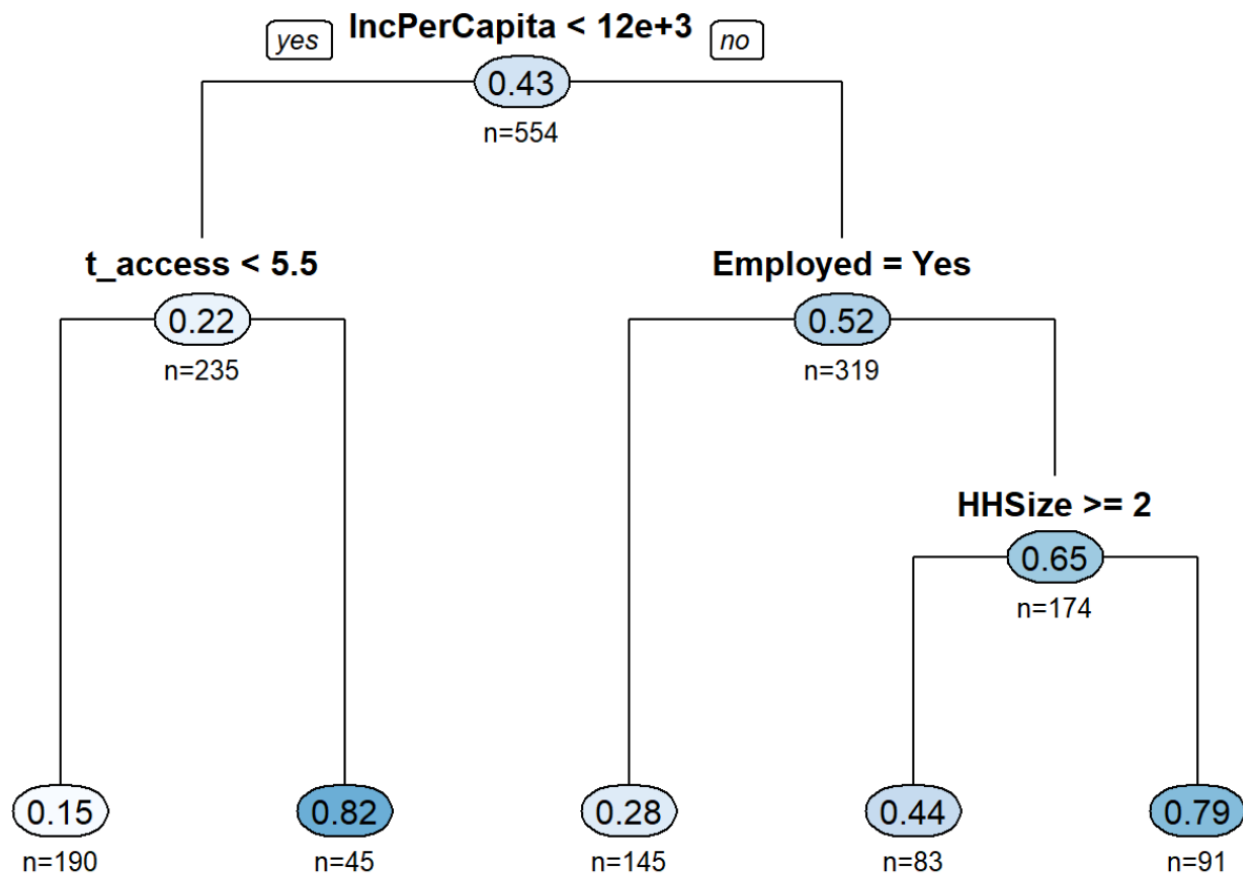
vs



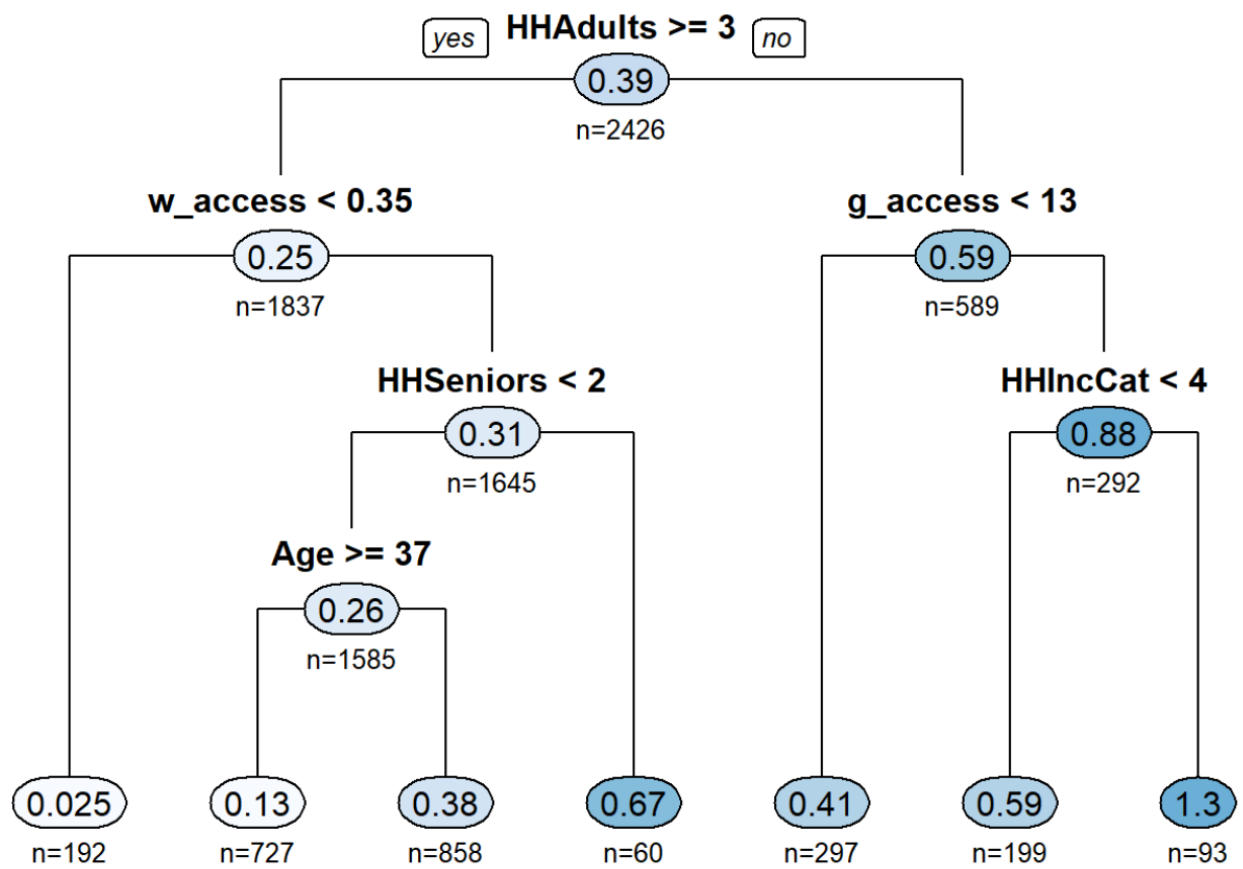
N_HBSR

The social/recreational models are sensitive to variables like accessibility, employment, per capita income, age, and presence of children.

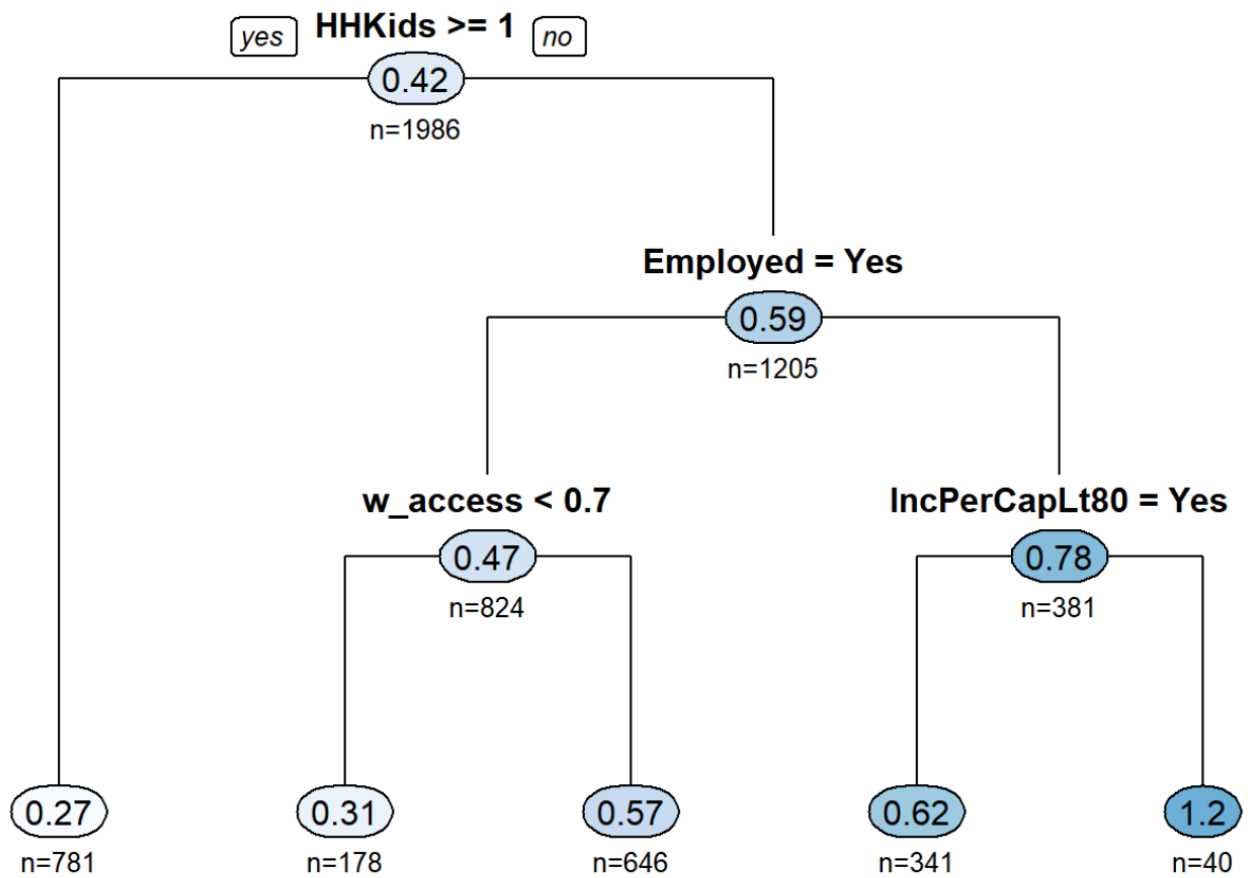
v0



vi



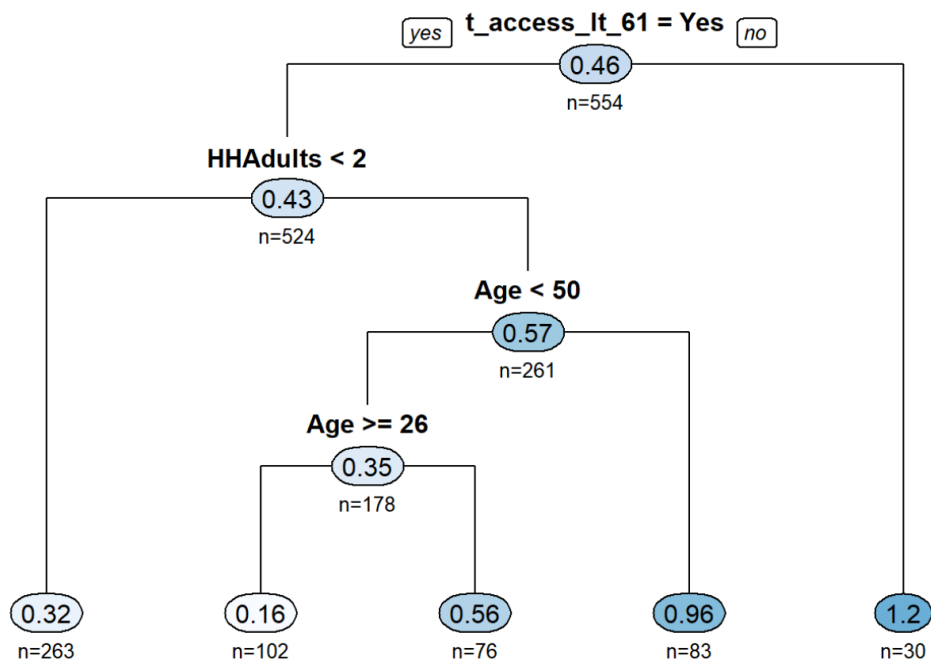
vs



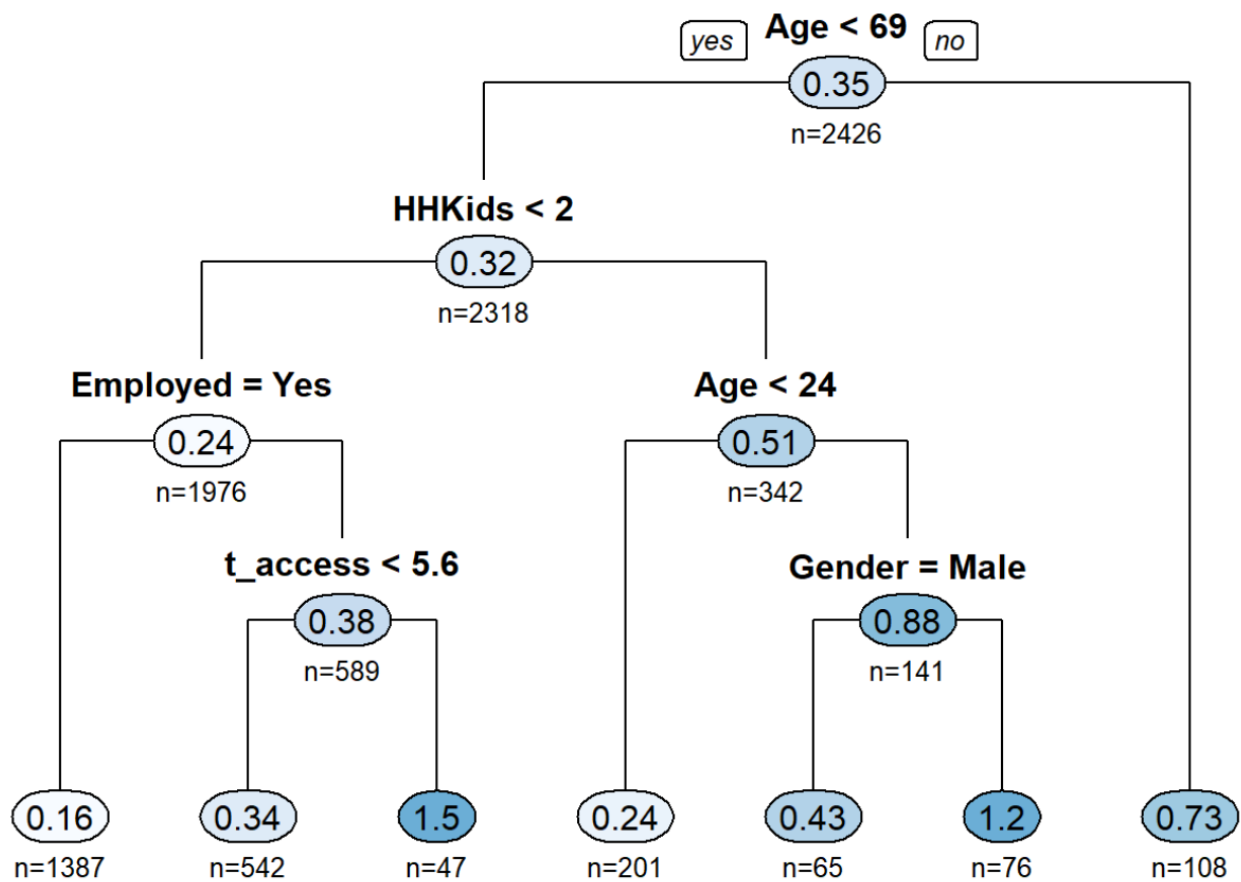
N_HBO

These models are sensitive to variables like age, gender, accessibility, and employment status.

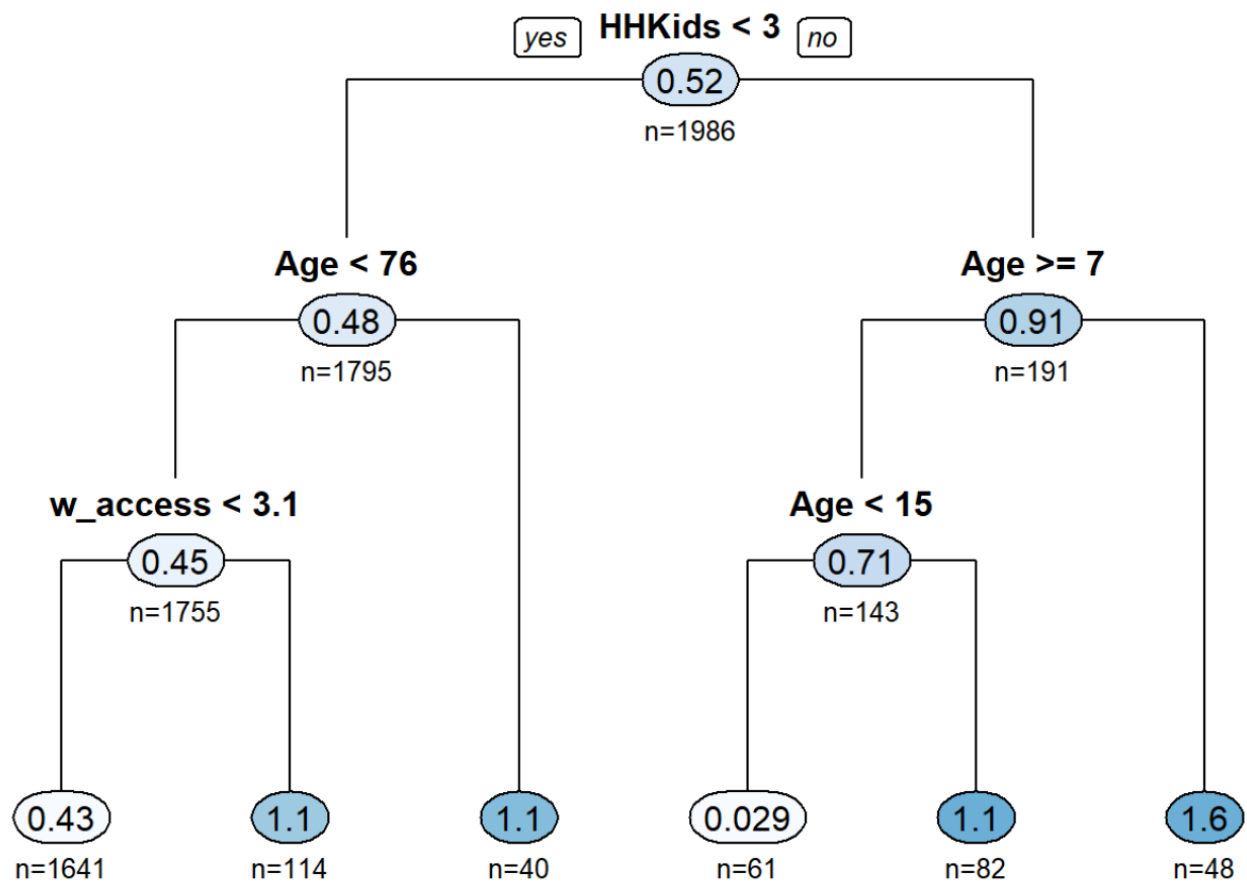
v0



vi

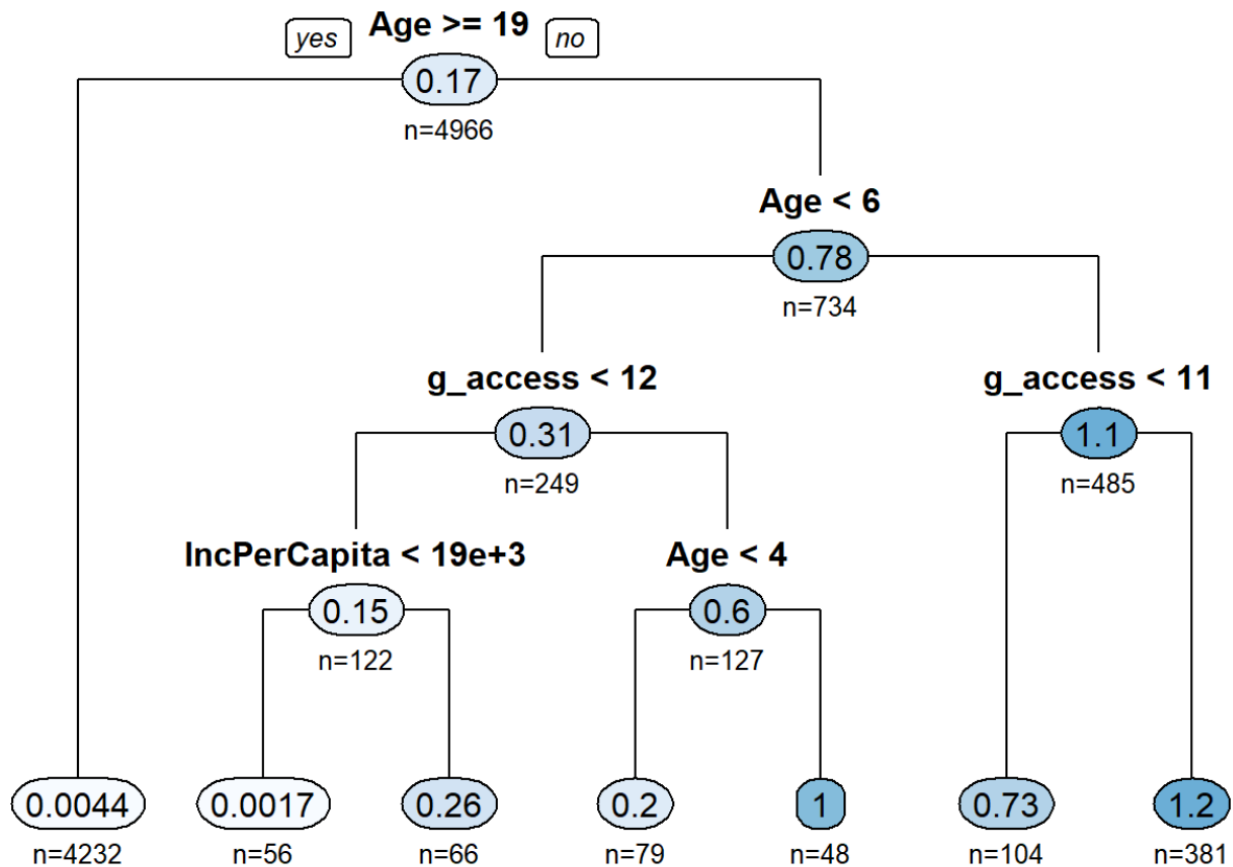


VS



N_HBSCH

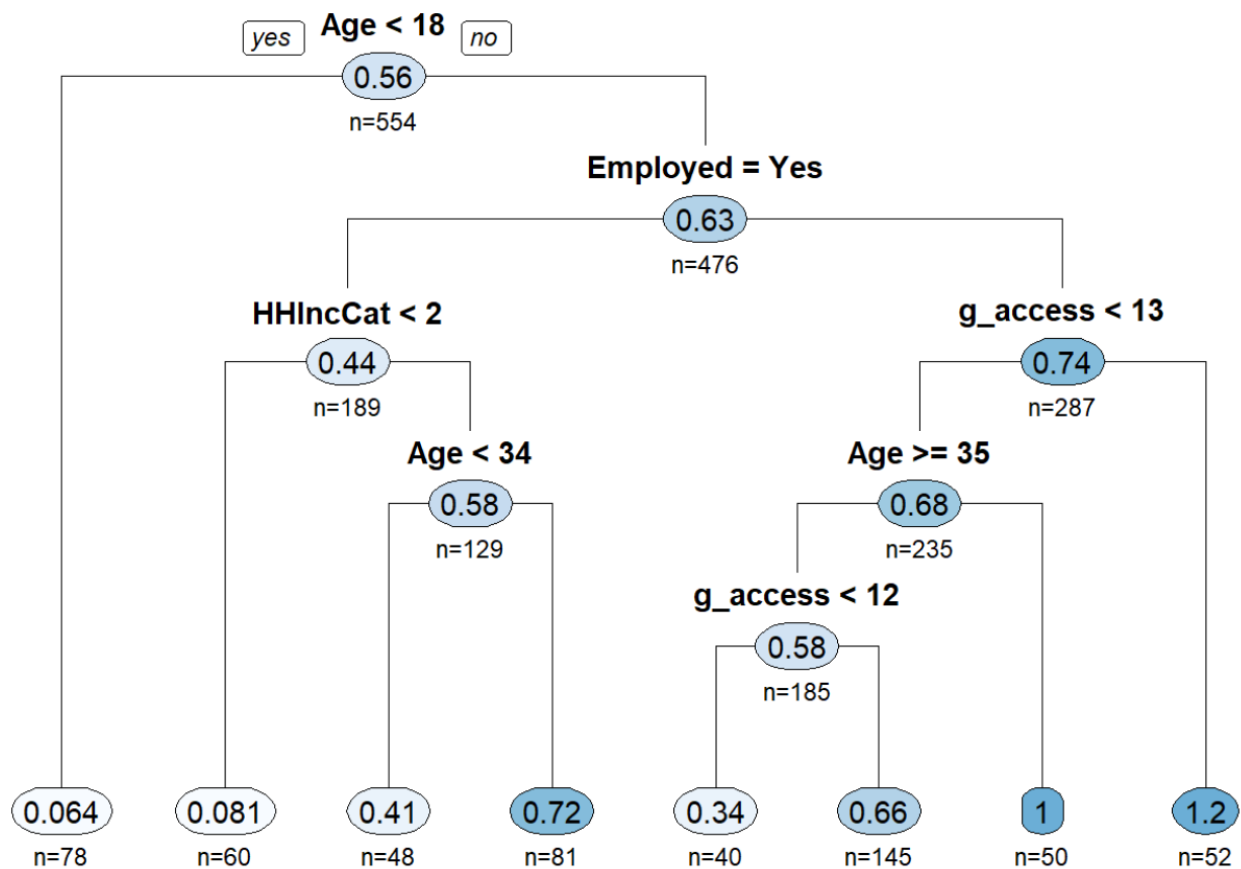
The most important predictor for these model is the age of the person, but other factors like accessibility and income do influence the number of school trips made. This purpose is not stratified by market segment.



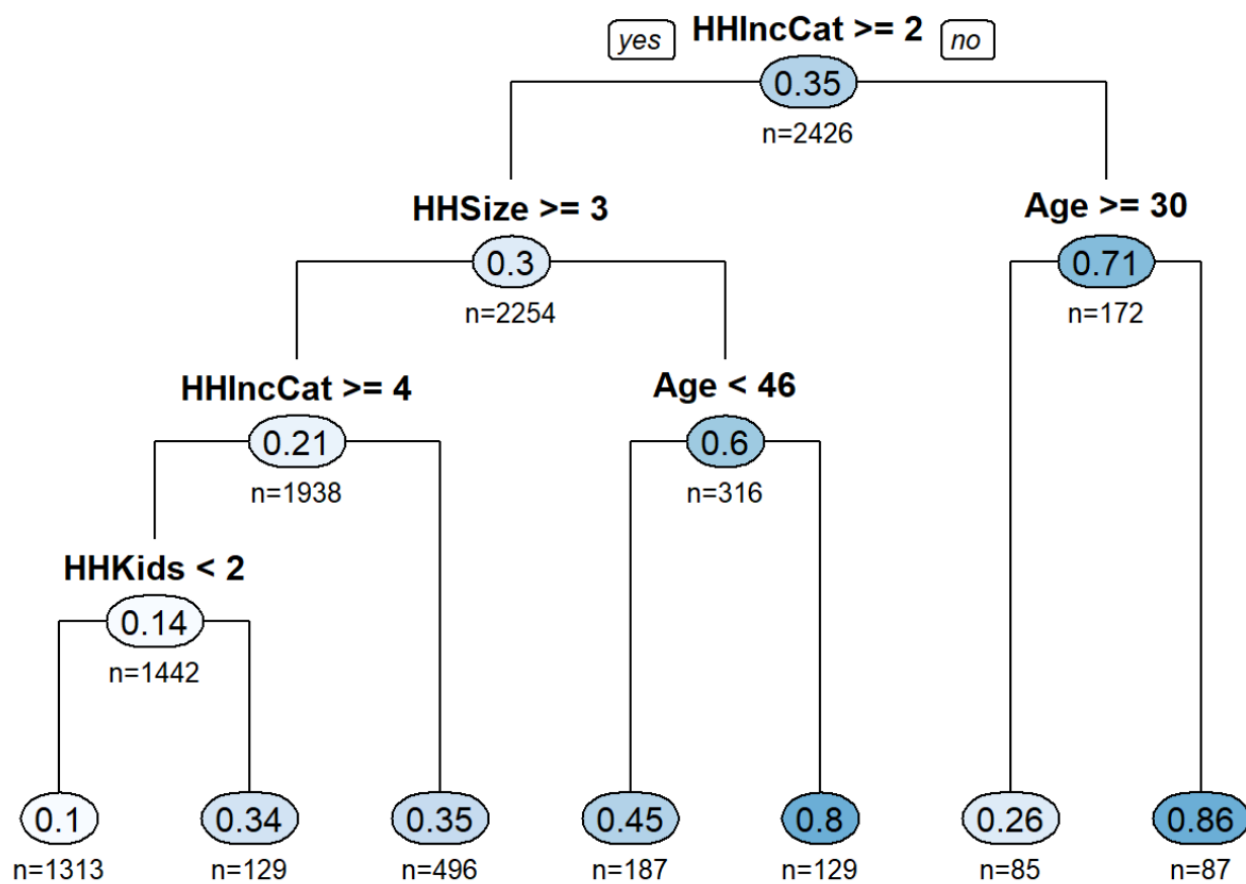
N_HBSHP

Shopping trips are influenced by employment status, accessibility, age, income, along with the presence of kids and seniors.

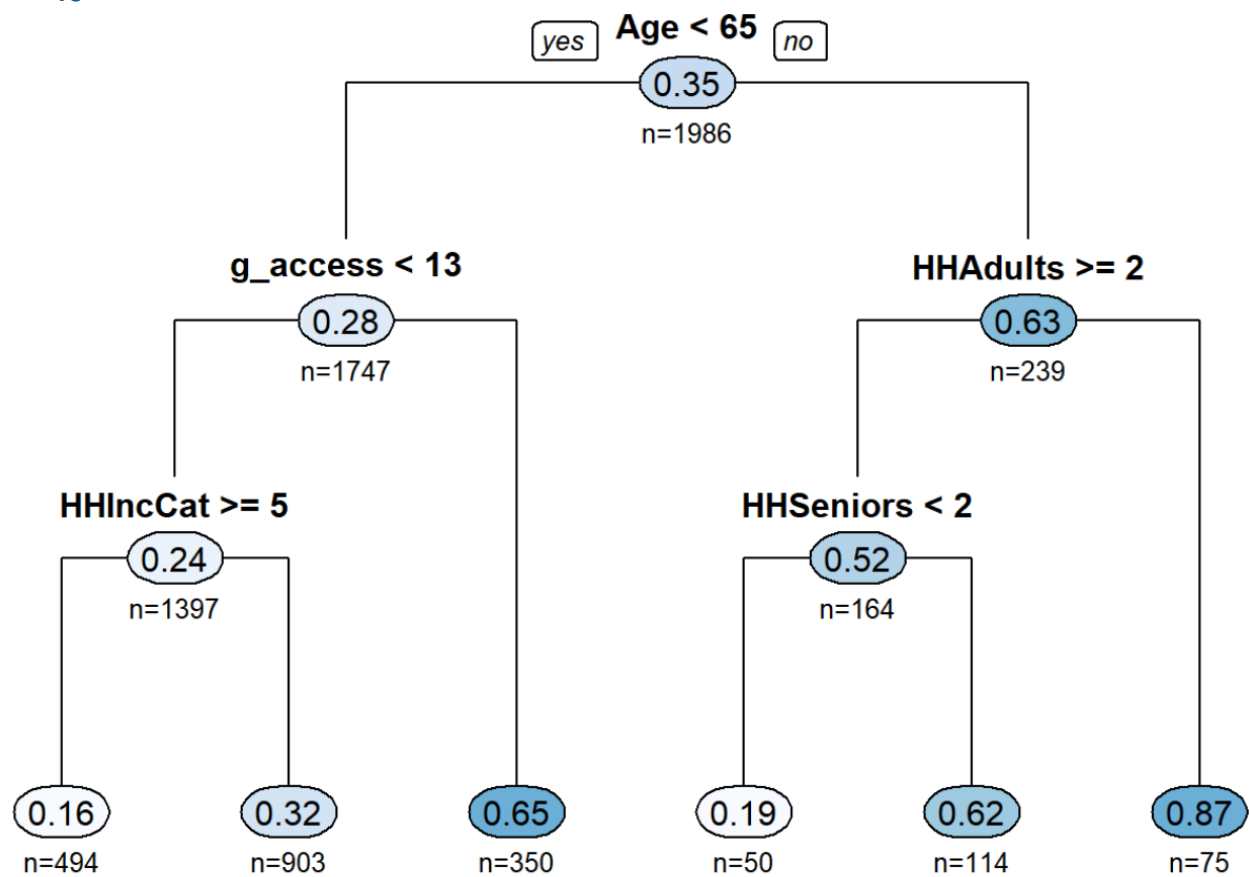
v0



vi



VS



Calibration

Calibrating the production rates is done to ensure that the final model is producing the same number of trips per person on average as the survey. One complicating factor in this comparison is that the survey and model socio-economic data have different total populations.

- Survey: 476,187
- Model: 523,542

As a consequence, the survey trip totals are increased by 1.1 and then compared to model results. The table below shows the production model results compared back to the total trips in the survey (factored up). While some purposes are higher or lower than the survey, the model does a good job of predicting total trips even before calibration.

Trip Type	Observed	Modelled	Difference	%Difference
N_HBO	237,004	271,029	34,026	14
N_HBSCH	91,037	105,942	14,905	16
N_HBSHP	190,471	182,342	-8,130	-4
N_HBSR	215,651	204,147	-11,504	-5
W_HBO	111,315	118,982	7,668	7
W_HBW	232,059	240,432	8,374	4
All	1,077,536	1,122,874	45,338	4

The calibration factors below are the ratio of the observed to modeled trips. These are applied by trip type and auto sufficiency segment to ensure total trip making matches the survey (based on trip weight).

Trip Type	segment	Factor
N_HBO	v0	0.78
N_HBO	vi	1.35
N_HBO	vs	0.76
N_HBSCH		0.86
N_HBSHP	v0	2.10
N_HBSHP	vi	1.50
N_HBSHP	vs	0.80
N_HBSR	v0	1.86
N_HBSR	vi	1.56
N_HBSR	vs	0.85
W_HBO	v0	0.87
W_HBO	vi	1.33
W_HBO	vs	0.84
W_HBW	v0	1.27
W_HBW	vi	1.38
W_HBW	vs	0.79

Non-motorized Mode Choice

After resident productions are estimated, the model removes non-motorized trips (bike and walk) using a binary logit model. For each trip purpose, the estimated coefficients are shown along with the adjusted rho-squared and the calibration constant applied to match survey shares of non-motorized trips.

W_HBW

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
HH Kids	0.066	2.8107		X
HH Adults	0.047	-1.3555		X
Vehicle per Adult	0.254	-11.657		X
Walk Access	0.073	1.2553		X
Constant	0.270	-3.7151		X
Rho^2	0.66			

Calibration Constant	0.60	X
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W_HBO

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
HH Kids	-0.290	-2.213		X
HH Adults	0.149	2.2412		X
Vehicle per Adult	-3.807	-7.922		X
Walk Access	0.093	0.7413		X
Constant	-1.156	-2.528		X
Rho^2	0.69			

Calibration Constant	1.03	X
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N_HBO

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
Vehicle per Adult	-0.490	-2.9348		X
Walk Access	0.470	5.9330		X
Senior	-0.942	-2.5293		X
Constant	-2.786	-11.476		X
Rho^2	0.56			

Calibration Constant	0.31	X
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N_HBSCH

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
HH Kids	0.320	4.4758		X
HH Adults	-0.527	-3.325		X
Vehicle per Adult	-1.060	-3.888		X
Walk Access	0.609	4.2519		X
Constant	-1.843	-3.051		X
Rho^2	0.48			

Calibration Constant	0.19	X
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N_HBSHP

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
Vehicle per Adult	-3.054	-10.159		X
Walk Access	0.253	2.6459		X
Senior	-0.661	-2.0033		X
Constant	-1.315	-5.0432		X
Rho^2	0.65			

Calibration Constant	0.66	X
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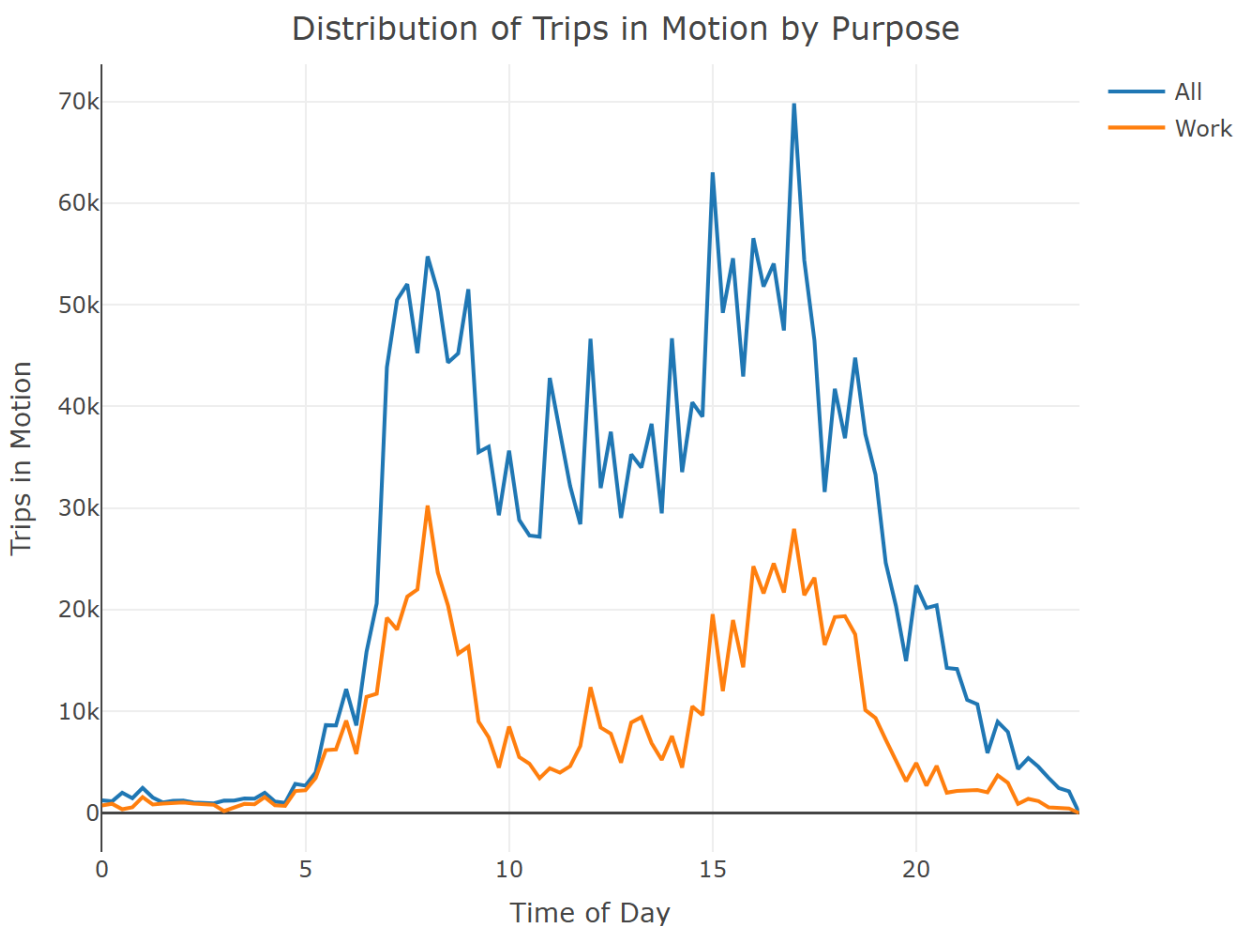
N_HBSR

Variable	Coefficient	t-Statistic	Alternatives	
			moto	non-moto
HH Kids	0.369	6.6602		X
HH Adults	-0.117	-1.481		X
Vehicle per Adult	-2.378	-9.723		X
Walk Access	0.453	4.8056		X
Senior	-0.612	-1.515		X
Constant	-1.816	-5.326		X
Rho^2	0.63			

Calibration Constant	0.46	X
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Time of Day

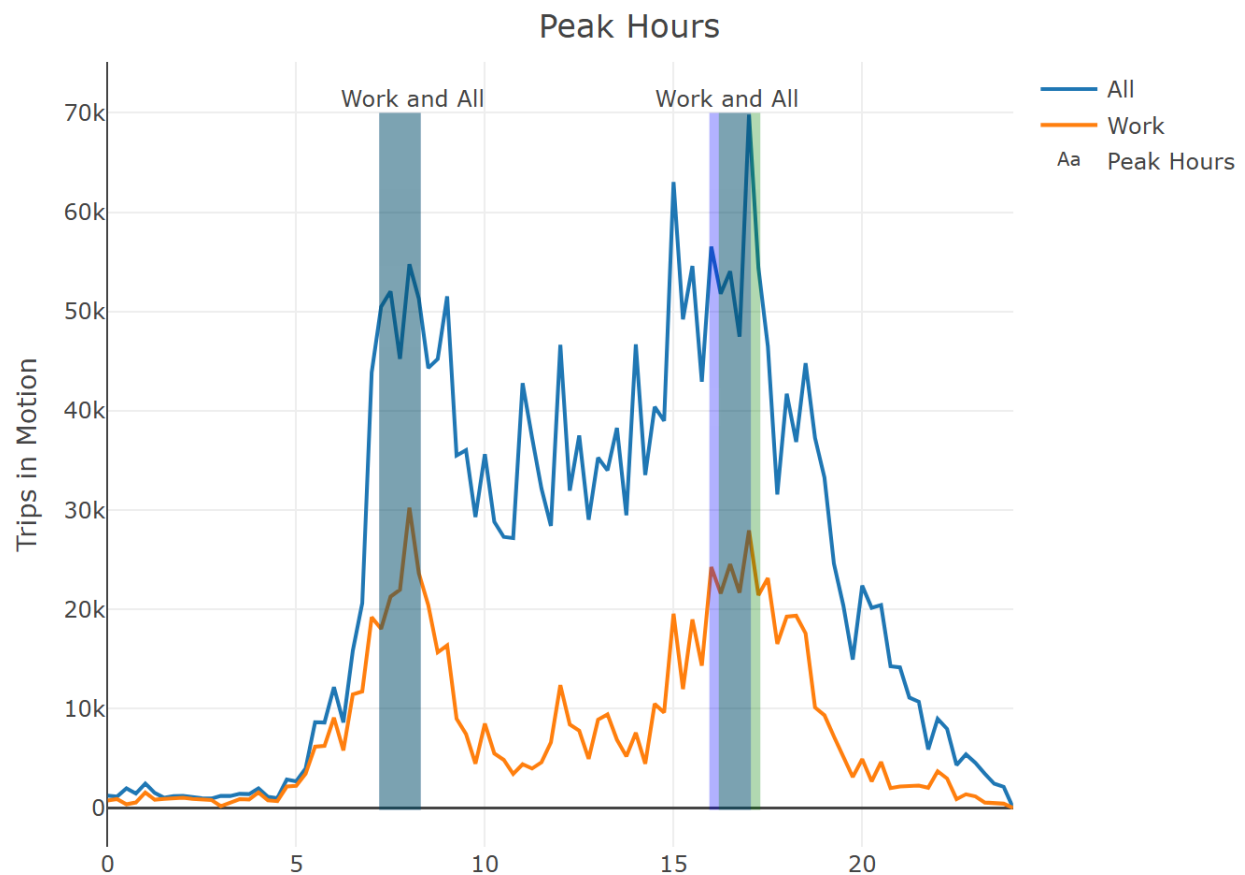
Peak period determination is based on an analysis of the trips in motion throughout the day. First, the day is broken up into 15-minute increments. A trip from the household survey is said to be “in motion” if any portion of the trip occurs within the 15-minute bin. As a result, a single trip can be counted in multiple bins. Determining the peak period considers the distribution of all trips as well as the distribution of trips on work tours individually. The chart below shows these two distributions, which look as expected with the AM peak being shorter and more condensed than the PM.



The next step is to determine the peak hour for all trips and those on work tours. In the tables below, the AM and PM peak hours by type are presented. At a minimum, the AM and PM peak periods should contain the respective peak hour for work trips and all trips.

Trip Purpose	Peak Start	Peak End
Work	07:15:00	08:15:00
All	07:15:00	08:15:00

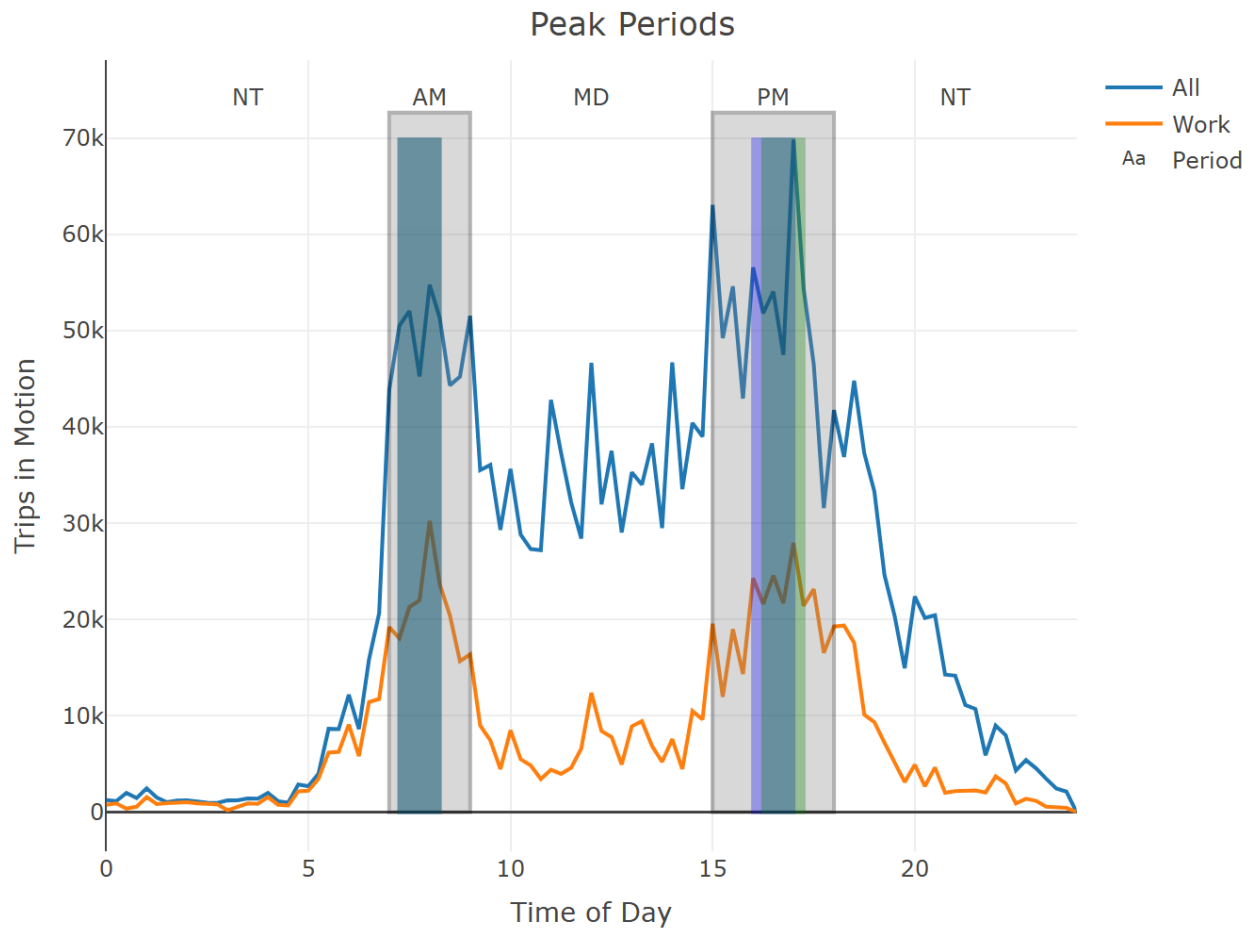
Trip Purpose	Peak Start	Peak End
Work	16:00:00	17:00:00
All	16:15:00	17:15:00



Based on the trips in motion profile, there are four distinct periods of the day. AM and PM peaks have the highest intensity of trip making, followed by mid-day, with the overnight period containing the fewest trips in motion. Boundaries for these periods are defined such that the variance of trips in motion within periods is smallest while variance between periods is largest.

The final period definitions are shown in the table below including the mid-day (MD) period. The remaining hours of the day are captured in the night (NT) period. In the chart, the final period is represented by the gray rectangle.

Period	From	To	Hours
AM	07:00:00	09:00:00	2
MD	09:00:00	15:00:00	6
PM	15:00:00	18:00:00	3
NT	18:00:00	07:00:00	13



Time of day choice models

In the Reno travel demand model, the motorized home-based productions are assigned one of the four time periods based on Multinomial Logit (MNL) models estimated from the travel survey. This replaced fixed factors from the previous model. The choice models are superior to fixed shares because they allow the time-of-day splits to be dependent on household and person characteristics. This procedure leverages data from the population and household tables.

Time of day models were estimated for W_HBW, W_HBO_N_HBSHp, N_HBO, N_HBSr purposes. The time-of-day splits for the N_HBSch (school) were based on fixed shares.

The remainder of the section shows the estimated models along with the variables, coefficients and t-statistics. In the tables below, all models have four alternatives AM, MD, PM and NT and the 'X' indicates alternatives where the variable (row) is applicable.

For these models, the alternative specific constants (ASCs) are adjusted post estimation to match weighted survey targets. The ASCs in this table show the estimated values prior to ASC adjustments.

W_HBW (Home based work trips on a work tour)

Beta	Coefficient	t statistic	W_HBW TOD Model			
			AM	MD	PM	NT
ASCs				-4.0708	0.1283	-0.3362
ASC t-statistics				-2.58	1.24	-2.90
Access to Employment SOV	0.3493	2.49		X		
Access to Transit	0.0793	3.53	X			
High Income > 100k	0.4328	4.72	X			
Presence of Kids in HH (AM)	-0.4858	-3.59	X			
Presence of Kids in HH (NT)	0.4527	3.72				X
Zero Auto HH	-0.6652	-2.40	X			
Low Income < 75k	0.5210	5.02				X
Sample size	2630					
Rho^2	0.037					

A few salient observations:

- People in high-income households favor AM whereas people in low-income households favor the NT period.
- Presence of kids in the household tends to push the departure times later than the AM peak
- Accessibility to employment (auto) and transit accessibility at home improve the probability of choosing an earlier period.
- Estimated ASCs clearly show a propensity for peak period trips.
- The low value of Rho-squared indicates that perhaps the model constants are heavily influencing the choice of the time-period. Nevertheless, the strong t-statistics of most of the coefficients indicate some explanatory behavior. The rho-squared on non-mandatory trips TOD choice models tend to be higher

W_HBO (Home based other trips along a work tour)

Beta	Coefficient	t statistic	W_HBO TOD Model			
			AM	MD	PM	NT
ASCs				-1.0146	0.2828	0.3820
ASC t-statistics				-1.24	0.35	0.48
Access to Nearby Attractions SOV	0.1735	2.49		X	X	X
Presence of Kids in HH (AM)	0.4866	3.53	X			
Presence of Kids in HH (NT)	-1.0712	4.72				X
Presence of Kids in HH (PM)	-0.9257	-3.59			X	
Vehicle Sufficient HH	0.2333	-2.40		X		
Low Income < 75k	-0.3347	5.02				X
Sample size	1078					
Rho^2	0.128					

- People in households with kids prefer the AM period and this could be reflection of school drop-offs in the AM and pick-ups during the MD period.
- Persons in vehicle sufficient HHs prefer MD period for W_HBO trips
- The rho-squared shows a significant improvement over W_HBW. Note that these trips are still other trips on a work tour.

N_HBO (Home based other trips on a non-work tour)

Beta	Coefficient	t statistic	N_HBO TOD Model			
			AM	MD	PM	NT
ASCs				0.4325	0.4095	-0.6151
ASC t-statistics				1.01	4.58	-1.43
HH with kids (AM)	0.5542	5.32	X		X	
High Income (> 100k)	0.5498	3.83	X			
Access Nearby SOV (OP)	0.0843	1.90		X		X
Senior	0.8939	5.67		X		
Female	0.1846	1.86		X		
Employed	0.2020	1.52				X
Vehicle Insufficient HH (NT)	0.2728	2.02				X
Sample size	1743					
Rho^2	0.103					

- HH with kids make N_HBO trips in the AM or PM, likely due to corresponding school drop-off and after school activities
- Seniors and Females have a propensity to make MD trips, owing to the non-mandatory nature of these trips.
- Vehicle insufficient HHs prefer trips in the NT period, likely when vehicles are available.

N_HBSR (Home based social recreation trips on a non-work tour)

Beta	Coefficient	t statistic	N_HBSR TOD Model			
			AM	MD	PM	NT
ASCs				1.6385	-3.4530	-0.4540
ASC t-statistics				9.37	-2.72	-0.43
Access General SOV (NT)	0.2089	2.46				X
Access General SOV (PM)	0.4186	4.06			X	
Female	0.3716	1.83	X			
Presence of Kids in HH	0.5230	4.45				X
Non Worker	0.5167	4.21		X		
Senior (AM)	0.9748	3.72	X			
Senior (MD)	0.6255	3.41		X		
Vehicle Insufficient HH (NT)	-0.3928	-3.16				X
Vehicle Insufficient HH (PM)	-0.2583	-1.84			X	
Child	0.2388	1.33			X	
Sample size	1616					
Rho^2	0.119					

- Non workers have a propensity to avoid the AM peak.
- Seniors prefer earlier part of the day.
- Children may have activities after school during the PM period.

N_HBSHp

Beta	Coefficient	t statistic	N_HBSHp TOD Model			
			AM	MD	PM	NT
ASCs				2.6330	2.4620	3.5747
ASC t-statistics				2.93	2.58	4.81
Gini Simpson Index	5.9457	4.17	X			
Access Nearby SOV (MD)	0.2675	4.32		X		
Access Nearby SOV (PM)	0.2043	2.87			X	
Senior	0.6109	3.44		X		
Non Worker	0.0561	0.39		X		
HH with Kids (NT)	0.4384	2.60				X
HH with Kids (PM)	0.3331	2.03			X	
Worker	0.6816	4.18				X
Vehicle Insufficient HH (NT)	0.2388	1.52				X
Single Parent	0.4069	1.41			X	
Sample size	1232					
Rho^2	0.195					

- As expected, shopping trips tend to occur mostly in the off-peak periods for workers, households with kids, single parents and vehicle insufficient households.
- A high Gini-Simpson index indicates that the person is residing in a diverse zone with mixed land use types, perhaps facilitating a quick shopping trip in the AM.

N_HBSch (Home based school trips on a non-work tour)

The model is a fixed share model corresponding to the following ASCs and splits:

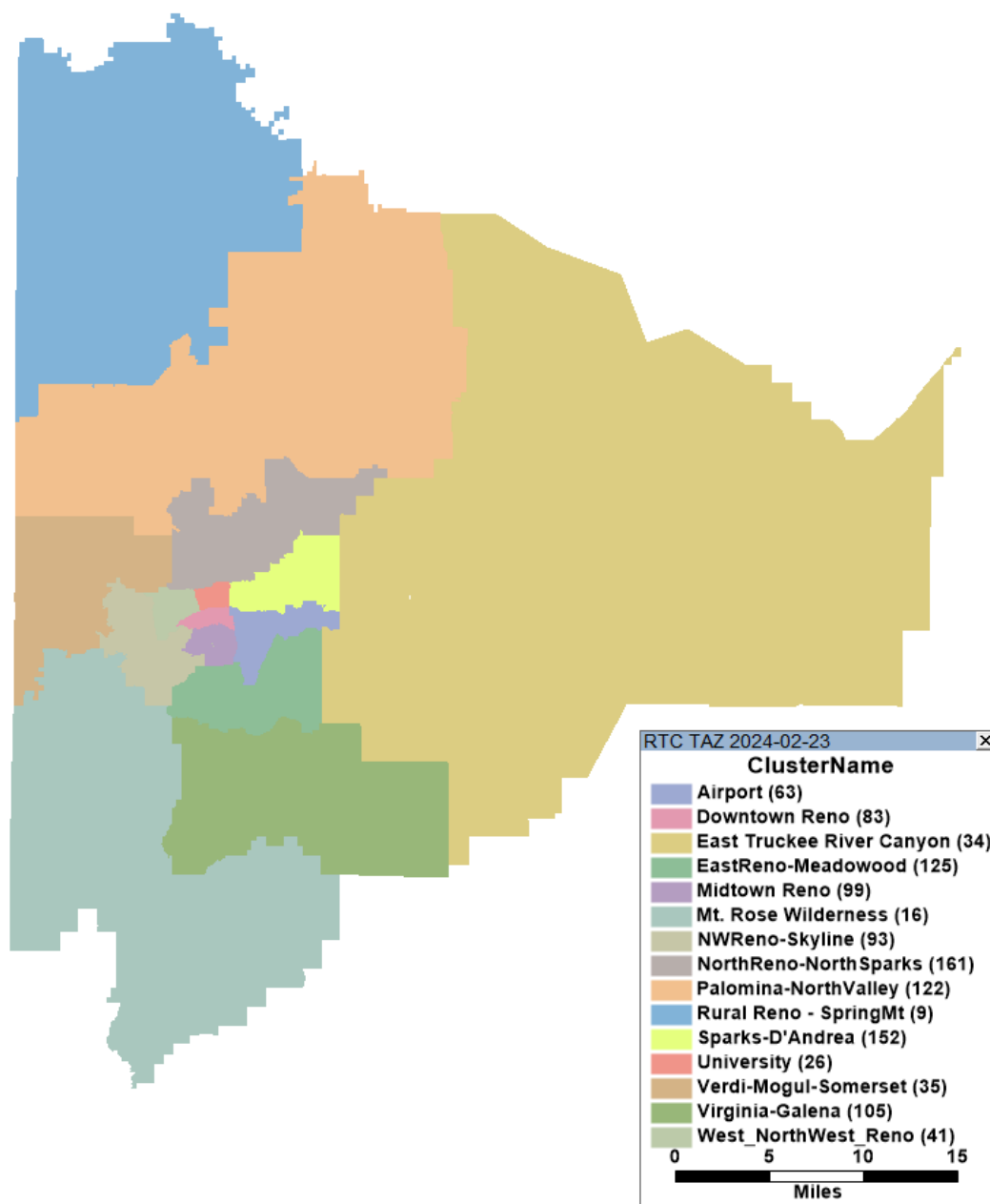
Beta	N_HBSch TOD Model (Constants)			
	AM	MD	PM	NT
ASCs	0.0000	-0.4777	-0.4816	-1.5579
Shares	40.84%	25.33%	25.23%	8.60%

During the model run, once the TODs have been established, the productions are aggregated by purpose, period and market segment for input into the destination and mode choice models.

Destination Choice

Locations in the Reno travel demand model are nested into a two-stage behavioral process: a cluster (spatially aggregated from the zones) is first chosen, followed by a second choice of a zone from within the selected cluster. This process has many advantages, such as enhanced accuracy on a dimension that influences other critical outcomes such as mode choice.

The fifteen clusters are illustrated in the map below:



Home-based trip purposes

The destination choice specifications for the six home-based trip purposes are split across two tables. The first table below contains the cluster constants, intra-cluster constants, and other variables related to geography, level of service and accessibility:

Variable		Trip Purpose											
		W_HBW		N_HBSCH		N_HBSHP		N_HBSR		N_HBO		W_HBO	
		Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic
Cluster ASCs	Downtown	-3.550	-4.8			-1.690	-2			-0.733	-5.32		
	University	-2.590	-3.51			-1.690	-2						
	Midtown	-3.080	-3.69	1.390	3.48	-1.060	-1.28			-0.661	-4.52		
	Airport	-3.190	-3.56	1.230	3.25	1.530	1.87	1.380	4.05	-1.100	-1.81		
	West - NorthWest	-2.040	-2.53	1.540	3.26	0.558	0.63			0.823	2.44		
	Sparks - D'Andrea	-2.620	-3.18	1.230	3.25	1.230	1.5	0.854	2.02				
	NW Reno - Skyline	-1.200	-1.58	1.560	3.51	1.900	2.31	1.710	4.6	0.735	2.45	0.837	1.43
	East Reno - Meadowood	-2.140	-2.7			0.918	1.1	1.190	3.07			1.070	2.4
	Virginia - Galena	-2.510	-3.15	1.480	2.96	-0.709	-0.78	0.986	2.05				
	North Reno - Sparks	-2.950	-3.53	0.629	1.64	0.867	1.04	1.440	4.05	0.729	2.95	1.280	3.3
	Palomina - North Valley	-2.430	-3.35			0.330	0.35			-0.616	-3.29	1.260	2.59
	Verdi - Mogul - Somerset	-2.730	-2.65			0.330	0.35			-0.616	-3.29	1.260	2.59
	East Truckee	-0.559	-0.76										
	Rural Spring Mt												
	Mt Rose	-0.554	-0.59										
Intra-cluster constants	Downtown	0.889	4.61			0.657	1.6	0.654	2.58	0.420	1.81		
	University	0.641	2.26			0.657	1.6	1.190	3.07	1.210	4.24	1.190	2.44
	Midtown	1.250	7.62			0.589	1.81			0.422	1.84	0.435	1.1
	Airport					0.962	0.77					1.420	2.65
	West - NorthWest	1.210	4.88			0.483	0.89			0.622	1.37	0.711	0.97
	Sparks - D'Andrea	0.692	2.68			0.928	3.43	0.680	3.14	1.050	4.94	1.570	4.22
	NW Reno - Skyline	1.050	4.86			0.502	1.21	0.373	1.14	0.769	2.43	1.170	1.88
	East Reno - Meadowood	0.640	3.75			0.608	1.71	0.698	2.48			0.590	1.54
	Virginia - Galena	1.460	6.03			1.900	3.6			0.383	1.2	1.310	2.19
	North Reno - Sparks	0.312	1.2			0.208	0.65	0.455	1.75			1.020	2.54
	Palomina - North Valley					1.970	3.4						
	Verdi - Mogul - Somerset					1.970	3.4						
	East Truckee	-3.270	-5.77					1.130	1.42				
	Rural Spring Mt	-2.600	-1.79					1.130	1.42				
	Mt Rose							1.130	1.42				
Variables	Intra-Cluster			0.682	3.5								
	Time	-0.156	-24.01	-0.222	-9.17	-0.238	-14.84	-0.218	-18.45	-0.197	-21.27	-0.201	-11.06
	Time in excess of 20 minutes					0.094	2.62	0.157	5.67	0.113	6.04	0.095	2.37
	Time in excess of 30 minutes			0.256	5.95								
	Transit accessibility	0.022	1.03										
	Transit accessibility * v0					0.212	3.56	0.246	3.61	0.092	1.93	0.364	3.95
	Walk accessibility												
	Walk accessibility * IZ	0.132	1.24										
	Walk accessibility * IZ * v0					0.636	3.96	0.164	0.84	0.305	1.82	0.259	1.41
	Intrazonal	0.680	3.02							0.147	0.71	0.259	1.16

Many intra-cluster effects are significant, which indicate a propensity to remain relatively close to home for these trip purposes. A further intra-zonal effect is in play for the 'work' and 'other' purposes. Time has the expected negative impact on location choice, though its influence attenuates beyond a certain threshold for many purposes. Destinations with higher transit and walk accessibilities are more attractive to the zero-car segment and on intrazonal trips.

The second table (shown below) contains the effects related to the nests and the double constraint effects via the size terms:

Variable		Trip Purpose											
		W_HBW		N_HBSCH		N_HBSHP		N_HBSR		N_HBO		W_HBO	
		Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic
Size terms	Retail	-0.146	-0.14			-1.930	-5.53	-1.660	-2.66	-2.310	-4.64	0.038	0.04
	Retail * IncCat <= 3	0.215	0.21							-3.250	-5.22	-2.040	-1.84
	Industry											-1.190	-1.14
	Office												
	Office * IncCat >= 4	-1.280	-1.14										
	Service_RateHigh							-1.670	-2.25				
	Service_RateLow							-4.110	-3.71	-1.960	-4.04		
	Service_RateHigh + Service_RateLow	-0.238	-0.23										
	HHPOP							-4.930	-9.6	-7.990	-3.02	-6.470	-3.61
Nest coefficients	K-12 school enrollment			0.000						-2.670	-5.25		
	Size coefficient	0.500	-19.4	0.597	-4.2	1.000		0.388	-13.03	0.586	-13.63	0.342	-15.33
	Downtown	1.000		1.000		1.000		0.908	-1.29	1.000		0.861	-2.4
	University	1.000		1.000		1.000		1.000		1.000		1.000	
	Midtown	0.895	-1.28	0.368	-7.53	1.000		1.000		1.000		0.849	-2.39
	Airport	1.000		0.490	-5.79	0.620	-6.14	0.515	-6.81	1.000		0.775	-2.92
	West - NorthWest	0.684	-3.61	0.245	-7.31	0.725	-2.12	0.824	-1.76	0.417	-7.64	0.711	-2.21
	Sparks - D'Andrea	0.673	-4.23	0.490	-5.79	0.643	-5.99	0.680	-4.03	0.733	-7.3	0.829	-2.57
	NW Reno - Skyline	0.570	-9.39	0.701	-1.96	0.527	-7.53	0.540	-6.93	0.658	-4.86	0.712	-2.09
	East Reno - Meadowood	0.802	-3.25	0.866	-0.58	0.750	-3.91	0.664	-4.25	0.899	-3.19	0.622	-3.84
	Virginia - Galena	0.817	-2.68	0.602	-2.43	1.000		0.752	-2.33	0.816	-3.32	1.000	
	North Reno - Sparks	0.870	-1.43	0.713	-2.36	0.905	-1.13	0.578	-7.85	0.709	-5.13	0.575	-5.57
	Palomina - North Valley	1.000		1.000		0.613	-5.13	0.852	-1.83	1.000		0.723	-2.44
	Verdi - Mogul - Somerset	0.833	-0.65	1.000		0.613	-5.13	0.852	-1.83	1.000		0.723	-2.44
	East Truckee	1.000		1.000		1.000		1.000		1.000		0.796	-0.95
	Rural Spring Mt	1.000		1.000		1.000		1.000		1.000		0.796	-0.95
	Mt Rose	0.297	-3.88	1.000		1.000		1.000		1.000		0.796	-0.95
Rho-squared		0.139		0.255		0.208		0.162		0.158		0.235	

Numerous nest coefficients were statistically different from the default of 1.0, which validates the nested approach to destination choice. Various employment totals, population numbers, and school enrollments ensure that the model sends the right number of trips to each zone.

Non-home-based trip purposes

Destinations for the mode-specific non-home-based trip purposes are determined via multinomial choice models with the following specifications:

Variable		Trip Purpose							
		N_NHB_Auto		W_NHB_Auto		NHB_Transit		NHB_WalkBike	
		Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic	Coefficient	t statistic
Variables	(Total) Time	-0.187	-22.06	-0.266	-28.58	-0.057	-8.7		
	Time in excess of 30 minutes	0.128	5.89						
	Time in excess of 20 minutes			0.241	13.92				
	Distance							-0.381	-5.24
	Distance > 1 mile							-1.460	-6.08
	Transit accessibility					0.114	1.49		
	Walk accessibility * v0							0.416	3.24
	Intra-cluster	0.698	9.57	0.741	10.52			1.710	8.84
	Intrazonal	0.988	6.95	1.210	11.51			2.070	10.58
Size terms	Retail	-0.426	-0.63	0.846	1.04			-1.700	-1.33
	Office			-0.586	-0.69			-3.440	-1.27
	Service_RateHigh	-1.990	-1.91						
	Service_RateLow	-2.590	-2.89	-1.840	-2.09	3.400	0.87	-2.570	-1.87
	K-12 school enrollment	-2.250	-3.27	-2.150	-2.63			-2.510	-1.81
	Size coefficient	0.434	-18.6	0.383	-25.84	0.212	-15.83	0.403	-8.08
Rho-squared		0.155		0.212		0.177		0.381	

The models are sensitive to a range of network and level-of-service effects including distance and mode-specific travel times and accessibilities. Distances to potential destinations (and those over a mile) are less attractive for the walk/bike segment. Intra-cluster and intra-zonal effects are strong for auto and non-motorized modes. Walk accessibility at the destination makes destinations more desirable when there are no cars in the household. Travel time has the expected negative impact, but its impact attenuates beyond a certain threshold for the drive mode. Attractions depend on a variety of employment (retail, office, service) and school enrollment totals.

Mode Choice

This document summarizes the mode choice (MC) models estimated using travel survey and other data from the Reno region. These are nested logit models that predict which modes of travel will be used based on factors like travel time, transit fare, and transit headway. Mode choice models play a crucial role in the system of travel demand models, not only directly but also through logsums that are then used to inform destination choices. For more information on mode choice models, see the TFRsource website (https://tfrsource.org/topics/Mode_choice.html).

W_HBW (home-based work trips along a work tour)

Variable	Coefficient	t statistic	Modes					
			SOV	HOV2	HOV3	Bus	TNC	Shuttle
Constant (SOV)	3.6900	17.97	X					
Constant (HOV2)	2.7000	7.06		X				
Constant (HOV3)	1.2800	1.85			X			
Constant (Bus)	0.8250	1.76				X		
Constant (TNC)	3.0600	8.32					X	
Time/IVTT	-0.0106	-3.20	X	X	X	X	X	X
Toll (SOV)	-0.0388	-3.20	X					
Toll (HOV2)	-0.0194	-3.20		X				
Toll (HOV3+)	-0.0118	-3.20			X			
High-income HH	0.2890	3.13	X					
Low-income HH	2.0300	5.51				X		
Vehicle-sufficient HH	0.9980	4.86	X					
Vehicle-insufficient HH	0.6160	2.95					X	
Zero-vehicle HH	2.3200	5.75				X		
SOV Distance * AOC (\$/mile) (SOV)	-0.0388	-3.20	X					
HOV Distance * AOC (\$/mile) (HOV2)	-0.0194	-3.20		X				
HOV Distance * AOC (\$/mile) (HOV3+)	-0.0118	-3.20			X			
Bus walk time (access + egress + transfer)	-0.0302	-2.57				X		
Bus wait time (initial + transfer)	-0.0212	-3.20				X		
Bus fare	-0.0535	-3.20				X		
Bus dwell time	-0.0106	-3.20				X		
TNC wait time	-0.0893	-3.26					X	
TNC fare	-0.0388	-3.20					X	
Auto nest coefficient	0.7390	-1.56	X	X	X			
Rho-squared			0.532					

Tolls and distance have reducing negative impacts on the auto modes with increasing occupancy. High-income and vehicle-sufficient household members are more likely to drive alone, while low-income and zero-vehicle household members see a significant shift to bus. Vehicle insufficiency increases the use of TNC options. Time and fare variables have the expected negative effects.

N_HBSCH (home-based school trips along a non-work tour)

Variable	Coefficient	t statistic	Modes				
			SOV	HOV2	HOV3	Bus	School Bus
Constant (SOV)	-2.0500	-6.03	X				
Constant (HOV2)	-1.0400	-3.25		X			
Constant (HOV3)	-0.9220	-2.89			X		
Constant (School Bus)	-1.2600	-3.32					X
Time/IVTT	-0.0290		X	X	X	X	
Time (School Bus)	-0.0920	-4.02					X
Vehicle-insufficient HH	1.0800	4.67					X
Zero-vehicle HH	3.0900	5.53				X	
Bus walk time (access + egress + transfer)	-0.0725					X	
Bus wait time (initial + transfer)	-0.0580					X	
Bus fare	-0.1462					X	
Bus dwell time	-0.0290					X	
Auto nest coefficient	0.7000		X	X	X		
Rho-squared	0.170						

Time and fare variables have the expected negative effects. Students from vehicle-insufficient households ride the school bus more, while those from zero-vehicle households ride the bus.

N_HBSHP (home-based shop trip along a non-work tour)

Variable	Coefficient	t statistic	Modes				
			SOV	HOV2	HOV3	Bus	TNC
Constant (SOV)	1.0600	2.37	X				
Constant (HOV2)	1.5700	3.54		X			
Constant (HOV3)	1.0600	2.37			X		
Constant (Bus)	-0.1270	-0.22				X	
Toll (SOV)	-0.0341	-2.74	X				
Toll (HOV2)	-0.0171	-2.74		X			
Toll (HOV3+)	-0.0103	-2.74			X		
SOV Distance * AOC (\$/mile) (SOV)	-0.0341	-2.74	X				
HOV Distance * AOC (\$/mile) (HOV2)	-0.0171	-2.74		X			
HOV Distance * AOC (\$/mile) (HOV3+)	-0.0103	-2.74			X		
Time/IVTT	-0.0093	-2.74	X	X	X	X	
Time (TNC)	-0.0294	-2.45					X
Vehicle-sufficient HH	0.9260	9.08	X				
Zero-vehicle HH	3.4500	12.78				X	
Bus walk time (access + egress + transfer)	-0.0233	-2.74				X	
Bus wait time (initial + transfer)	-0.0187	-2.74				X	
Bus fare	-0.0470	-2.74				X	
Bus dwell time	-0.0093	-2.74				X	
TNC fare	-0.1076	-2.45					X
Auto nest coefficient	0.7000		X	X	X		
Rho-squared	0.290						

Tolls and distance have reducing negative impacts on the auto modes with increasing occupancy. Vehicle-sufficient household members are more likely to drive alone, while zero-vehicle household members see a significant propensity for bus. Time and fare variables have the expected negative effects.

N_HBSR (home-based social/recreation trip along a non-work tour)

Variable	Coefficient	t statistic	Modes				
			SOV	HOV2	HOV3	Bus	TNC
Constant (SOV)	1.0900	2.32	X				
Constant (HOV2)	1.8400	3.97		X			
Constant (HOV3)	1.7200	3.70			X		
Constant (Bus)	-1.1100	-1.46				X	
Toll (SOV)	-0.0443	-3.06	X				
Toll (HOV2)	-0.0221	-3.06		X			
Toll (HOV3+)	-0.0134	-3.06			X		
SOV Distance * AOC (\$/mile) (SOV)	-0.0443	-3.06	X				
HOV Distance * AOC (\$/mile) (HOV2)	-0.0221	-3.06		X			
HOV Distance * AOC (\$/mile) (HOV3+)	-0.0134	-3.06			X		
Time/IVTT	-0.0121	-3.06	X	X	X	X	
Time (TNC)	-0.0439	-3.31					X
Low-income HH	1.4600	3.42				X	
Vehicle-sufficient HH	1.1500	12.20	X				
Vehicle-insufficient HH	0.8470	2.17				X	
Zero-vehicle HH (Bus)	3.4400	8.88				X	
Zero-vehicle HH (TNC)	3.2100	9.69					X
Bus walk time (access + egress + transfer)	-0.0303	-3.06				X	
Bus wait time (initial + transfer)	-0.0242	-3.06				X	
Bus fare	-0.0610	-3.06				X	
Bus dwell time	-0.0121	-3.06				X	
TNC fare	-0.1606	-3.31					X
Auto nest coefficient	0.7000		X	X	X		
Rho-squared	0.294						

Tolls and distance have reducing negative impacts on the auto modes with increasing occupancy. Vehicle-sufficient household members are more likely to drive alone, while low-income, vehicle-insufficient and zero-vehicle household members see a significant shift to bus. The absence of vehicles in the household also increases the likelihood of TNC options. Time and fare variables have the expected negative effects.

N_HBO (home-based other trip along a non-work tour)

Variable	Coefficient	t statistic	Modes				
			SOV	HOV2	HOV3	Bus	TNC
Constant (SOV)	1.7000	6.24	X				
Constant (HOV2)	1.9400	7.25		X			
Constant (HOV3)	1.6100	5.98			X		
Constant (Bus)	-0.9060	-1.72				X	
Toll (SOV)	-0.0221	-2.34	X				
Toll (HOV2)	-0.0110	-2.34		X			
Toll (HOV3+)	-0.0067	-2.34			X		
SOV Distance * AOC (\$/mile) (SOV)	-0.0221	-2.34	X				
HOV Distance * AOC (\$/mile) (HOV2)	-0.0110	-2.34		X			
HOV Distance * AOC (\$/mile) (HOV3+)	-0.0067	-2.34			X		
Time/IVTT	-0.0060	-2.34	X	X	X	X	X
Low-income HH	1.0500	2.47				X	
Vehicle-sufficient HH	0.6520	8.22	X				
Zero-vehicle HH	2.7900	11.22				X	
Bus walk time (access + egress + transfer)	-0.0151	-2.34				X	
Bus wait time (initial + transfer)	-0.0121	-2.34				X	
Bus fare	-0.0304	-2.34				X	
Bus dwell time	-0.0060	-2.34				X	
TNC wait time	-0.0393	-1.10					X
TNC fare	-0.0221	-2.34					X
Auto nest coefficient	0.7000		X	X	X		
Rho-squared			0.256				

Tolls and distance have reducing negative impacts on the auto modes with increasing occupancy. Vehicle-sufficient household members are more likely to drive alone, while low-income and zero-vehicle household members see a significant shift to bus. Time and fare variables have the expected negative effects.

W_HBO (home-based other trip along a work tour)

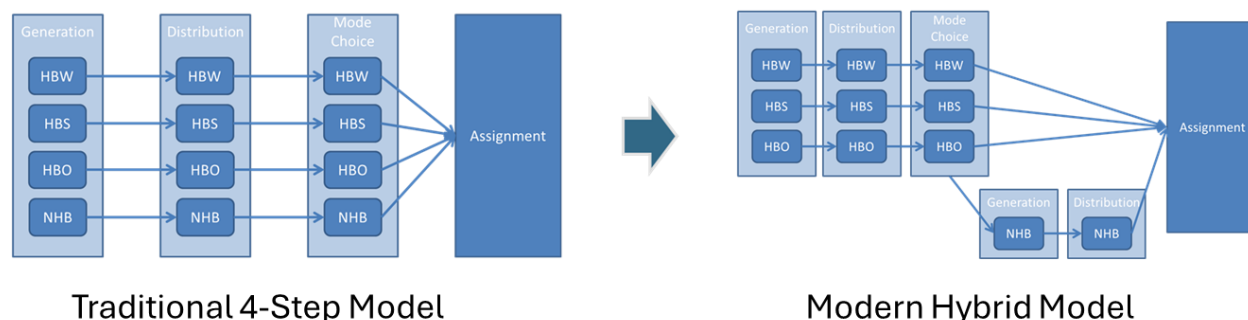
Variable	Coefficient	t statistic	Modes				
			SOV	HOV2	HOV3	Bus	TNC
Constant (SOV)	2.4100	7.79	X				
Constant (HOV2)	1.5300	4.99		X			
Constant (HOV3)	0.9190	2.92			X		
Constant (Bus)	-0.3360	-0.50				X	
Toll (SOV)	-0.0662	-2.94	X				
Toll (HOV2)	-0.0331	-2.94		X			
Toll (HOV3+)	-0.0075	-2.94			X		
SOV Distance * AOC (\$/mile) (SOV)	-0.0662	-2.94	X				
HOV Distance * AOC (\$/mile) (HOV2)	-0.0331	-2.94		X			
HOV Distance * AOC (\$/mile) (HOV3+)	-0.0075	-2.94			X		
Time/IVTT	-0.0181	-2.94	X	X	X	X	X
High-income HH	0.3080	2.73	X				
Low-income HH	1.5500	2.53				X	
Vehicle-insufficient	1.0700	3.68					X
Zero-vehicle HH (Bus)	3.5400	7.18				X	
Zero-vehicle HH (TNC)	1.0600	1.70					X
Bus walk time (access + egress + transfer)	-0.0453	-2.94				X	
Bus wait time (initial + transfer)	-0.0362	-2.94				X	
Bus fare	-0.0913	-2.94				X	
Bus dwell time	-0.0181	-2.94				X	
TNC fare	-0.0662	-2.94					X
Auto nest coefficient	0.7000		X	X	X		
Rho-squared	0.456						

Tolls and distance have reducing negative impacts on the auto modes with increasing occupancy. High-income household members are more likely to drive alone, while low-income and zero-vehicle household members see a significant shift to bus. Vehicle insufficiency and absence of household vehicles increases the use of TNC options. Time and fare variables have the expected negative effects.

The non-home-based trip purposes have the travel mode built into them and hence do not require separate mode choice models.

Non-Home-Based

There are many problems related to non-home-based trips in traditional trip-based models arising from the fact that they are disconnected from the home-based trips with which they comprise complete tours. In order to properly represent non-home-based trips, two spatial distribution or destination/spatial choice models are required to account for both the trip's origin location and destination location. The four-step model architecture is fundamentally flawed because it produces non-home-based trips from only one trip distribution or spatial choice model.



To address these problems, the model adopts an alternative approach with a simple change to the structure of the trip-based model, running the non-home-based model components after and conditional on the home-based model components instead of in parallel and independently of them as in the traditional four-step model. This relatively simple structural change significantly improves the model's ability to represent non-home-based trips and their response to land use changes and transportation infrastructure investments. Running a NHB distribution or destination choice model after and conditional on home-based destination choices in this approach, provides the required second spatial distribution model to properly model both the origin and destination of NHB trips.

In this approach, NHB trips are generated separately by mode based on home-based mode choices. This essentially provides information about whether a traveler has a car with them and allows the model, despite its trip-based form, to ensure a reasonable consistency of modes on tour.

Available modes include:

- SOV
- HOV2
- HOV3+
- Auto Pay
- Transit
- Non-motorized

The model coefficients below are the result of multiple linear regression with a forced intercept at zero. In addition, the model estimation will be scaled up to a predicted regional total. As a result of these two factors, the displayed r-squared values are not as meaningful. Instead, the value of the coefficients is in determining the relative effect that various home-based trip types have on non-home-based trip generation.

When reviewing the coefficients below, note their logical consistency: SOV NHB trips are most likely to result when the HB trip is SOV or HOV. NHB walk trips can be made when a person drives from home, but is more much more likely if they walk. These results greatly improve the NHB models compared to traditional trip-based construction.

In the tables below, any “alpha” and “gamma” terms refer to boosting coefficients used. See the section on boosting for an explanation of those terms.

W_NHBO

These are non-home-based trips made on a work tour not related to work. In other words, neither end of the trip is work or home.

SOV

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_sov	W_HBO_sov	0.0897	0.0079	11.2855	0.0000
W_HBW_sov	W_HBW_sov	-0.0092	0.0058	-1.5937	0.1111
r_sq	NA	0.1326	NA	NA	NA

HOV2

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_hov2	W_HBO_auto	0.0102	0.0022	4.6312	0
W_HBO_hov3	W_HBO_auto	0.0102	0.0022	4.6312	0
W_HBO_sov	W_HBO_auto	0.0102	0.0022	4.6312	0
r_sq	NA	0.2330	NA	NA	NA

HOV3+

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_hov2	W_HBO_auto	0.0027	0.0013	2.0586	0.0397
W_HBO_hov3	W_HBO_auto	0.0027	0.0013	2.0586	0.0397
W_HBO_sov	W_HBO_auto	0.0027	0.0013	2.0586	0.0397
r_sq	NA	0.2330	NA	NA	NA

Auto Pay

No significant presence of this trip type in the survey, and it is not an important source of travel demand. No model was estimated.

Transit

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_bus	W_HBO_bus	0.0368	0.0043	8.5550	0
W_HBO_walkbike	W_HBO_walkbike	0.0242	0.0040	6.0579	0
r_sq	NA	0.0176	NA	NA	NA

Non-motorized

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_walkbike	W_HBO_walkbike	0.0188	0.0036	5.2108	0
r_sq	NA	0.0133	NA	NA	NA

W_NHBW

These are non-home-based trips made on a work tour where one trip end is work or work-related.

SOV

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_hov2	W_HBO_hov2	0.7462	0.0364	20.5072	0
W_HBO_hov3	W_HBO_hov3	0.4522	0.0799	5.6598	0
W_HBO_sov	W_HBO_sov	0.6283	0.0209	30.1282	0
W_HBW_hov2	W_HBW_auto	0.1637	0.0145	11.2657	0
W_HBW_hov3	W_HBW_auto	0.1637	0.0145	11.2657	0
W_HBW_sov	W_HBW_auto	0.1637	0.0145	11.2657	0
r_sq	NA	0.3111	NA	NA	NA

HOV2

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_hov2	W_HBO_hov	0.1376	0.0225	6.1309	0
W_HBO_hov3	W_HBO_hov	0.1376	0.0225	6.1309	0
W_HBO_sov	W_HBO_sov	0.1493	0.0132	11.3168	0
r_sq	NA	0.2342	NA	NA	NA

HOV3+

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_hov2	W_HBO_hov	0.0200	0.0037	5.3335	0.000
W_HBO_hov3	W_HBO_hov	0.0200	0.0037	5.3335	0.000
W_HBW_hov3	W_HBW_hov3	0.0334	0.0267	1.2512	0.211
r_sq	NA	0.0688	NA	NA	NA

Auto Pay

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_auto_pay	W_HBO_auto_pay	0.2440	0.0144	16.9478	0
r_sq	NA	0.0137	NA	NA	NA

Transit

term	estimated_as	estimate	std.error	statistic	p.value
W_HBO_bus	W_HBO_bus	0.6725	0.0137	49.2453	0
r_sq	NA	0.0095	NA	NA	NA

Non-motorized

term	estimated_as	estimate	std.error	statistic	p.value
W_HBW_walkbike	W_HBW_walkbike	0.2112	0.0112	18.8617	0
r_sq	NA	0.0451	NA	NA	NA

N_NHBSHP

These are non-home-based trips made on a non-work tour with one trip end for shopping.

SOV

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_sov	N_HBO_sov	0.2384	0.0161	14.8456	0
N_HBSHP_hov2	N_HBSHP_auto	0.1130	0.0118	9.5638	0
N_HBSHP_hov3	N_HBSHP_auto	0.1130	0.0118	9.5638	0
N_HBSHP_sov	N_HBSHP_auto	0.1130	0.0118	9.5638	0
N_HBSR_hov2	N_HBSR_auto	0.0839	0.0115	7.2967	0
N_HBSR_hov3	N_HBSR_auto	0.0839	0.0115	7.2967	0
N_HBSR_sov	N_HBSR_auto	0.0839	0.0115	7.2967	0
alpha	NA	0.0038	0.6392	-10.8695	0
gamma	NA	2.5523	0.2886	8.8450	0
r_sq	NA	0.4549	NA	NA	NA

HOV2

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_hov2	N_HBO_hov	0.2186	0.0144	15.1538	0.0000
N_HBO_hov3	N_HBO_hov	0.2186	0.0144	15.1538	0.0000
alpha	NA	0.1533	0.5671	-3.9832	0.0001
gamma	NA	0.8574	0.2562	3.3471	0.0008
r_sq	NA	0.1389	NA	NA	NA

HOV3+

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_hov2	N_HBO_hov2	-0.0017	0.0074	-0.2234	0.8232
N_HBSHP_hov3	N_HBSHP_hov	0.6680	0.0111	59.9399	0.0000
alpha	NA	0.5607	0.2433	-2.4386	0.0148
gamma	NA	0.2647	0.1099	2.4088	0.0161
r_sq	NA	0.1429	NA	NA	NA

Auto Pay

term	estimated_as	estimate	std.error	statistic	p.value
N_HBSHP_auto_pay	N_HBSHP_auto_pay	0.3032	0.0288	10.5336	0.0000
alpha	NA	1.0652	0.1204	0.4506	0.6523
gamma	NA	-0.0276	0.0544	-0.5080	0.6115
r_sq	NA	0.1037	NA	NA	NA

Transit

term	estimated_as	estimate	std.error	statistic	p.value
N_HBSHP_bus	N_HBSHP_bus	0.6522	0.0094	69.2889	0
r_sq	NA	0.0703	NA	NA	NA

Non-motorized

term	estimated_as	estimate	std.error	statistic	p.value
N_HBSHP_walkbike	N_HBSHP_walkbike	0.4006	0.0147	27.3263	0
r_sq	NA	0.0660	NA	NA	NA

N_NHBO

These non-home-based trips are those where neither end is shopping. In effect, these are all other non-home-based trips made on non-work tours. Visiting friends or family at their home is an example of this trip type.

SOV

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_hov2	N_HBO_auto	0.0943	0.0098	9.6196	0
N_HBO_hov3	N_HBO_auto	0.0943	0.0098	9.6196	0
N_HBO_sov	N_HBO_auto	0.0943	0.0098	9.6196	0
N_HBSR_hov2	N_HBSR_auto	0.0816	0.0107	7.6090	0
N_HBSR_hov3	N_HBSR_auto	0.0816	0.0107	7.6090	0
N_HBSR_sov	N_HBSR_auto	0.0816	0.0107	7.6090	0
alpha	NA	0.0088	0.6110	-9.7293	0
gamma	NA	2.1653	0.2759	7.8473	0
r_sq	NA	0.2933	NA	NA	NA

HOV2

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_hov2	N_HBO_auto	0.1086	0.0079	13.7771	0.0000
N_HBO_hov3	N_HBO_auto	0.1086	0.0079	13.7771	0.0000
N_HBO_sov	N_HBO_auto	0.1086	0.0079	13.7771	0.0000
N_HBSHP_sov	N_HBSHP_sov	0.0366	0.0167	2.1865	0.0289
N_HBSR_hov2	N_HBSR_hov	0.0893	0.0106	8.4648	0.0000
N_HBSR_hov3	N_HBSR_hov	0.0893	0.0106	8.4648	0.0000
alpha	NA	0.0075	0.6226	-9.5635	0.0000
gamma	NA	2.2363	0.2812	7.9539	0.0000
r_sq	NA	0.3064	NA	NA	NA

HOV3+

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_hov2	N_HBO_auto	0.0451	0.0056	8.0997	0.0000
N_HBO_hov3	N_HBO_auto	0.0451	0.0056	8.0997	0.0000
N_HBO_sov	N_HBO_auto	0.0451	0.0056	8.0997	0.0000
N_HBSHP_hov2	N_HBSHP_auto	0.0069	0.0062	1.1141	0.2653
N_HBSHP_hov3	N_HBSHP_auto	0.0069	0.0062	1.1141	0.2653
N_HBSHP_sov	N_HBSHP_auto	0.0069	0.0062	1.1141	0.2653
alpha	NA	0.0459	0.5107	-7.1966	0.0000
gamma	NA	1.4169	0.2308	6.1401	0.0000
r_sq	NA	0.2621	NA	NA	NA

Auto Pay

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_auto_pay	N_HBO_auto_pay	0.0828	0.0098	8.4440	0
N_HBSHP_auto_pay	N_HBSHP_auto_pay	0.1937	0.0153	12.6285	0
N_HBSR_auto_pay	N_HBSR_auto_pay	1.0688	0.0230	46.4451	0
r_sq	NA	0.0339	NA	NA	NA

Transit

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_bus	N_HBO_bus	0.0063	0.0047	1.3289	0.1840
N_HBO_walkbike	N_HBO_walkbike	0.0336	0.0066	5.0741	0.0000
N_HBSR_bus	N_HBSR_bus	0.0088	0.0039	2.2307	0.0258
r_sq	NA	0.0267	NA	NA	NA

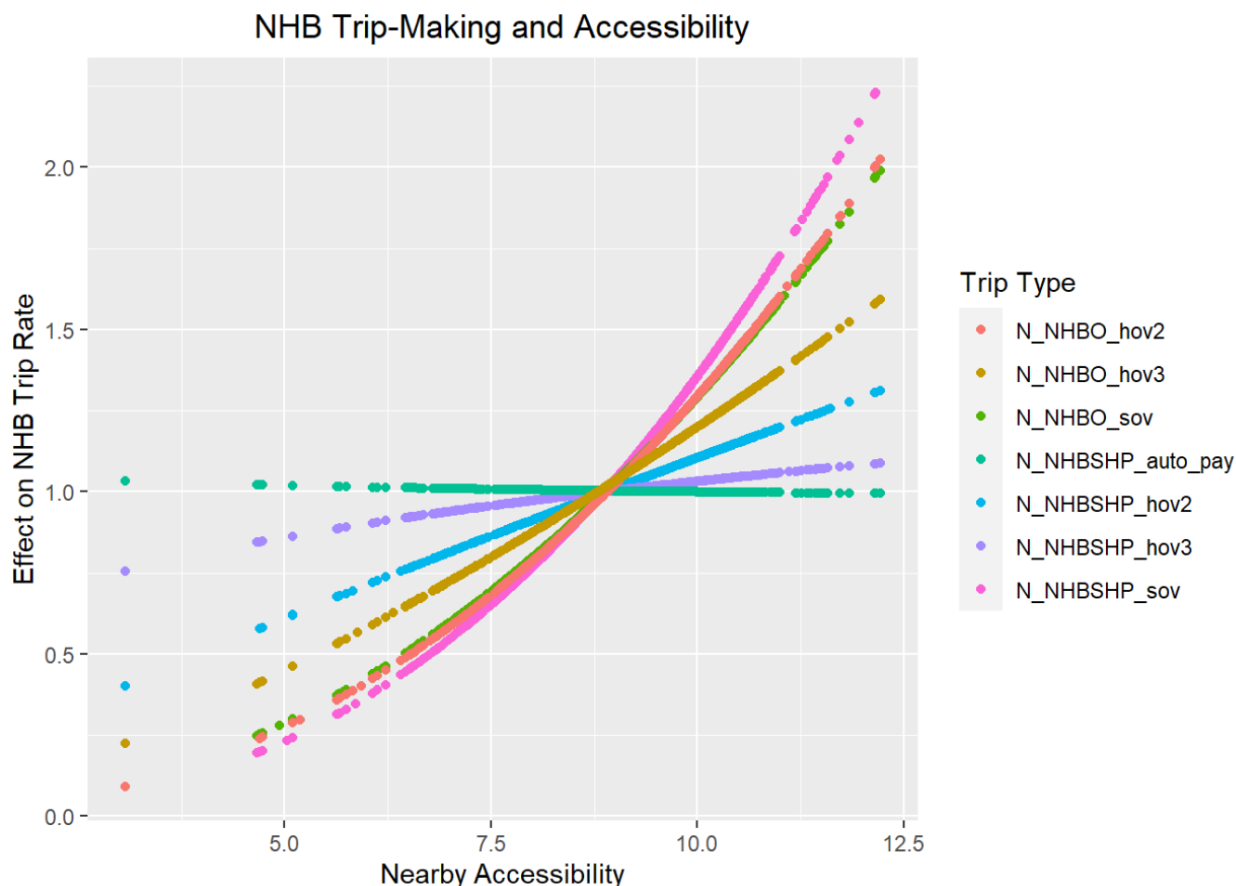
Non-motorized

term	estimated_as	estimate	std.error	statistic	p.value
N_HBO_walkbike	N_HBO_walkbike	0.0381	0.0094	4.0727	0
r_sq	NA	0.0176	NA	NA	NA

Boosting

“Boosting” is an approach borrowed from machine learning where the errors of a previous model are used to estimate a second model. The models shown above illustrate the link between home- and non-home-based trips by mode, but intuition and experience tells us that accessibility should also influence NHB trip making. Travel to a central business district is more likely to lead to further trips compared to traveling to a rural zone.

The chart below shows how accessibility impacts each trip type and mode combination. The y-axis is a simple factor. When the factor is 1, the trip rates will be the same as displayed in the tables above. A y-value of 0.5 means the trip rates will be reduced by 50 percent. Conversely, a y-value of 1.5 means that trip rates are increased by 50 percent. With this additional factor, the model will understand the role that accessibility plays in NHB trip generation.



NHB Time of Day Adjustment

The independent feedback by time of day in the model presents a unique challenge for NHB. NHB and HB trips have very different time of day patterns (see Time of Day documentation), but NHB generation is simply rates multiplied by HB trip ends. Without correction, this would mean that the NHB trips would have the same distribution as HB trips. To correct this, Caliper calculated adjustment factors by tour type, mode, and time of day. This was done by comparing raw model outputs back to the observed NHB trips from the survey.

The easiest way to understand the table below is by highlighting one row as an example. For SOV trips on work tours, these factors move NHB trips out of the AM and NT periods and into the MD and PM. Another way of saying it is that, compared to HB trips, NHB trips are less likely in the AM/NT and more likely in the MD/PM. The large mid day factor reflects lunch trips and other mid-day activities. The relative size of the AM and PM periods confirms what we know: that stops are more likely to be made on the way home compared to the way to work (e.g. picking up groceries).

Tour Type	Mode	AM	MD	NT	PM
NonWork	HOV2	0.74	1.38	2.07	1.54
NonWork	HOV3	1.66	1.07	2.18	2.67
NonWork	SOV	0.60	1.50	0.43	0.72
NonWork	Transit	1.19	3.13	1.42	1.35
NonWork	Nonmotorized	1.33	3.20	1.14	0.53
NonWork	Auto Pay	1.00	1.22	2.33	2.86
Work	Auto Pay	0.53	5.93	3.57	5.44
Work	HOV2	0.78	2.74	0.38	1.34
Work	HOV3	2.00	3.90	1.55	4.69
Work	SOV	0.53	2.58	0.38	1.06
Work	Transit	0.29	2.11	2.25	0.79
Work	Nonmotorized	0.44	5.39	2.07	2.10

University Travel

The sub-model for university travel is important to the Reno region to capture the impact of the University of Nevada, Reno on the transportation system.

The Reno Regional University Model simulates the on- and off-campus travel of students of the region's major university. It relies on socio-economic data inputs from the TAZ layer that designates the locations of the student population as well as the building locations on the various campuses to allocate the university trips.

For student travel, the most important distinction between students is whether they live on or off campus. On-campus students are expected to make more trips, given their ability to easily travel back and forth to their dorm/apartment rooms between classes/lunch. In addition, many of those trips are expected to start and end on campus, thus likely producing shorter trip lengths. The typical student demographic is On-campus students are more likely to be undergraduate students who are enrolled at the university full-time, with more classes. Graduate students are more likely to reside off-campus and have different travel patterns than their on-campus counterparts. Therefore, the market segmentation for the university model is stratified by on-campus and off-campus students as well as by trip purpose.

For the university model, trip generation estimates the number of university trips by market segment (on-campus and off-campus students) and purpose (6 purposes, as explained later) as a function of observed/available land use characteristics. Trip distribution uses a gravity formulation with distance as the measure of impedance and a multinomial logit mode choice model is then used to segment trips by travel mode. Time of day or "diurnal" factors are used to segment trips by time and day and convert the outcomes from production-attraction to origin-destination format. The university trips are then combined with trips from other model components prior to carrying out highway and transit assignments.

The balance of this document specifies each model component, presenting empirical evidence for each specification. As local data for the UNR student population was unavailable, coefficients are asserted, based on evidence for other travel purposes. For these asserted models, calibration targets were developed from the available university survey data in other areas and calibrated to match observed local outcomes (e.g., traffic in the vicinity of campus).

Model Segmentation

The model design reflects the desire for the university model to understand potential differences in travel for on-campus and off-campus students so that it is better able to predict behavior for individual markets than it would be for all students combined. Therefore, the model structure segments trips into the following purposes:

- **Home-Based-Campus (Home-Campus or UHC):** from off-campus home locations to buildings on campus or from on-campus dorms to buildings on campus
- **Home-Based-Other (Home-Other or UHO):** from off-campus home locations to other locations not on campus or from on-campus dorms to locations not on campus
- **Campus-Based-Other (Campus-Other or UCO):** from campus to locations off campus
- **On-Campus (On-campus or UC1):** to and from locations on the SAME campus
- **University student Other-Other (Other-to-Other or UOO):** from and to locations not associated to a campus

The structure segments the home-based trip purposes, Home-Campus and Home-Other, by on- and off-campus students. On-campus students are expected to make more trips, with many of those trips

expected to start and end on campus. On-campus students are more likely to be undergraduate students who are enrolled at the university full-time. Graduate students are more likely to reside off-campus.

The first model step is [trip generation](#), which estimates the number of trips by market segment and purpose as a function of observed land use characteristics. The second model step is trip distribution, which uses a common [gravity formulation](#) with distance as the measure of impedance. A [multinomial logit mode choice model](#) is then used to segment trips by travel mode. Time of day or “diurnal” factors are used to segment trips by time and day and convert the outcomes from production-attraction to origin-destination format. The university trips are then combined with trips from other model components prior to carrying out roadway and transit assignment.

An innovative feature of the model design is the generation step for the Other-to-Other trip purpose. To maintain consistency across travel model, these trips are generated directly from the Home-based Other and Campus-based Other trip ends. The generation step is mode specific. For example, if all the Home-based Other and Campus-based Other trips select automobile, then only automobile Other-to-Other trips will be generated.

University Trip Productions

The selected production rates for four of the five trip types or purposes (all except Other-to-other) are presented in the table below. For home-based trip purposes, separate rates are presented based on student home location (on- or off-campus). The production rates for home-based trips are trips per student. For campus-based trips, the production rates are per 1000 square feet of campus building space.

Home Location	Trip Purpose	Production Rate	Unit
On-campus	Home-Campus	1.08	Student
On-campus	Home-Other	1.54	Student
Off-campus	Home-Campus	0.23	Student
Off-campus	Home-Other	1.12	Student
All-Students	On-campus	10.05	Campus Building SF (in 000s)
All-Students	Campus-Other	3.60	Campus Building SF (in 000s)

Trip production rates by mode were used to estimate productions for Other-to-other trips. More specifically, the Other-to-other productions, as described previously, were a function of Home-based Other and Campus-to-Other attractions. Like the resident models, this approach ensures consistency between the “Other” trip ends by mode, i.e., it is not possible to have only automobile Home-Other and Campus-Other movements and only bicycle Other-to-other movements. The choice to generate trips by mode for this purpose was driven by the desire to maintain model consistency with tours.

The table below summarizes the Home-Other and Campus-Other trips by mode: (1) Auto, which includes both drive and carpool; (2) Transit; (3) Walk; and (4) Bicycle.

Trip Purpose	Mode	Trip Rate
Home-Other and Campus-Other	Auto	0.283
Home-Other and Campus-Other	Walk	0.003
Home-Other and Campus-Other	Transit	0.013
Home-Other and Campus-Other	Bicycle	0.024

University Trip Attractions

Trip attraction rates for off-campus attractions (Home-Other and Campus-Other) are applied to zonal characteristics including the off-campus student population and the number of retail jobs. Home-Other attractions are estimated separately for on-campus and off-campus students.

variable	Home-Other On-Campus	Home-Other Off-Campus	Campus- Other
intercept	5.5856	100.1907	73.7073
retail_employment	0.0247	0.0394	0.0563
student_off_campus	0.0292	0.224	0.4086

For on-campus attractions (Home-based Campus, Campus-to-campus), the attractions are distributed among campus zones proportional to the distribution of the square footage of university buildings.

University Trip Distribution

The second step of the University Model development process is to develop a trip distribution model. In this step, the TAZ level productions and attractions estimated in the trip generation step are allocated to TAZ pairs. The allocation is based on a gravity model that takes into account the distance between each of the TAZ pairs.

A distance-based gravity model is used to distribute the university trips from production zones to attraction zones. It was assumed that the friction factors are a function of distance (along the roadway network) and that they follow a gamma distribution, as below:

Purpose	a	b	c
Home-Campus - Off-campus students	5	0.8	0.1
Home-Other- Off-campus students	5	0.9	0.1
All other	5	1.1	0.1

University Mode Choice

The third step in the University Model development process is the mode choice model. During the mode choice step, trips by zonal pair are distributed across available modes. A multinomial mode choice model was used to split trips between the University Model's four modes - walk, bike, transit and car based on time, cost and modal preferences. The model assumed a very simple utility function based only on the in-vehicle and out-of-vehicle travel times by mode and the cost of each mode (if any). The same parameters were used for all home- and campus-based trips, although asserted constants were used to vary mode split by purpose.

Coefficient	Value
IVTT	-0.02
OVTT	-0.03
Cost	-0.20

University Time-of-Day

The fixed factors shown below were applied to split out university trips by purpose into the model's time-of-day periods.

	Home-Campus	Home-Other	Campus-Other	On-Campus
AM	0.1000	0.1213	0.1343	0.1276
MD	0.4975	0.4332	0.5983	0.6688
PM	0.1858	0.1970	0.2674	0.2036
NT	0.2167	0.2485	0.0000	0.0000

External Travel

Introduction

Travel from outside the Reno regional model's geographic boundary to locations inside the boundary are referred to as "external-internal" or "EI" trips. Movements from inside the boundary to outside are "internal-external" or "IE" trips. Movements through but not stopping within the region are referred to as "external-external" or "EE" trips. Together, these travel patterns are referred to here as "EE" and "IE/EI" travel. This document discusses the development of these components of the regional model.

Base year auto external travel patterns were developed from Caliper's Transography data, drawing on connected vehicle data from April of 2022. A total of 119,559 external trips were observed in the data, of which 3,970 were through trips and 115,589 were inbound/outbound trips.

External-External (Through) Trips

External-external or through trips are estimated by applying iterative proportional fitting (also known as raking or the RAS algorithm and sometimes mistakenly identified with the similar Fratar process) to grow the estimated base year external-external ("seed") matrix to estimated future marginal sums (the total trips to/from each station). The base year auto and truck matrices are shown below.

	3001	3002	3003	3004	3005	3006	3007	3008	3009	3010	3011	3012
3001	0.0	17.7	39.0	3.7	0.0	207.6	68.8	1.4	34.8	6.7	402.2	1.6
3002	14.3	0.0	1.2	0.0	0.0	4.4	7.0	0.0	4.1	1.1	54.9	1.1
3003	29.2	0.8	0.0	0.3	0.0	3.4	33.8	3.4	27.5	3.5	178.2	1.6
3004	8.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	7.4	0.0
3005	0.0	0.0	0.0	0.0	0.0	4.9	13.7	11.2	0.0	0.0	0.0	0.0
3006	150.6	3.1	5.4	0.0	3.1	0.0	13.1	146.4	1.8	0.7	45.5	5.4
3007	103.2	12.2	29.8	0.0	16.9	9.3	0.0	2.2	0.0	0.0	7.4	0.0
3008	0.0	0.0	1.1	0.0	12.1	181.1	13.3	0.0	0.0	0.0	0.0	0.0
3009	34.1	4.8	17.0	0.0	0.0	3.4	1.0	0.0	0.0	0.0	0.0	0.0
3010	9.4	1.7	8.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.0	9.0
3011	377.9	46.4	184.6	2.1	0.0	85.4	12.2	2.5	2.2	15.7	0.0	25.6
3012	4.1	0.0	2.5	0.0	0.0	0.0	0.0	0.0	0.0	18.7	20.2	0.0

Figure 6. Base Year (2022) Auto External-External Trips

	3001	3002	3003	3004	3005	3006	3007	3008	3009	3010	3011	3012
3001	0.0	0.0	0.0	1.9	0.0	258.4	0.0	0.0	0.0	0.0	0.0	0.0
3002	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	0.0	0.0	0.0
3003	0.0	0.0	0.0	1.8	0.0	43.4	0.0	0.0	0.0	0.0	0.0	0.0
3004	8.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.1	0.0
3005	0.0	0.0	0.0	0.0	0.0	5.9	0.0	0.0	0.0	0.0	0.0	0.0
3006	196.4	4.7	35.7	0.0	5.4	0.0	17.0	18.4	6.4	5.1	16.7	0.4
3007	0.0	0.0	0.0	0.0	0.0	22.3	0.0	0.0	0.0	0.0	0.0	0.0
3008	0.0	0.0	0.0	0.0	0.0	27.1	0.0	0.0	0.0	0.0	0.0	0.0
3009	0.0	0.0	0.0	0.0	0.0	6.5	0.0	0.0	0.0	0.0	0.0	0.0
3010	0.8	0.2	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
3011	0.0	0.0	0.0	0.2	0.0	23.2	0.0	0.0	0.0	0.0	0.0	0.0
3012	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.0	0.0

Figure 7. Base Year (2022) Truck External-External Trips

The total external-external trips by station for autos and trucks for a given scenario are given by simply apply the percent auto EE and percent truck EE to the total external station volume provided by the user in the SE data file. The base year data is provided below:

Station	AWDT	% Auto EE	% Auto IEEI	% Trk EE	% Trk IEEI
3001	32000	4.7	72.4	1.5	21.4
3002	4600	3.8	90.6	0.2	5.4
3003	6300	9.1	78.4	1.3	11.2
3004	870	2.5	57.3	1.6	38.6
3005	1250	5.0	80.0	0.9	14.1
3006	9400	9.3	45.5	7.5	37.7
3007	11550	3.0	87.1	0.3	9.6
3008	4350	8.6	80.9	1.0	9.4
3009	5100	2.6	88.7	0.3	8.5
3010	2450	3.4	85.5	0.4	10.7
3011	44000	3.4	93.9	0.1	2.6
3012	5350	1.7	97.3	0.0	1.0

External-Internal / Internal-External (Inbound/Outbound) Trips

The number of IE/EI trips are controlled by the volumes at the external station after external travel has been subtracted (using the table above). These trips are distributed to internal zones using a gravity model. The number of trips at each external station are treated as the productions. The attractions at each internal zone are estimated as a simple function of the zone's population and employment:

$$\text{Inbound/Outbound Attractions} = 0.1551 \times \text{Population} + 0.0059 \times \text{Employment}$$

The gravity model uses a gamma function for the friction factors with parameters varying based on the type of external station as categorized by the average length of trip it serves.

Type	a	b	c
Long	5	0.25	0.1
Medium	5	0.6	0.1
Short	5	1.3	0.1

The external stations were categorized as follows:

Station	Type
3001	Medium
3002	Medium
3003	Long
3004	Medium
3005	Medium
3006	Long
3007	Short
3008	Medium
3009	Medium
3010	Medium
3011	Long
3012	Medium

External Trip Time-of-Day

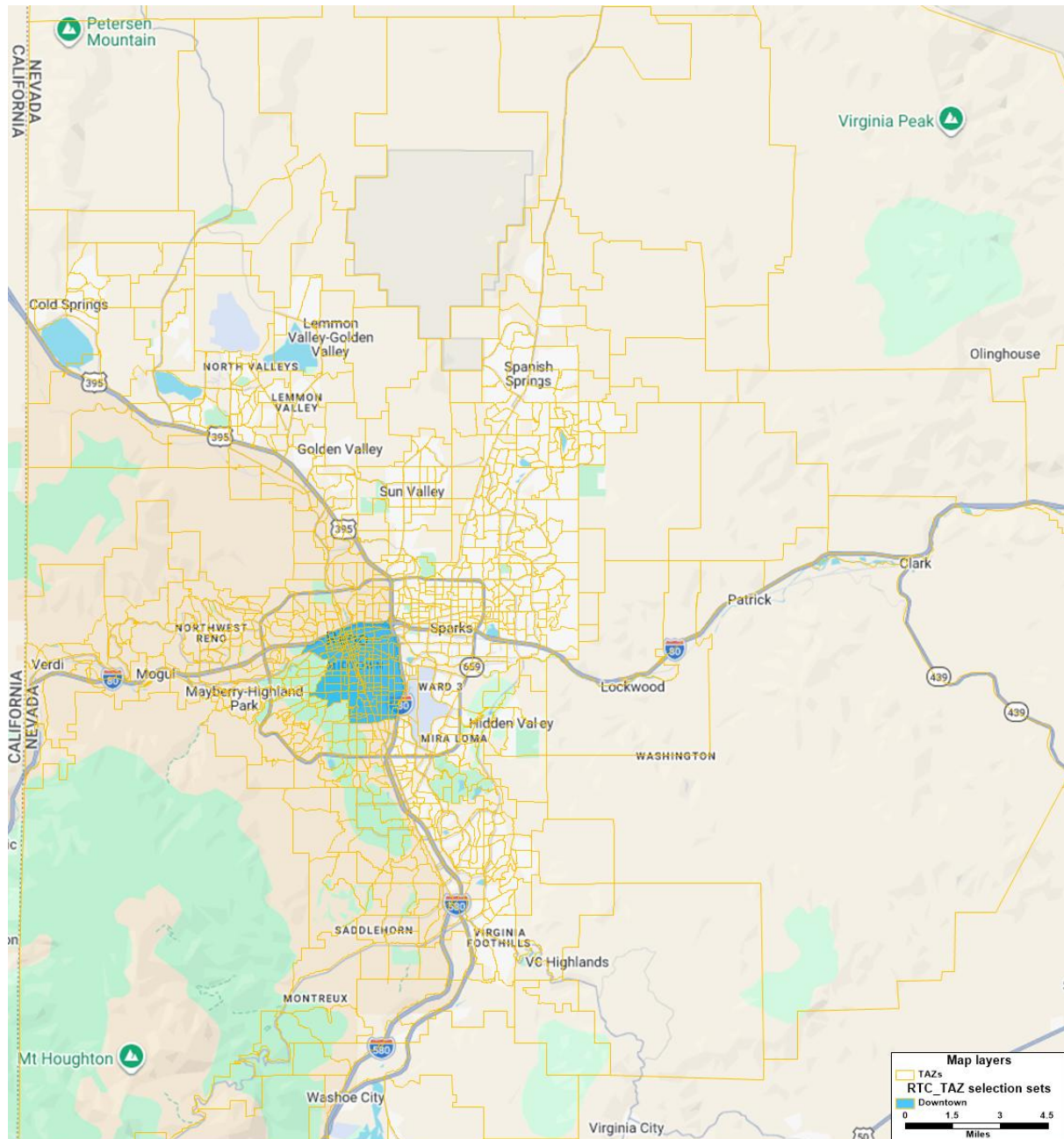
The fixed factors shown below were applied to split out external trips into the model's time-of-day periods:

Period	EE Auto	EE Truck	EI/IE
AM	11.80%	15.50%	12.72%
MD	48.06%	46.30%	40.46%
PM	18.84%	14.05%	22.71%
NT	21.30%	24.15%	24.12%

Visitor Travel

Visitor population synthesis methodology

The visitor model begins with a simple population synthesis. The number of occupied hotel rooms is estimated using occupancy rates from the Reno-Sparks Convention and Visitors Authority (RSCVA) and the Nevada Commission on Tourism (NCOT) for Downtown (68%) and elsewhere (58%). The Downtown area is highlighted blue in the map below based on the definitions provided by RSCVA. The population synthesis uses the visitor survey as its seed data and draws one travel party from the survey based on its weights for every occupied hotel room. This ensures that the synthesized visitor population has the same makeup as the visitor's survey respondents.



Visitor trip generation

The visitor models operate on trip purposes that reflect the special nature of this market segment. Visitors may be in the region for business or as tourists. Their trips may further be either directly to or from their lodging or be generated through intermediate stops along their daily tours. In addition, tourists' lodging-based activity purposes are divided into Entertainment/Social, Gaming/Dining, and Other. Non-lodging-based trips are described by travel mode. The various combinations are summarized in the table below:

Trip Type	Description			
	Visit Type	Trip End Type	Activity Purpose	Mode
BLB	Business	Lodging-Based	Work	
TLBES	Tourist	Lodging-Based	Entertainment/Social	
TLBGD	Tourist	Lodging-Based	Gaming/Dining	
TLBOTH	Tourist	Lodging-Based	Other	
BNLB SOV	Business	Non-Lodging-Based		SOV
BNLB HOV	Business	Non-Lodging-Based		HOV
BNLB TNC	Business	Non-Lodging-Based		TNC
BNLB BUS	Business	Non-Lodging-Based		Bus
BNLB WLK	Business	Non-Lodging-Based		Walk
TNLB SOV	Tourist	Non-Lodging-Based		SOV
TNLB HOV	Tourist	Non-Lodging-Based		HOV
TNLB TNC	Tourist	Non-Lodging-Based		TNC
TNLB BUS	Tourist	Non-Lodging-Based		Bus
TNLB WLK	Tourist	Non-Lodging-Based		Walk

Lodging-based trip rates

The trip production rates for lodging-based visitor travel are estimated using simple decision trees expressed in the equations summarized in the table below:

Trip Type	Rule	Rate
BLB	Purpose < 2	0.00
BLB	Purpose >= 2 and Income < 100000 and PartySize < 2	1.46
BLB	Purpose >= 2 and Income >= 100000 and PartySize < 2	1.33
BLB	Purpose >= 2 and Income < 100000 and PartySize >=2	0.22
BLB	Purpose >= 2 and Income >= 100000 and PartySize >=2	0.91
TLBES	Purpose >= 2	0.00
TLBES	Purpose < 2 and Income < 50000 and PartySize < 3	0.17
TLBES	Purpose < 2 and Income >= 50000 and Income < 100000 and PartySize < 3	0.15
TLBES	Purpose < 2 and Income >=100000 and PartySize < 3	0.33
TLBES	Purpose < 2 and Income < 50000 and PartySize >= 3	0.26
TLBES	Purpose < 2 and Income >= 50000 and Income < 100000 and PartySize >= 3	0.37
TLBES	Purpose < 2 and Income >=100000 and PartySize >= 3	0.46
TLBGD	Purpose >= 2	0.00
TLBGD	Purpose < 2 and Downtown = 1 and PartySize <3	0.62
TLBGD	Purpose < 2 and Downtown = 0 and PartySize <3	0.25
TLBGD	Purpose < 2 and Downtown = 1 and PartySize >=3	0.33
TLBGD	Purpose < 2 and Downtown = 0 and PartySize >=3	0.21
TLBOTH	Purpose >= 2	0.00
TLBOTH	Purpose < 2 and Income < 75000 and PartySize <3	0.11
TLBOTH	Purpose < 2 and Income >= 75000 and Income < 100000 and PartySize <3	0.34
TLBOTH	Purpose < 2 and Income >= 100000 and PartySize <3	0.33
TLBOTH	Purpose < 2 and Downtown = 1 and Income < 75000 and PartySize >=3	0.18
TLBOTH	Purpose < 2 and Downtown = 1 and Income >= 75000 and Income < 100000 and PartySize >=3	0.37
TLBOTH	Purpose < 2 and Downtown = 1 and Income >= 100000 and PartySize >=3	0.47
TLBOTH	Purpose < 2 and Downtown = 0 and PartySize >=3	0.21

The trip attraction equations for lodging-based visitor travel are summarized in the table below:

Trip Type	Rate	Variable
BLB	0.1241	HotelRms
BLB	0.2229	Retail
BLB	0.0071	TotalEmp
BLB	1.1461	NAICS72
TLBES	3.3976	NAICS71
TLBES	0.4225	NAICS72
TLBGD	3.3976	NAICS71
TLBGD	0.4225	NAICS72
TLBOTH	0.1661	HotelRms
TLBOTH	1.4435	Retail
TLBOTH	0.8890	NAICS72

Non-lodging-based trip rates

The trip production rates for non-lodging-based visitor travel are estimated using the simple equations summarized in the table below (which can be understood as transition probabilities):

Trip Type	Rate	Variable
BNLBSOV	0.1451	BLB_SOV
BNLBHOV	0.0333	BLB_HOV
BNLBTNC	0.0197	BLB_TNC
BNLBTNC	0.0030	BLB_BUS
BNLBBUS	0.0048	BLB_BUS
BNLBBUS	0.0005	BLB_WLK
BNLBWLK	0.0085	BLB_SOV
BNLBWLK	0.0085	BLB_HOV
BNLBWLK	0.0216	BLB_TNC
BNLBWLK	0.0143	BLB_BUS
BNLBWLK	0.0741	BLB_WLK
TNLBSOV	0.0034	TLBES_SOV
TNLBSOV	0.0015	TLBES_HOV
TNLBHOV	0.0324	TLBES_HOV
TNLBTNC	0.0057	TLBES_TNC
TNLBBUS	0.0014	TLBES_BUS
TNLBBUS	0.0005	TLBES_WLK
TNLBWLK	0.0117	TLBES_TNC
TNLBWLK	0.0146	TLBES_BUS
TNLBWLK	0.0489	TLBES_WLK
TNLBSOV	0.0046	TLBGD_SOV
TNLBSOV	0.0020	TLBGD_HOV
TNLBHOV	0.0515	TLBGD_HOV
TNLBTNC	0.0153	TLBGD_TNC
TNLBBUS	0.0029	TLBGD_BUS
TNLBBUS	0.0009	TLBGD_WLK
TNLBWLK	0.0131	TLBGD_TNC
TNLBWLK	0.0164	TLBGD_BUS
TNLBWLK	0.0548	TLBGD_WLK
TNLBSOV	0.0067	TLBOTH_SOV
TNLBSOV	0.0067	TLBOTH_HOV
TNLBHOV	0.0286	TLBOTH_HOV
TNLBTNC	0.0034	TLBOTH_TNC
TNLBTNC	0.0004	TLBOTH_WLK
TNLBBUS	0.0006	TLBOTH_TNC
TNLBBUS	0.0005	TLBOTH_BUS
TNLBBUS	0.0003	TLBOTH_WLK
TNLBWLK	0.0120	TLBOTH_TNC
TNLBWLK	0.0304	TLBOTH_WLK

In addition to the above rate equations, non-lodging-based productions are boosted by accessibility and other effects according to the table below:

Trip Type	Rate	Variable
BNLBSOV	0.0922	constant
BNLBSOV	1.3348	access_nearby_sov
BNLBHOV	0.0922	constant
BNLBHOV	1.3348	access_nearby_sov
BNLBTNC	0.7680	constant
BNLBTNC	0.2270	access_nearby_sov
BNLBWLK	0.5770	constant
BNLBWLK	0.1684	access_walk
TNLBSOV	1.0000	constant
TNLBSOV	0.1047	access_nearby_sov
TNLBHOV	1.0000	constant
TNLBHOV	0.1047	access_nearby_sov
TNLBTNC	0.4423	constant
TNLBTNC	0.4966	access_nearby_sov
TNLBWLK	0.5770	constant
TNLBWLK	0.1684	access_walk

Visitor destination choice

Lodging-based trips

The size variables for the destination choice models were estimated as traditional attraction models using zonal regression. The results are below:

$$\begin{aligned}
 \text{BLBA} &= 0.124136 * \text{HotelRms} \\
 &+ 0.222901 * \text{Retail} \\
 &+ 0.0071 * \text{TotalEmp} \\
 &+ 1.146074 * \text{NAICS72}
 \end{aligned}$$

$$\begin{aligned}
 \text{TLBESA} &= 3.397572 * \text{NAICS71} \\
 &+ 0.422516 * \text{NAICS72}
 \end{aligned}$$

$$\begin{aligned}
 \text{TLBGDA} &= 3.397572 * \text{NAICS71} \\
 &+ 0.422516 * \text{NAICS72}
 \end{aligned}$$

$$\begin{aligned}
 \text{TLBOTH A} &= 0.166057 * \text{HotelRms} \\
 &+ 1.443487 * \text{Retail} \\
 &+ 0.888962 * \text{NAICS72}
 \end{aligned}$$

Using the log of these attractions as the size variable, destination choice models were estimated from the visitor survey. The utility function for each trip type is given below (using V to indicate the systematic or non-stochastic part of the utility).

$$V_{\text{BLB}} = \ln(\text{BLBA}) + \text{MClogsum}_{\text{BLB}} + 0.389524 * \text{access_hotels} + 2.5 * \ln(\text{Time})$$

$$V_{\text{TLBES}} = 0.996179 * \ln(\text{TLBESA}) + \text{MClogsum}_{\text{TLBES}} + 0.438688 * \text{access_hotels} + 3.5 * \ln(\text{Time})$$

$$V_{TLBGD} = 0.996179 \cdot \ln(TLBGDA) + MClogsum_{TLBGD} + 0.438688 \cdot \text{access_hotels} - 0.3 \cdot \ln(\text{Time})$$

$$V_{TLBOTH} = \ln(TLBOTHA) + MClogsum_{TLBOTH} + 0.392097 \cdot \text{access_hotels} + 0.14 \cdot \text{Time}$$

In all cases the mode choice logsum was able to be included, but its nesting parameter had to be constrained to unity indicating destination and mode choice are simultaneous. However, in all cases either auto time or its logarithm were also significant as was accessibility to hotel rooms.

Non-lodging-based trips

Doubly-constrained gravity models were calibrated to distribute the NLB trips and roughly reproduce the observed trip length frequency distributions. The trips are distributed separately by mode using that mode's skims. Given the limited sample size, it was generally not possible to estimate gamma form impedance functions so inverse power functions were used. The calibrated beta parameters for the inverse power function are presented in below:

Trip Type	Beta
BNLB _{SOV}	4.0
BNLB _{HOV}	4.0
BNLB _{TNC}	4.0
BNLB _{BUS}	3.0
BNLB _{WLK}	2.0
TNLB _{SOV}	4.0
TNLB _{HOV}	4.0
TNLB _{TNC}	4.0
TNLB _{BUS}	3.5
TNLB _{WLK}	1.5

Visitor mode choice

Mode choice models were estimated for the LB trips from the visitor survey. The utility functions were composed of each mode's alternative specific bias constant, a term for the mode's fare, if any, and a term for travel time via that mode. SOV and HOV were nested under AUTO.

$$V = a + B_1 \cdot \text{fare} + B_2 \cdot \text{time}$$

The parameters are given below:

	BLB	TLBES	TLBGD	TLBOTH
AUTO nest	0.576486	0.576486	0.576486	0.576486
a – SOV	0.15	-2.50	-1.00	-1.50
a – HOV	0.14	0.15	1.10	0.20
a – TNC	0.14	-2.50	0.20	-1.00
a – BUS	-2.00	-1.50	-1.30	-0.90
a – WLK	5.50	5.50	1.30	2.50
B₁ (fare)	-0.012921	-0.012921	-0.012921	-0.012921
B₂ (time)	-0.090493	-0.090493	-0.090493	-0.090493

The modes of the non-lodging-based trips are already included in the corresponding trip purpose definitions.

Visitor time-of-day choice

Time-of-day and directional factors were estimated from the visitor survey.

	BLB		TLBES		TLBES		TLBOTH		BNLB	TNLB
	PA	AP	PA	AP	PA	AP	PA	AP		
AM	19.64%	0.00%	0.00%	0.00%	0.00%	0.00%	7.07%	0.00%	0.00%	0.84%
MD	14.29%	15.79%	36.11%	4.86%	20.28%	4.29%	32.07%	14.84%	50.00%	47.90%
PM	7.14%	28.95%	7.41%	20.83%	16.51%	7.14%	2.71%	25.00%	30.00%	31.94%
NT	8.93%	5.26%	6.48%	24.31%	13.21%	38.57%	8.15%	10.16%	20.00%	19.32%
Total	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	50.00%	100.00%	100.00%

Trucks and Commercial Vehicle Travel

Introduction

Trucks and commercial vehicles contribute significantly to congestion on major facilities in an urban area. In most instances, their travel behavior is distinct from residents of that urban area. To better capture commercial vehicle traffic in the Reno regional model, a separate set of commercial vehicle models were developed. These include trip generation, time of day, and distribution for three commercial vehicle types: commercial autos, vans, and pickups (CV), single unit trucks (SUT), and multi-unit trucks (MUT).

These classes are defined using the federal vehicle classification.

- CV - Classes 1-3
- SUT - Classes 5-7
- MUT - Classes 8-13

The models structure follows the Quick Response Freight Manual (QRFM) approach, but was re-calibrated by Wood Rodgers based on local truck count data.

Truck Trip Generation

Truck trip generation in the QRFM approach typically uses employment by industry variables, and this approach was retained for light commercial vehicles. However, there is mounting evidence that square footage variables are better predictors of the number of truck trips than the number of employees. Therefore, in the v1.2 update, Wood Rodgers re-calibrated the truck trip generation component to use these variables. The table below shows the rates for each vehicle class applied to each zonal variable to generate truck trips.

Field	CV	SUT	MUT
Industry	0.870	0	0
Retail	0.774	0	0
Office	0.383	0	0
Service_RateLow	0.383	0	0
Service_RateHigh	0.383	0	0
HH	0.218	0.019	0.007
IndManWar_Acres	0	6.693	2.962
IndOther_Acres	0	1.241	0.547
Retail_Acres	0	19.503	4.256
Office_Acres	0	2.185	0.259
Service_Acres	0	3.047	0.362

Special generators were also identified for three zones. The most significant was TAZ 755 southeast of the I-80/Sparks Blvd. interchange containing several major truck (and bus) depots or yards. The second was TAZ 1523 between I-580, Damonte Ranch Parkway and Bus/Alt US. 395, containing massive warehouses, and the third was TAZ 2103 containing the FedEx and Old Dominion Freight Lines facilities off of I-80 near Lockwood.

Truck Trip Distribution

Once generated, commercial vehicle and truck trips are distributed among zones using a gravity model. The gravity model uses a gamma function for the friction factors with parameters varying by truck type.

	a	b	c
CV	5	1	0.1
SUT	5	0.75	0.1
MUT	503	-0.1403	0.1133

Truck Time-of-Day

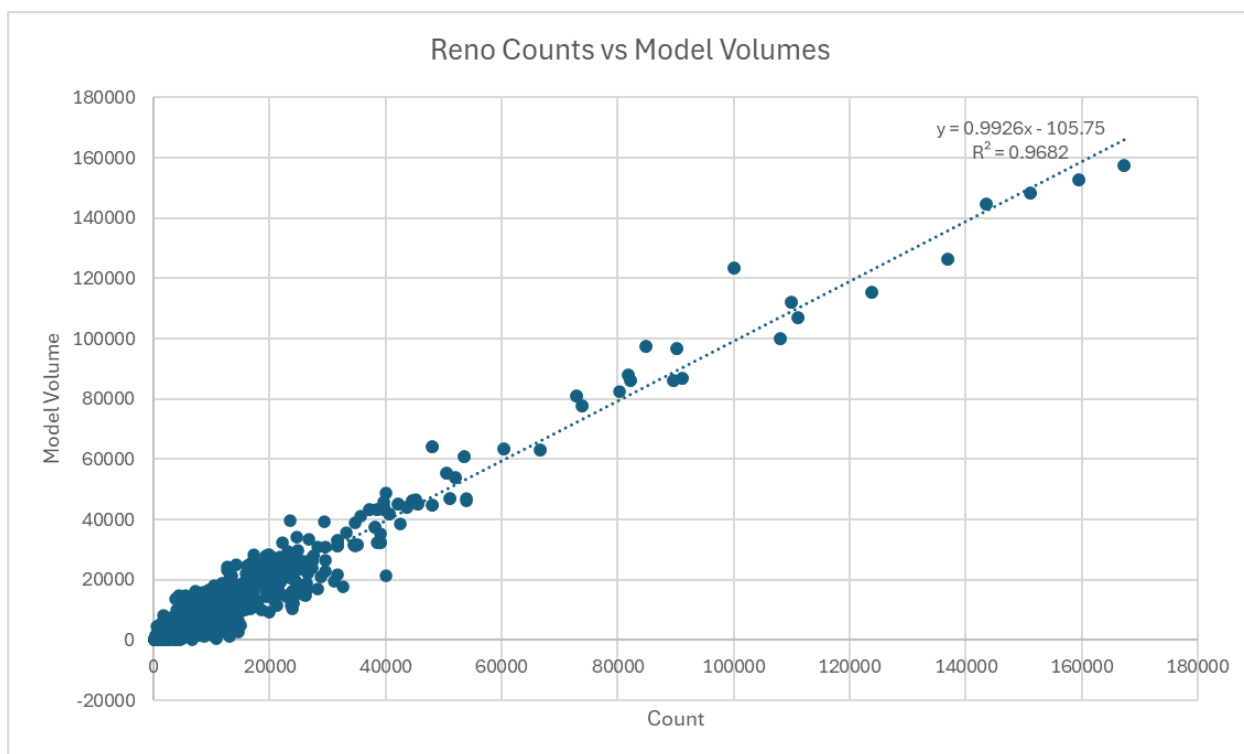
The fixed factors shown below were applied to split out truck and commercial vehicle trips into the model's time-of-day periods:

	CV	SUT	MUT
AM	0.091	0.097	0.106
MD	0.452	0.393	0.444
PM	0.125	0.133	0.145
NT	0.331	0.378	0.306

Validation

Once the behavioral models were estimated and calibrated using survey data, Caliper ran the full model and compared outputs to traffic counts (2023 counts). This data provided a check on model performance in the base year and built confidence that the model can be used in the future.

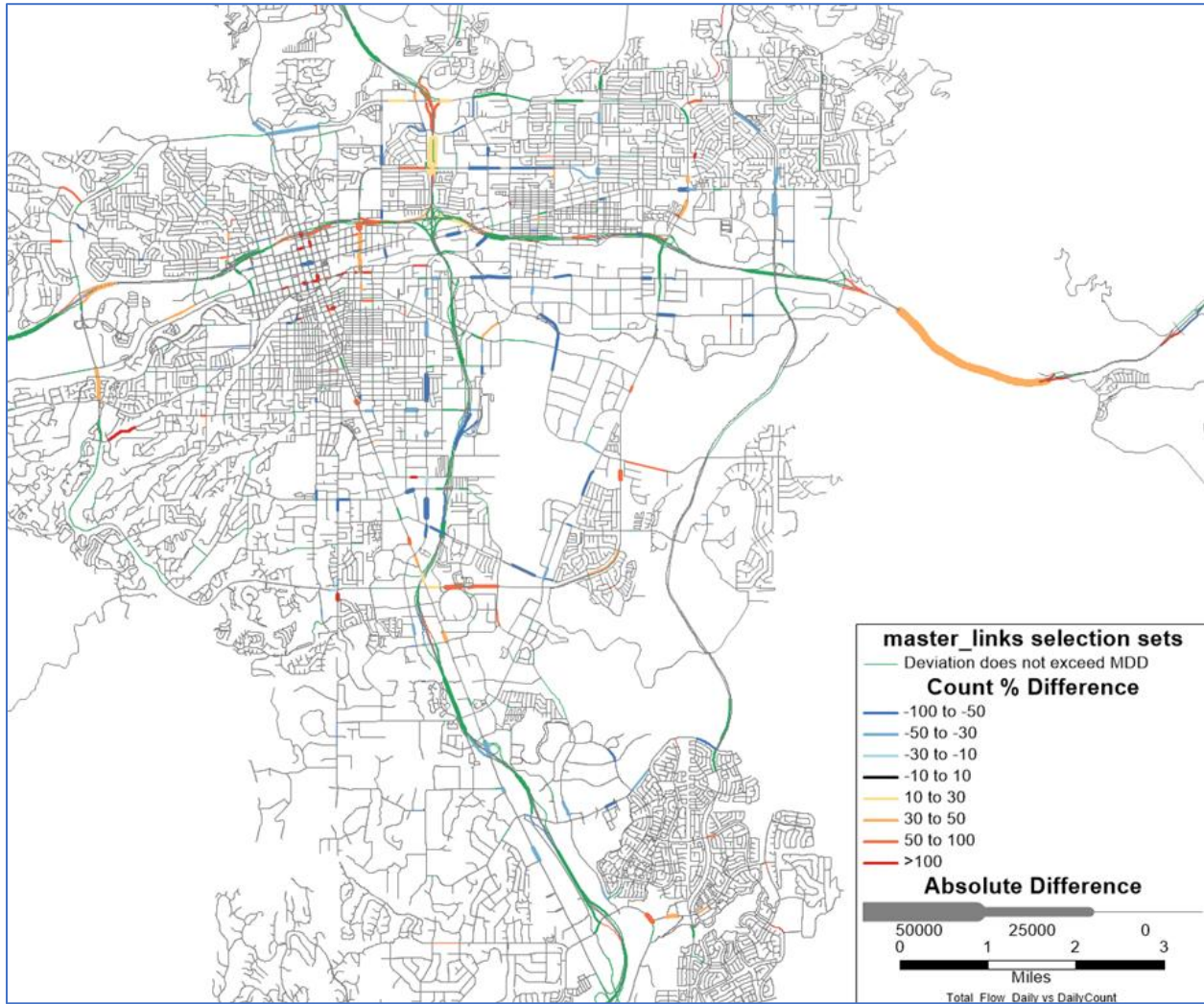
The new model validated notably better than its predecessor with lower overall root mean square error (RMSE) and higher coefficient of determination ($r^2 = 0.97$ in new model vs. $r^2 = 0.88$) despite validating to almost twice as many counts.

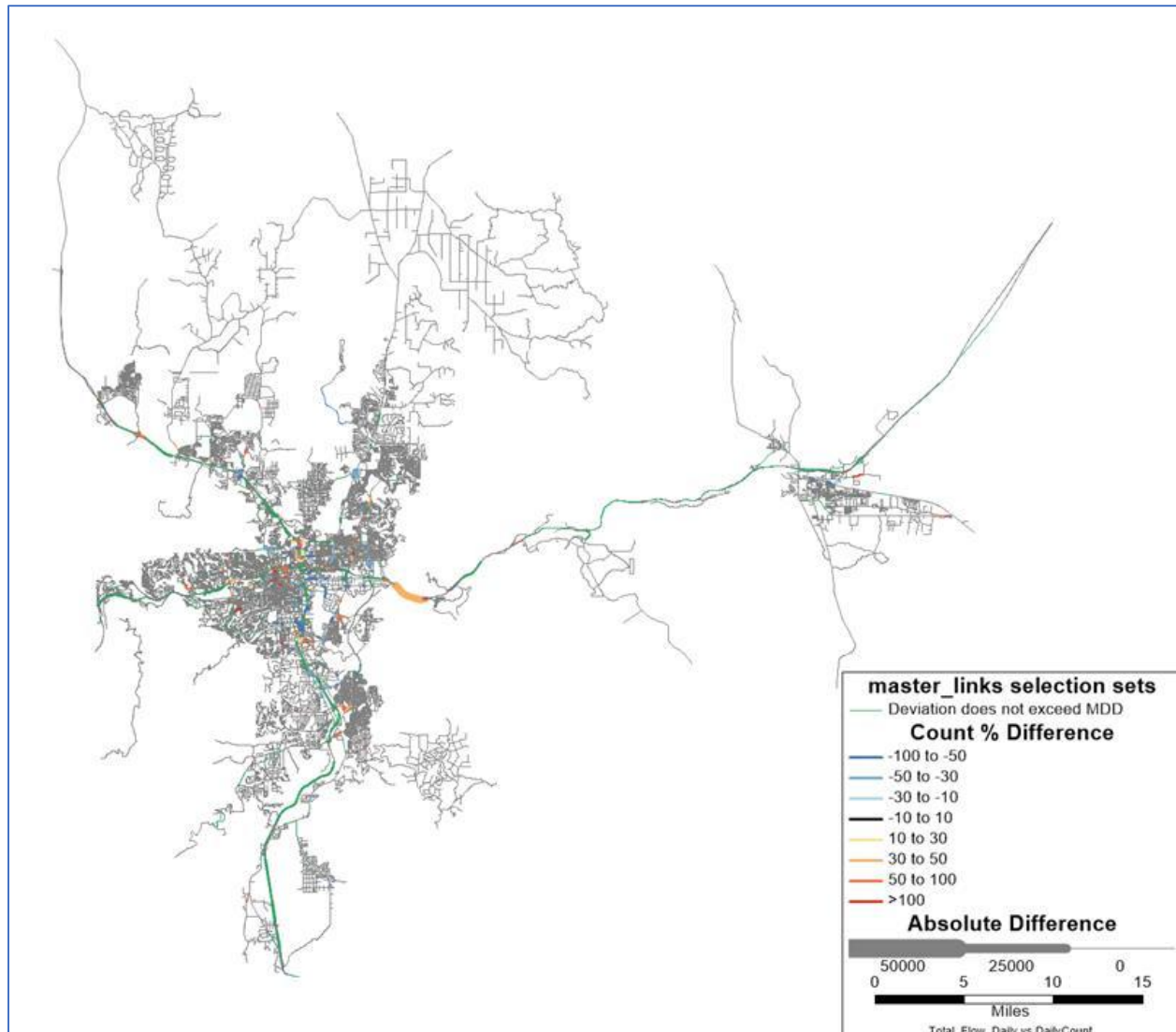


The table below presents the percent difference and percent root mean square error, which are measures of how well the model matches counts in aggregate. The table further breaks down this metric by volume group. The model matches count closely both overall and by volume group.

Volume Group	Counts	Avg. Count	Avg. Model	Avg. Error	RMSE
Less than 10,000	521	3,952	3,943	-0.2%	62.5%
10,000 - 25,000	200	15,817	15,131	-4.3%	36.6%
25,000 - 50,000	56	33,732	32,155	-4.7%	19.6%
50,000 - 100,000	18	71,611	74,582	4.2%	11.2%
Over 100,000	9	134,622	129,337	-3.9%	5.1%
All	804	11,955	11,676	-2.3%	34.8%

In addition to aggregate checks, Caliper performed link-level validation using maps like the one shown below. In this map, red colors show where the model is higher than counts while blue show where it is lower. Green indicates that the deviation is within the maximum desirable deviation range. No model with useful sensitivity will achieve a green color for all links. These maps guide the review process by showing where errors are largest and suggesting the presence of network or other errors.





Transit

Transit validation is done by checking total transit ridership to observed boardings and alightings. The table below shows unlinked trips (total boardings) for both model and observed. The observed boardings come from the 2024 APTA report.

Model: 18,490

Observed: 17,700

The model accurately predicts the aggregate level of transit usage.